



Flipsky ESC Tool V1.6
User Manual

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1 Overview

Flipsky ESC Tool is developed based on the FT ESC series, only adaptable for FT ESC series of Flipsky. The main function of this APP includes FT series ESCs and parameters settings, control mode setup, control test and data display, firmware updating, ESC restoration of factory settings, ESC reset and other functions.

Thanks for purchasing FT series ESC! In order to ensure the normal use of the product and avoid possible equipment damage, we recommend that you read this manual carefully before using and strictly follow the specified operating procedures. Our company will not be held responsible for any liability arising from any unauthorized modifications. Meanwhile, our company reserves the right to change the product design, appearance, performance and use requirements without notice.

2 Operation Requirements

2.1 Hardware Requirement

- FT Series ESC

Current Products including: FT85BS, FT85BS2.0, FT85BD, FT85RS, FT60BD, FT85KS, FT85GS, FT85CS_WP, FT110BS.

- Type-C USB cable

2.2 Software Requirement

- Windows 7 or above operation system is required;
- Windows 7 and Windows 8 need to install CDC port driver, or Flipsky ESC Tool cannot build communication with FT series ESC.
- Windows 10 or above have built-in CDC port driver.

2.2.1 System Driver Download and Related Files Description

Join QQ group to download related files: 1039697455。

Related folder including: (as Fig.2-1)

Fig.2-1. Folder content

名称	修改日期	类型	大小
Win7 or Win8 CDC Driver	2025/6/28 10:40	文件夹	
Flipsky_ESC_Tool_V1.6_20251014	2025/10/14 19:25	应用程序	95,654 KB
Flipsky ESC Tool V1.6 User Manual_...	2025/11/25 8:40	WPS PDF 文档	2,281 KB

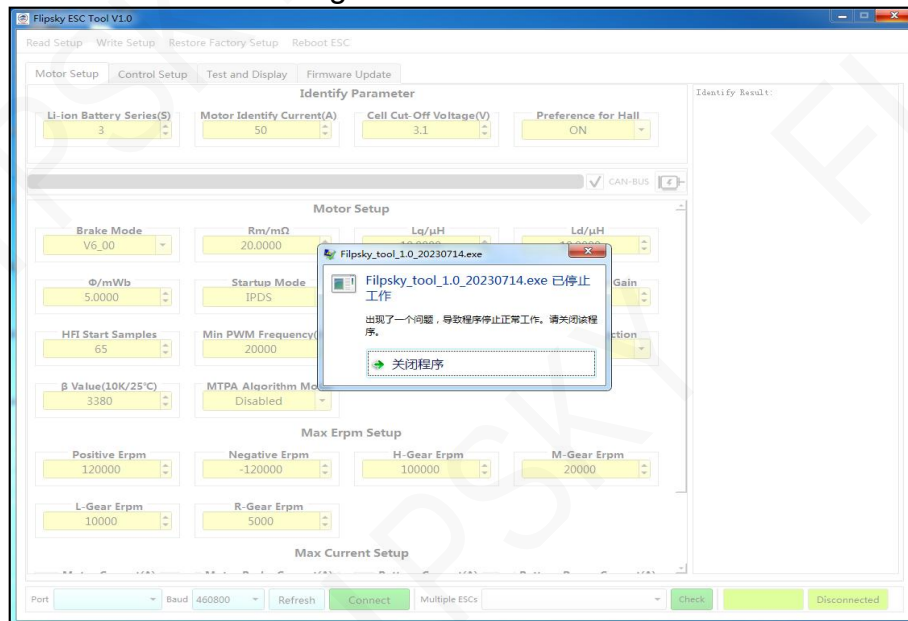
- Win7_or_ Win8 CDC Driver: Window 7 / Window 8 CDC port driver installation folder
- Flipsky ESC Tool V1.6 20250511.exe;
- Flipsky ESC Tool User Manual V1.6.

2.2.2 Windows 7 / Windows 8 Driver Installation

Take Windows 7 as an example:

The ESC might not connect with the ESC Tool for the first time if you use Windows 7. (as Fig.2-2) Please update / reinstall the port driver for your computer system to connect.

Fig.2-2. Fail to connect



Drive Update:

- Connect the ESC with computer via USB cable, right click the "Computer" and select "Device Manager" as Fig.2-3:

Fig. 2-3. Open Device Manager

Right-click the "Computer"



Select "Device Manager"

- When AT32 Virtual COM Port with a yellow icon shows as below(Fig.2-4), it means that you need to update the driver:
- Right-click the "AT32 Virtual COM Port" and select "Update Driver Software" as

Screenshot 2-4. COM port driver cannot be used with a yellow icon

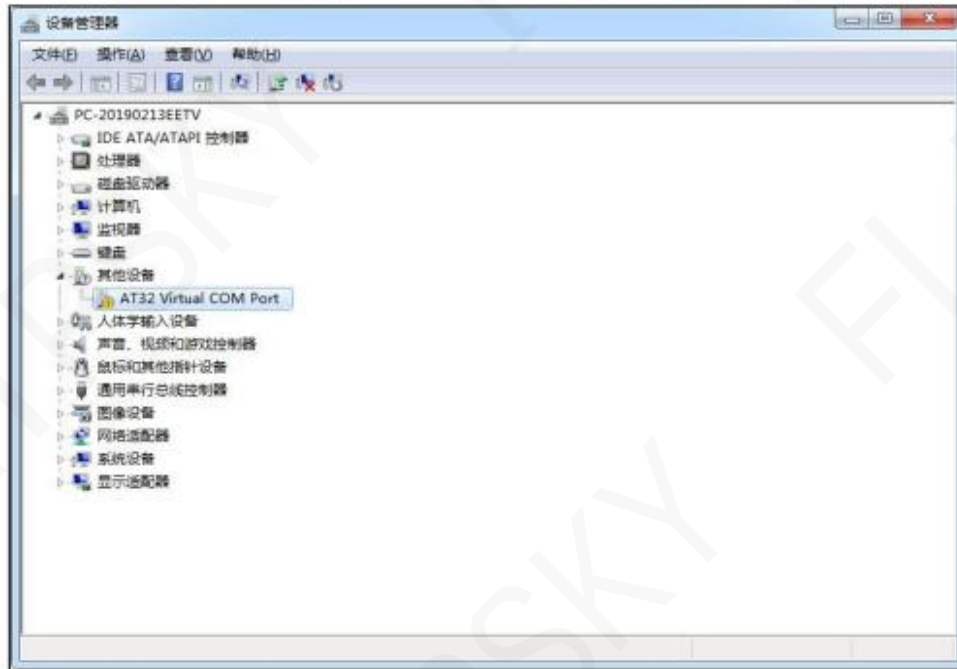
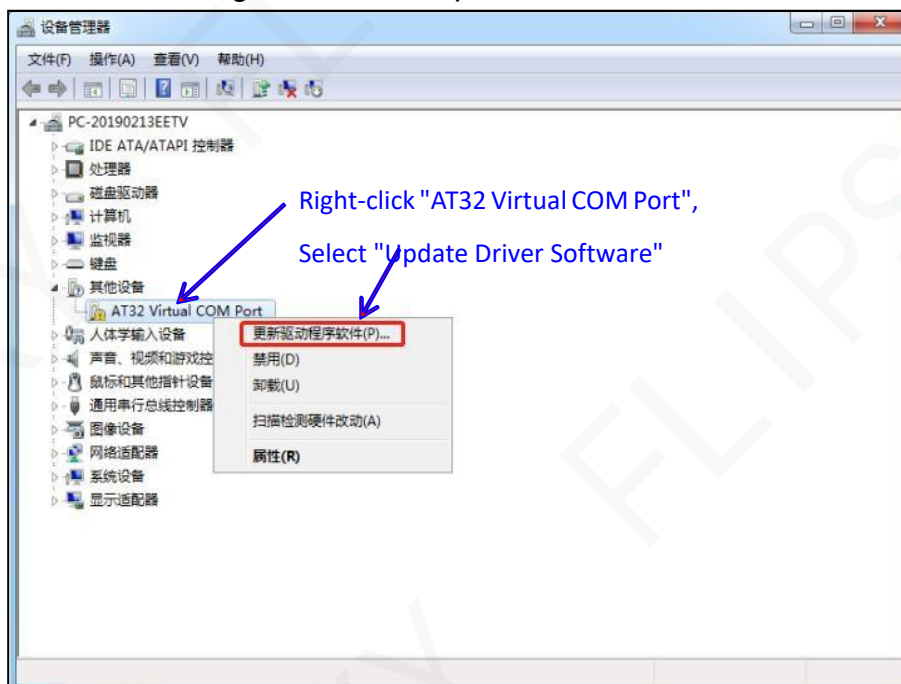
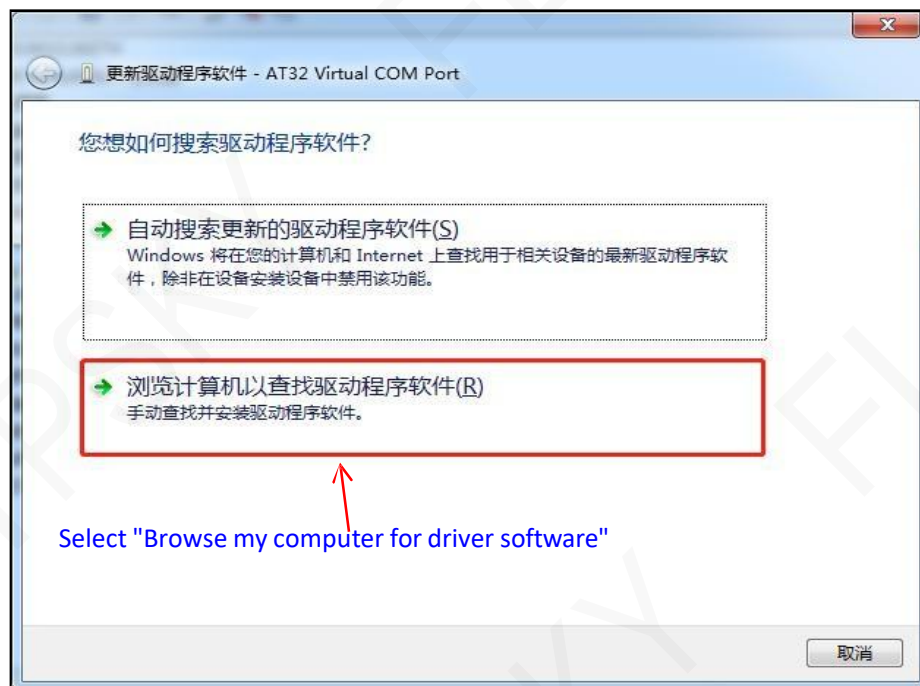


Fig.2-5.

Fig.2-5. Select "Update Driver Software"

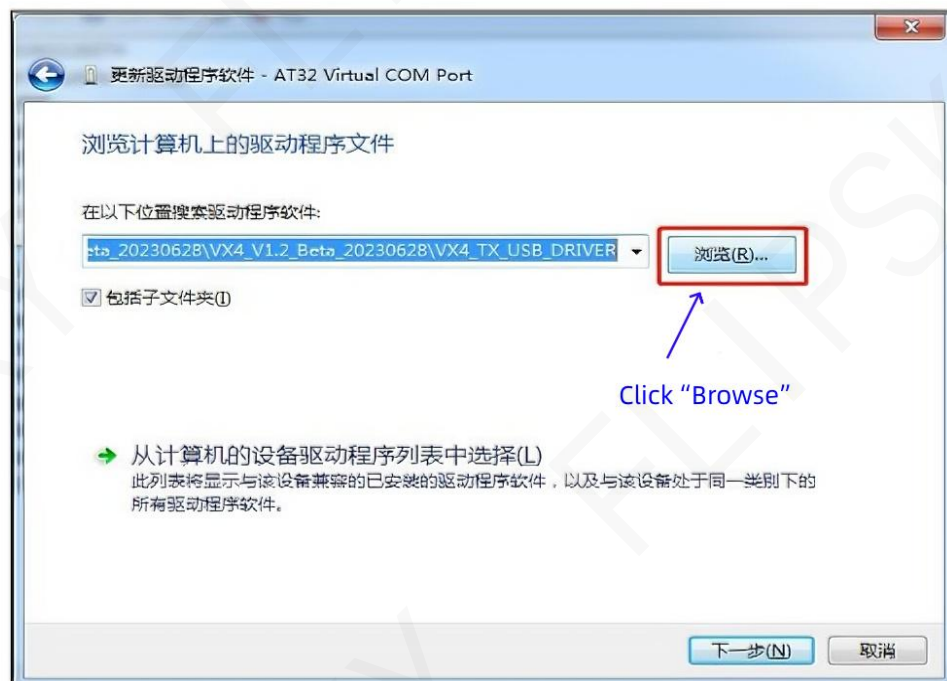


- Select "Browse my computer for driver software" as Fig.2-6:



- Click "Browse" to find the driver file as Screenshot 2-7 below .

Screenshot 2-7. Click "Browse" to find the location driver file



- Select right driver installation folder(the folder downloaded from the link above). Take Fig.2-8 as an example:

Screenshot 2-8. Select the driver installation folder



Find and Select "Win7_or_Win8 CDC Driver" Folder, and click "Next"

- If a security warning appears, select "Always install this driver software" as Fig.2-9 below:

Screenshot 2-9. Select "Always install this driver software"



Select "Always install this driver software"

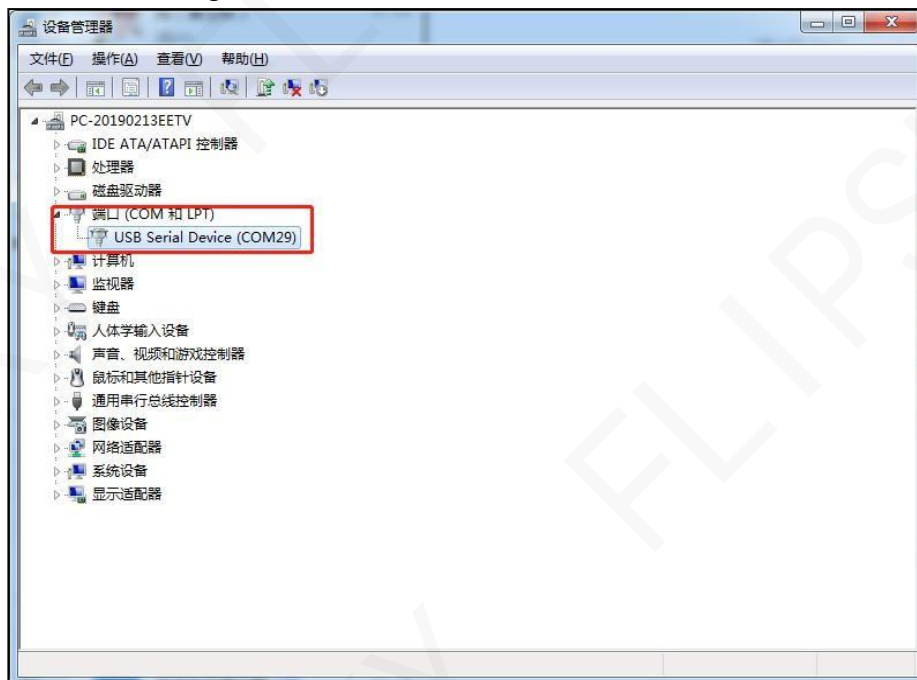
- The driver update finishes as Fig.2-10:

Fig.2-10. The Driver Installation Finish



- Enter the "Device Manager" and check whether yellow icon of the USB Serial Device disappears. If so, it means that the driver has been successfully updated as Fig.2-11:

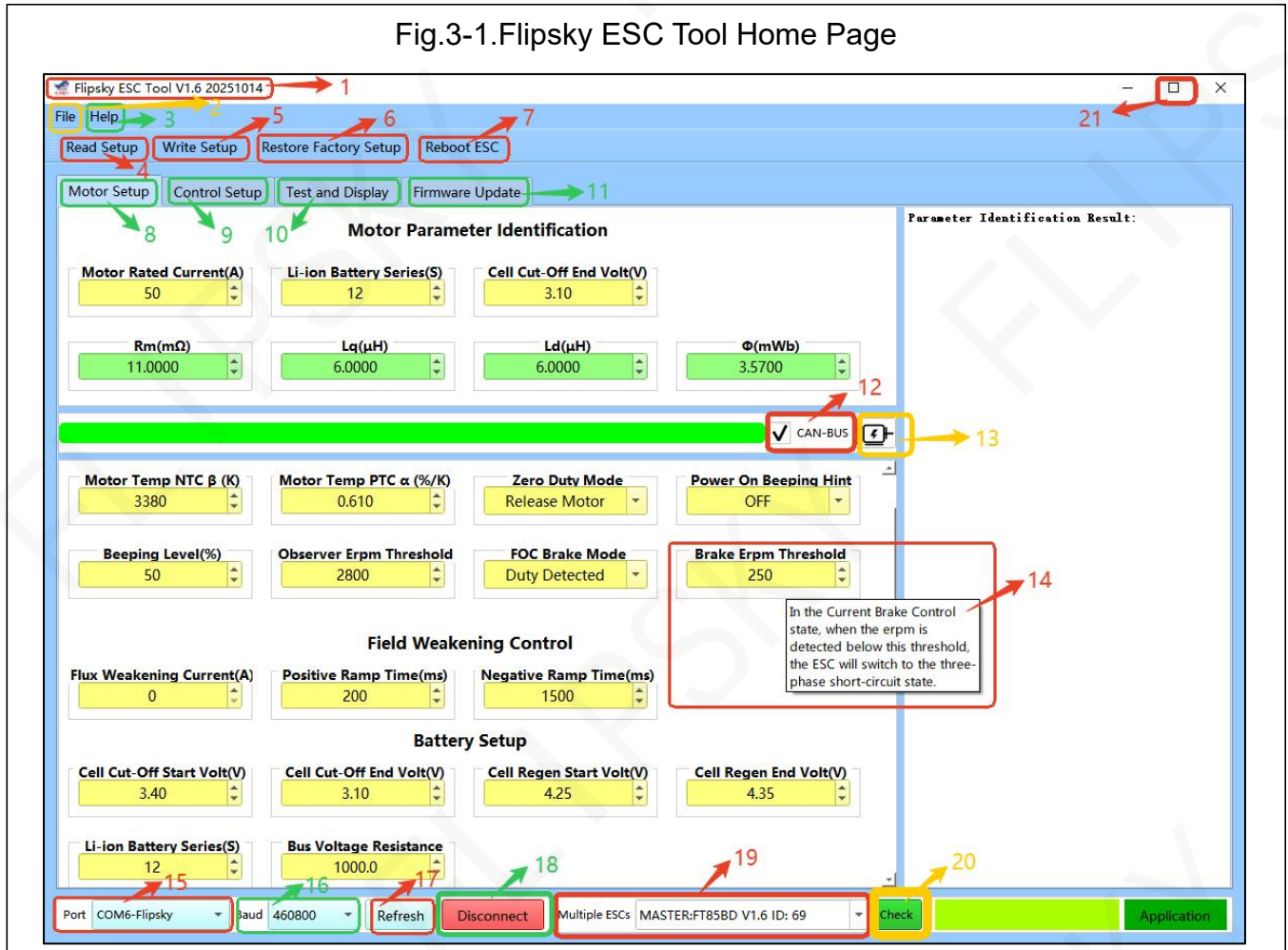
Fig.2-11. The USB Serial Device No Yellow Icon



3 Flipsky ESC Tool Home Page Introduction

Flipsky ESC Tool home page related controls introduction as Fig.3-1:

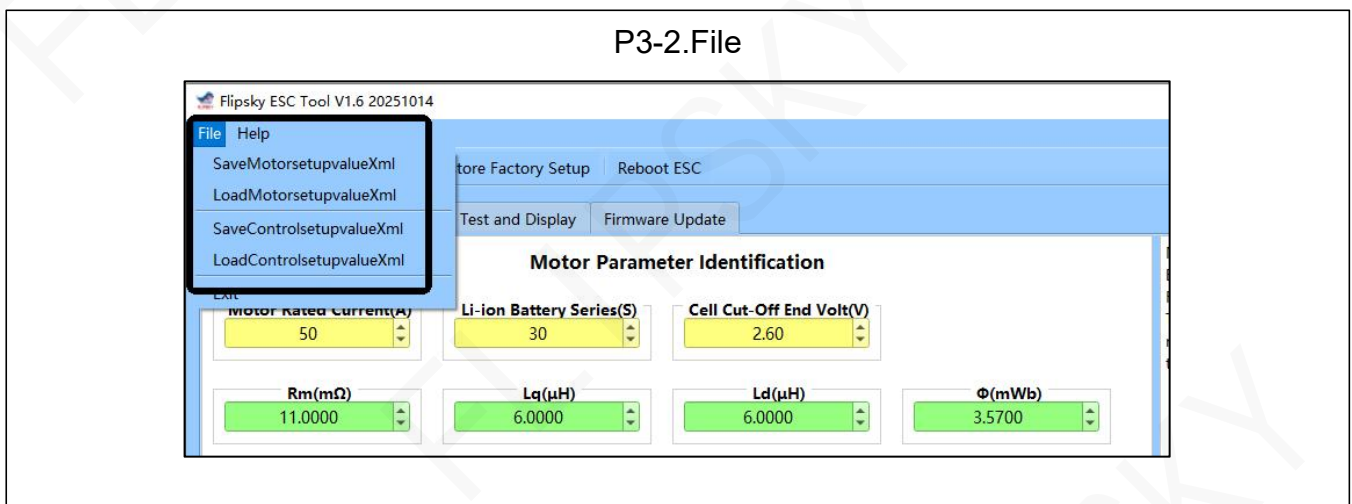
Fig.3-1.Flipsky ESC Tool Home Page



- (1) Flipsky ESC Tool V1.6 Version
- (2) File: Save/load configuration files
- (3) Help: Understand information about the ESC Tool
- (4) Read Setup: Read the current parameters of ESC
- (5) Write Setup: Write and save the current page parameters
- (6) Restore Factory Setup: Restore factory default setup, recover all ESC parameters to the initial default values
- (7) Reboot ESC: Click to re-connect the ESC
- (8) Motor Setup: Click to enter the motor setup
- (9) Control Setup: Click to enter the input control type setup
- (10) Test and Display: Click to enter the control test and data display interface
- (11) Firmware Update: Click to enter the firmware update interface

- (12) CAN-BUS: Click to identify multiple motor devices
- (13) Parameters Identification: Click to identify parameters
- (14) Explanation will be displayed when the mouse docks the parameters.
- (15) Port: Select COM
- (16) Baud: Select baud rate
- (17) Refresh: Detect and refresh Port
- (18) Connect / Disconnect: Connect/disconnect the ESC with the ESC Tool
- (19) Multiple ESCs: Check the model, number of ESC and Master and Slave ESC ID
- (20) Check: Search all ESCs on CAN bus
- (21) Maximize the window

■ File: Save/load configuration files



- Save Motor setup value Xml: Export and save current motor configurations
- Load Motor setup value Xml: Load motor configurations
- Save Control setup value Xml: Export and save current control configurations
- Load Control setup value Xml: Load control configurations

4 Build Communication

4.1 Build Communication of Flipsky ESC Tool and FT ESC

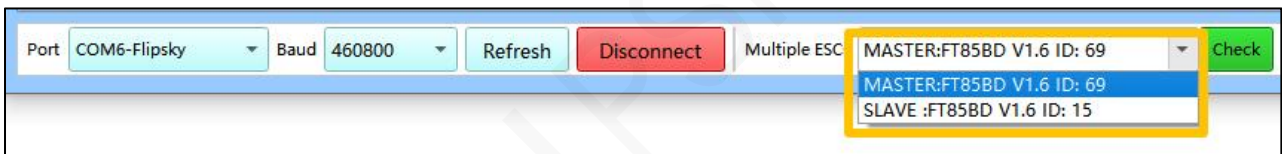
Power on the ESC and connect it with the computer via USB. Open the Flipsky ESC Tool V1.6 and the port for ESC ID (COM6-Flipsky) will be displayed. If no ID displays, please click "Refresh" to scan again as Fig.4-1.

Fig.4-1.Port display and select



Click "Connect" for ESC and ESC Tool connection. The ESC ID of corresponding model will appear on the screen as Fig.4-2.

Fig.4-2. Connection



The USB connection ESC is the master board, and all others connected via CAN bus are slave boards. Theoretically, 1 master board can be connected with a maximum of 253 slave CAN devices (except for master ESC ID, ESC ID=0 and ESC ID=255). While, it is recommended not using more than 8 slave boards in practice since too many CAN devices affect the working performance efficiency. Moreover, please remove the CAN terminal resistors except for the first and last ESCs' CANs. Set 1pcs Master ESC and 6pcs Slave ESCs as an example:

Master(Start)==Slave1==Slave2==Slave3==Slave4==Slave5==Slave6(End)

Remove the CAN terminal resistors of Slave1, Slave2, Slave3, Slave4 and Slave5, leaving only the CAN terminal resistors of Master (Start) and Slave6 (End).

4.2 CAN Devices Connection Error and Solution

If multiple ESCs are connected via CAN but one cannot communicate and the Flipsky ESC Tool detects that the number of ESC is less than actually connected, 3 possible causes are as below:

- 1) The slave ESC boards' CAN terminal resistors haven't been removed.
- 2) Different ESCs with same IDs
- 3) One of ESC's CAN baud rate is inconsistent with others.

Solution:

(1) Ensure that all CAN resistance are properly configured.(view Section 4.1)

(2) After step (1), unplug the CAN-BUS wires. Check the ESC ID one by one by connecting them with the ESC Tool separately via USB cable. If same IDs happen, revise them to make sure the uniqueness of the ESC ID.

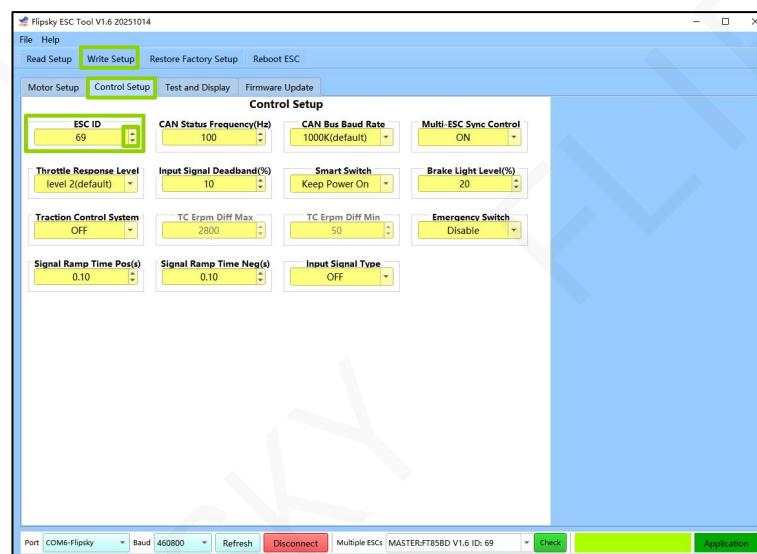
(3) If it has been verified that all ESCs have different ESC IDs and the CAN end resistance has been configured correctly, but you are still unable to detect the other ESCs on the CAN bus, you can continue to check whether the CAN baud rate setting is consistent with the others. Only you can communicate normally when you make sure that the CAN baud rate setup of all ESCs are the same.

(4) If still not succeed after the above trials, please restore all the ESCs default setup and try to reconnect.

4.2.1 ESC ID Setup

Setup steps are as Fig.4-3:

Fig.4-3.ESC ID setup



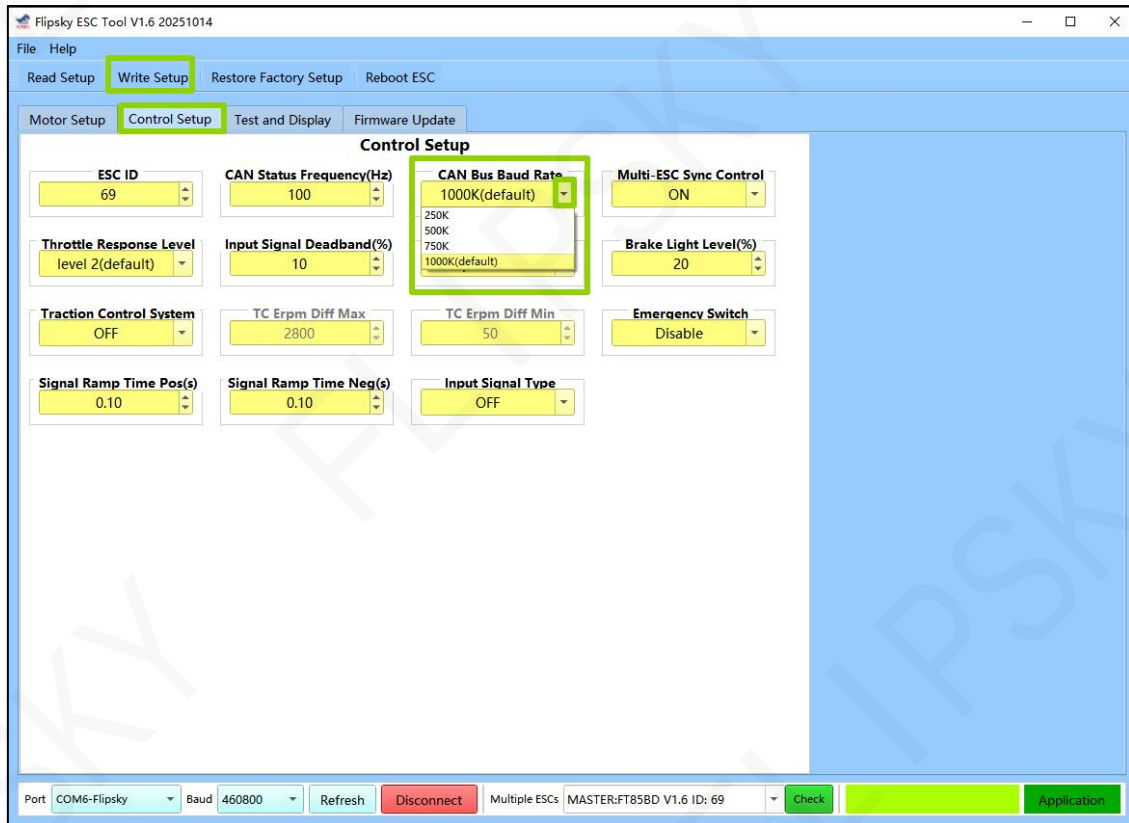
- Click “Control Setup” to enter the control setup interface
- Check the ESC ID, available range:1~254 (view details in section 7.1)
- Click "Write Setup" on the upper left to write the ESC ID into setup.

4.2.2 CAN Baud Rate Setup

Setting steps are as Fig.4-4:

- Click “Control Setup” to enter the control setup interface.
- Check the CAN Bus Baud Rate in"CAN Bus Baud Rate"(view details in Section 7.2)
- Click “Write Setup” on the upper left to save the setup(as Fig.4-4)

Fig.4-4.CAN Bus Baud Rate setup



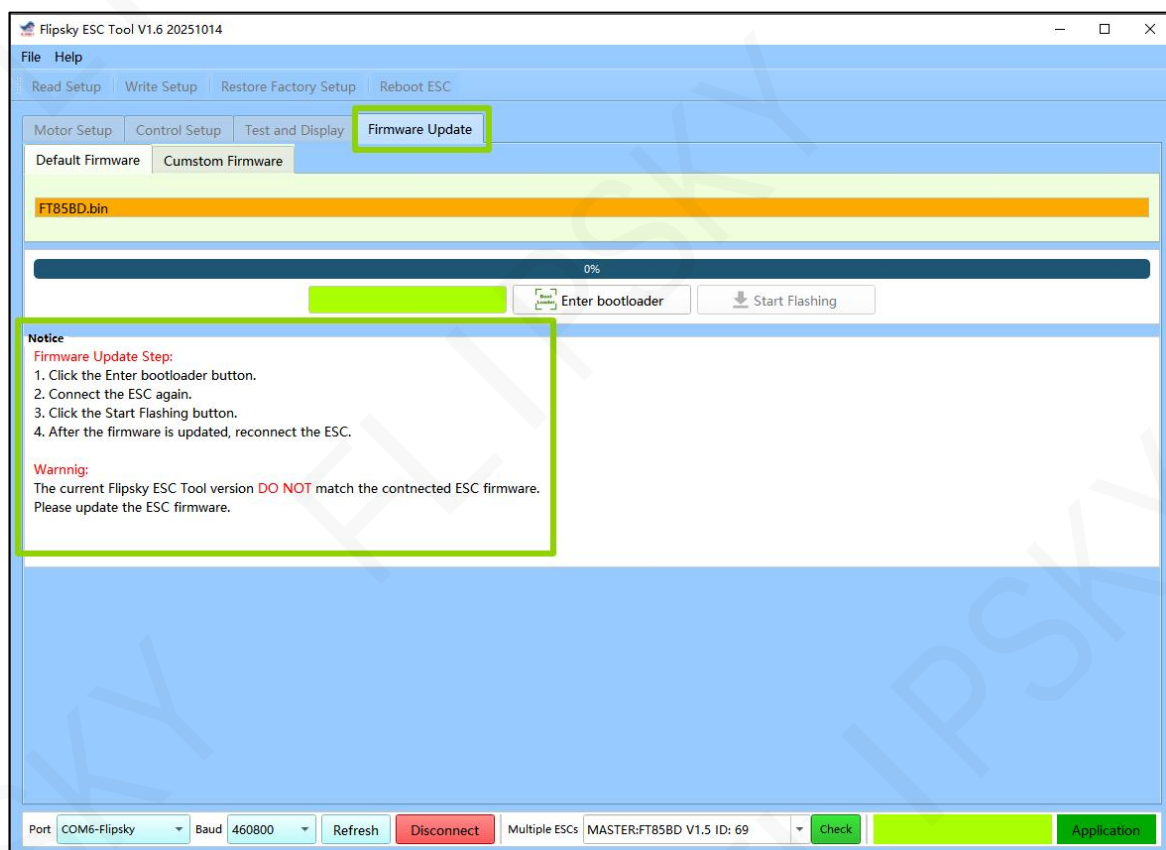
5 Firmware Update

5.1 FT Series ESC Firmware

Click “Firmware Update” to enter firmware update interface. It will automatically check if the ESC firmware version matches with the Flipsky ESC Tool. If not, please update the firmware as Fig.5-1.

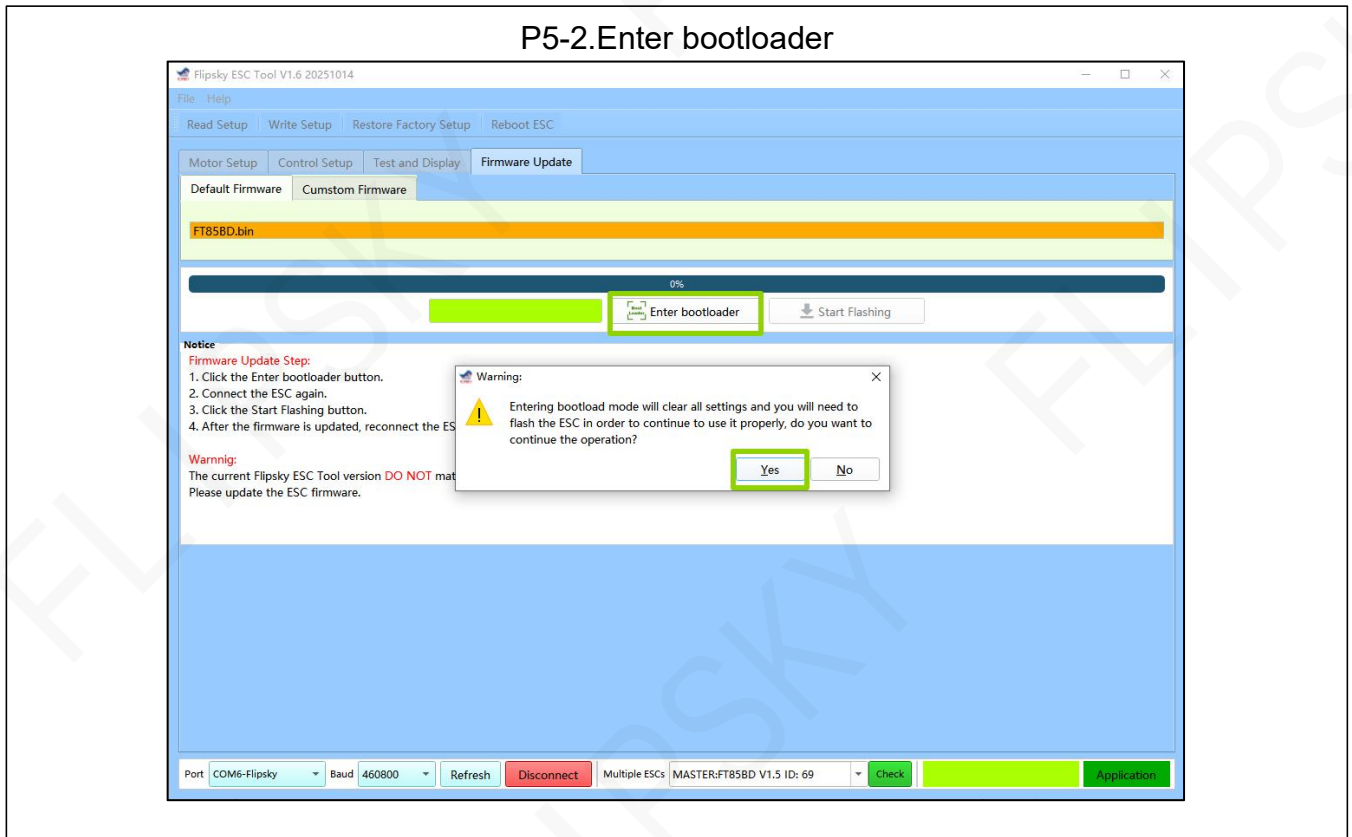
NOTE: When update firmware of dual ESC(like FT85BD), please update via USB cable of the master and slave independently.

Fig.5-1.Firmware update interface



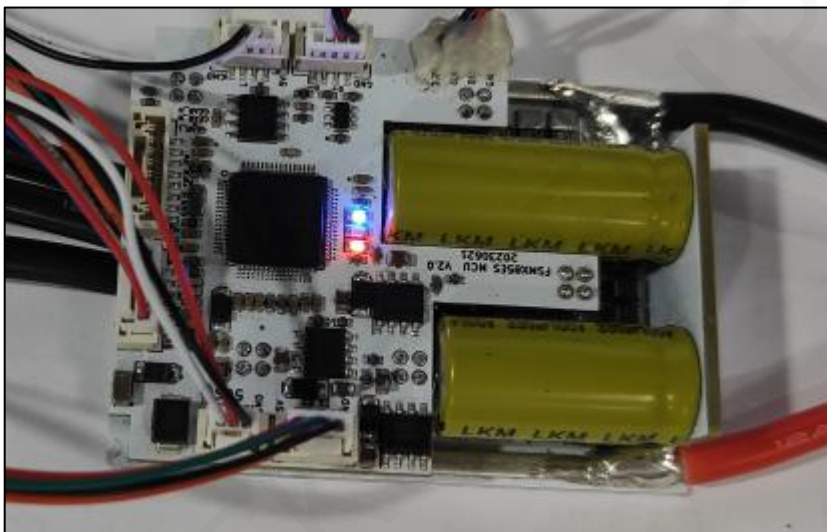
5.2 Firmware Update Steps

- Step 1, click “Enter bootloader” and “Yes” to enter the Bootloader mode as Fig.5-2:

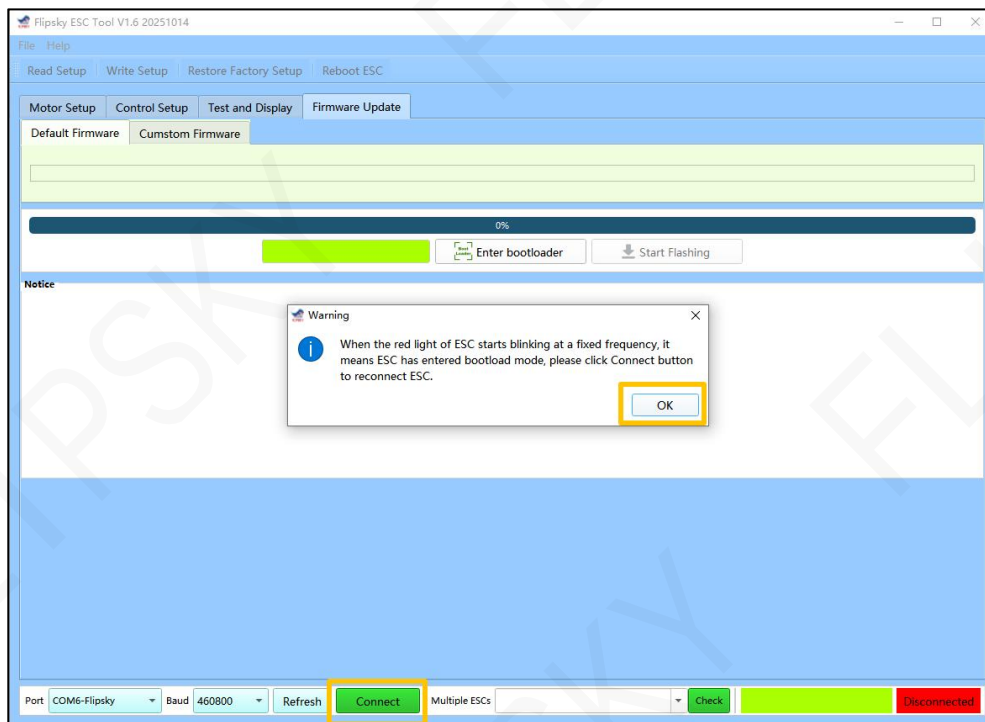


- Step 2: When the ESC red LED blinking at a fixed frequency, it means it has entered the Bootloader mode successfully (as Fig.5-3). Click "OK" and "Connect" to reconnect the ESC as Fig.5-4.

Fig.5-3.ESC enters the bootloader mode



P 5-4.Reconnect ESC



- Step 3: Click "Start Flashing", "Yes" to update corresponding firmware, as Fig.5-5, 5-6 below:

Fig.5-5.Start to update and download firmware

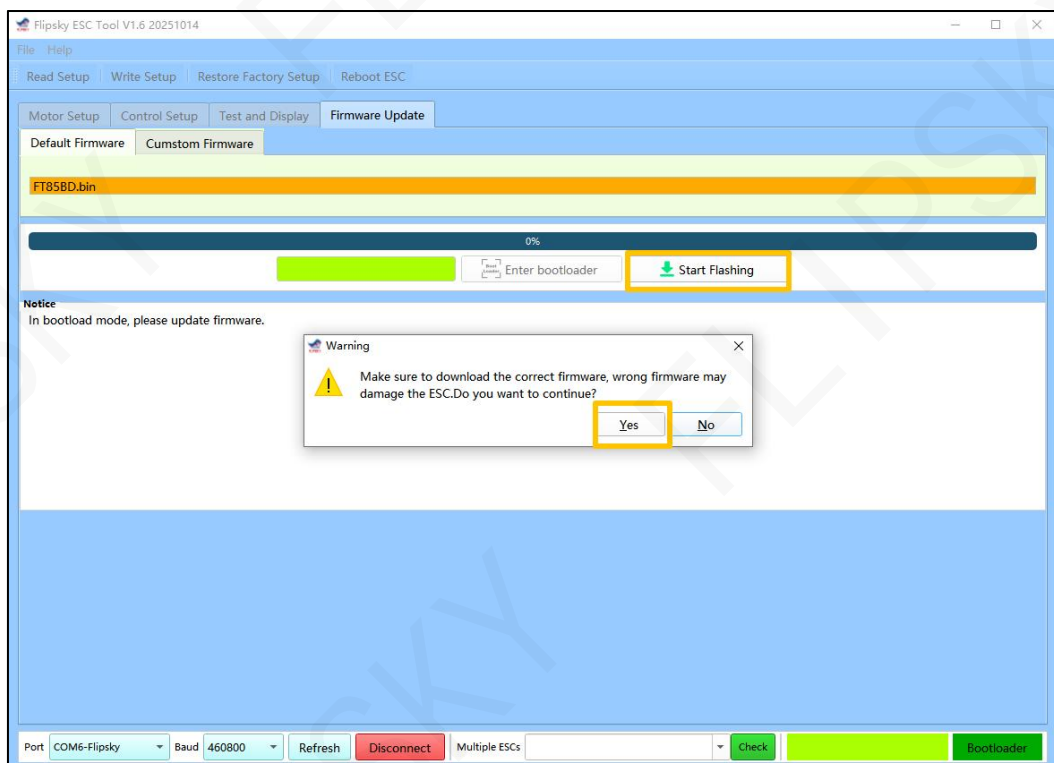
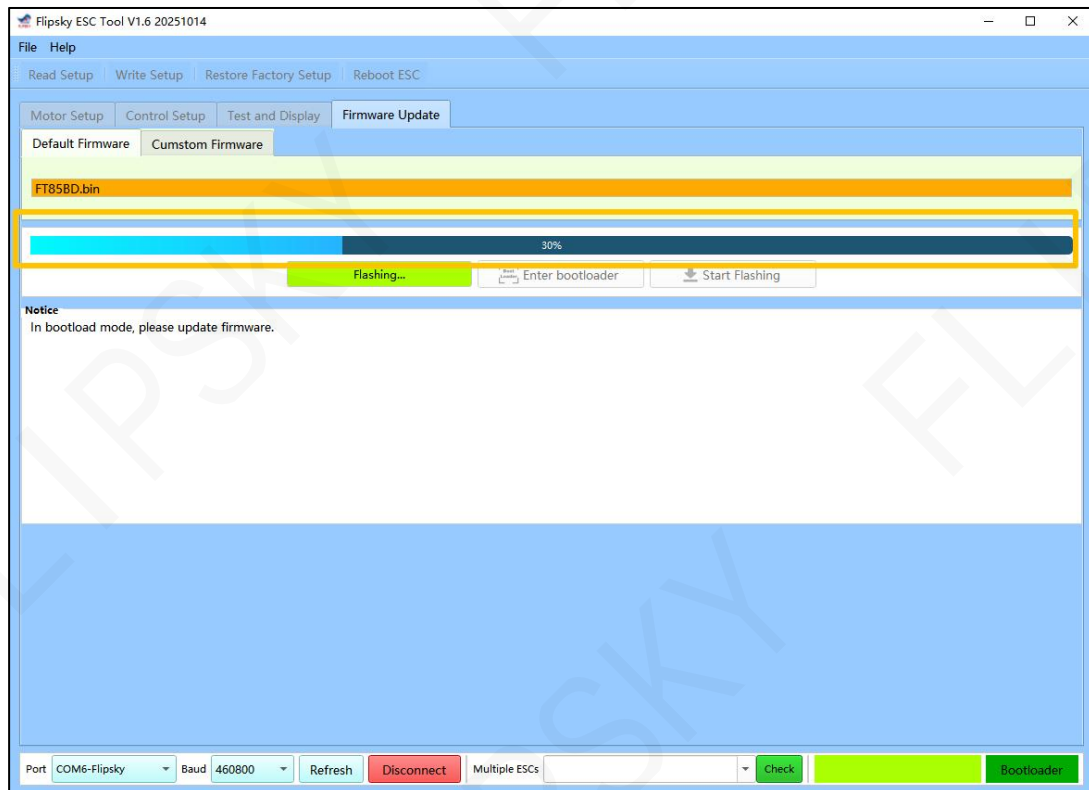
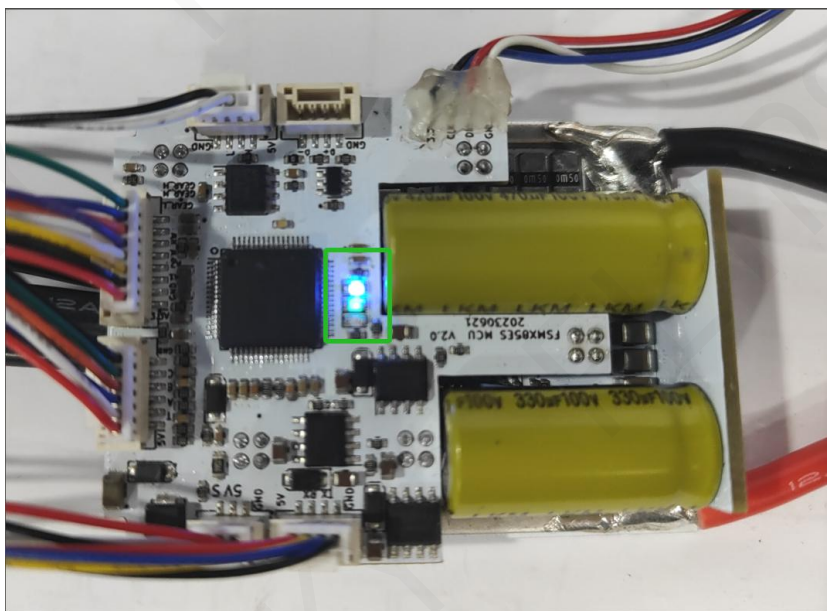


Fig.5-6.Downloading firmware



- When the ESC's green and blue LEDs keep on after the blinking of red and green LEDs, it means the firmware update succeeds(as Fig.5-7). Click "Connect" to reconnect the ESC.

Fig.5-7.Blue and green LEDs keep on



6 Motor Setup

Motor setup interface is as Fig.6-1:

Fig.6-1. Motor setup

Flipsky ESC Tool V1.6 20251014

File Help

Read Setup Write Setup Restore Factory Setup Reboot ESC

Motor Setup Control Setup Test and Display Firmware Update

Motor Parameter Identification

Motor Rated Current(A) 50 Li-ion Battery Series(S) 12 Cell Cut-Off End Volt(V) 3.10

Rm(mΩ) 31.1971 Lq(μH) 29.2474 Ld(μH) 18.4358 Φ(mWb) 5.2909

✓ CAN-BUS

ESC Basic Setup

Startup Mode HALL Motor Sensor Port Hall Sensors HFPI Saliency Gain 1.0 HFPI Voltage Gain 1.3

Rotation Direction Positive Min PWM Frequency(Hz) 9793 Max PWM Frequency(Hz) 14758 Motor Temp Sensor Type NTC 10K at 25°C

Motor Temp NTC β (K) 3380 Motor Temp PTC α (%/K) 0.610 Zero Duty Mode Release Motor Power On Beeping Hint OFF

Beeping Level(%) 50 Observer Erpm Threshold 2999 FOC Brake Mode Duty Detected Brake Erpm Threshold 250

Field Weakening Control

Flux Weakening Current(A) 0 Positive Ramp Time(ms) 200 Negative Ramp Time(ms) 1500

Port COM6-Flipsky Baud 460800 Refresh Disconnect Multiple ESCs MASTER:FT85BD V1.6 ID: 69 Check Application

6.1 Motor Parameters Identification

6.1.1 Preparations for Parameters Identification

Set "Motor Rated Current(A)", "Li-ion Battery Series(S)" and "Cell Cut-Off Voltage" before doing parameters identification as below Fig.6-2.

Fig.6-2.Three parameters setup before identification

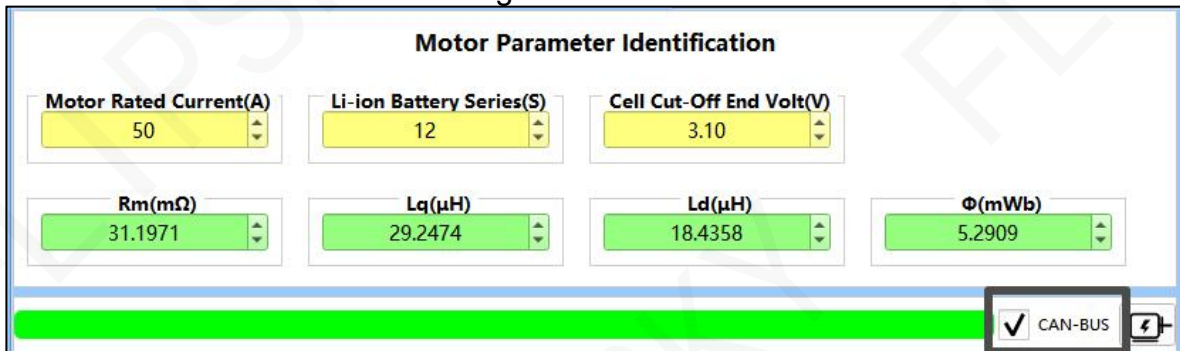


- **Motor Rated Current(A):** The maximum current of motor parameters identification(A). Please adjust the current according to the motor rated current. Default as 50A, available range: 5~500A.
- **Li-ion Battery Series(S):** Write the battery Lipos per battery specifications (1S=4.2V default: 12S; adjustable range: 3-100S)
- **Cell Cut-Off Voltage(V):** Battery voltage cutoff per cell setup to prevent the battery over discharge (default: 3.1V; adjustable range: 2.6 ~ 3.3V).

6.1.2 Parameter Identification Steps

- Step 1: Set "Motor Rated Current(A)", "Li-ion Battery Series(S)", "Cell Cut-Off Voltage(V)" accordingly.(as Fig.6-2)
- Step 2: When identifying 2 motors or above, please select "CAN-BUS"(if not, don't select) as Fig.6-3 as below.

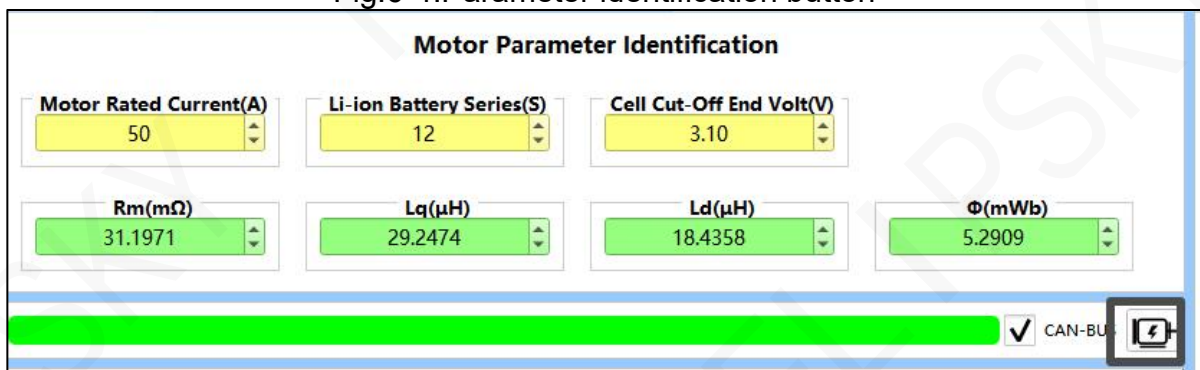
Fig.6-3.CAN-BUS



The screenshot shows the 'Motor Parameter Identification' window. It contains several input fields for motor parameters: Motor Rated Current(A) set to 50, Li-ion Battery Series(S) set to 12, Cell Cut-Off End Volt(V) set to 3.10, Rm(mΩ) set to 31.1971, Lq(μH) set to 29.2474, Ld(μH) set to 18.4358, and Φ(mWb) set to 5.2909. At the bottom right, there is a checkbox labeled 'CAN-BUS' which is checked, and a button with a lightning bolt icon.

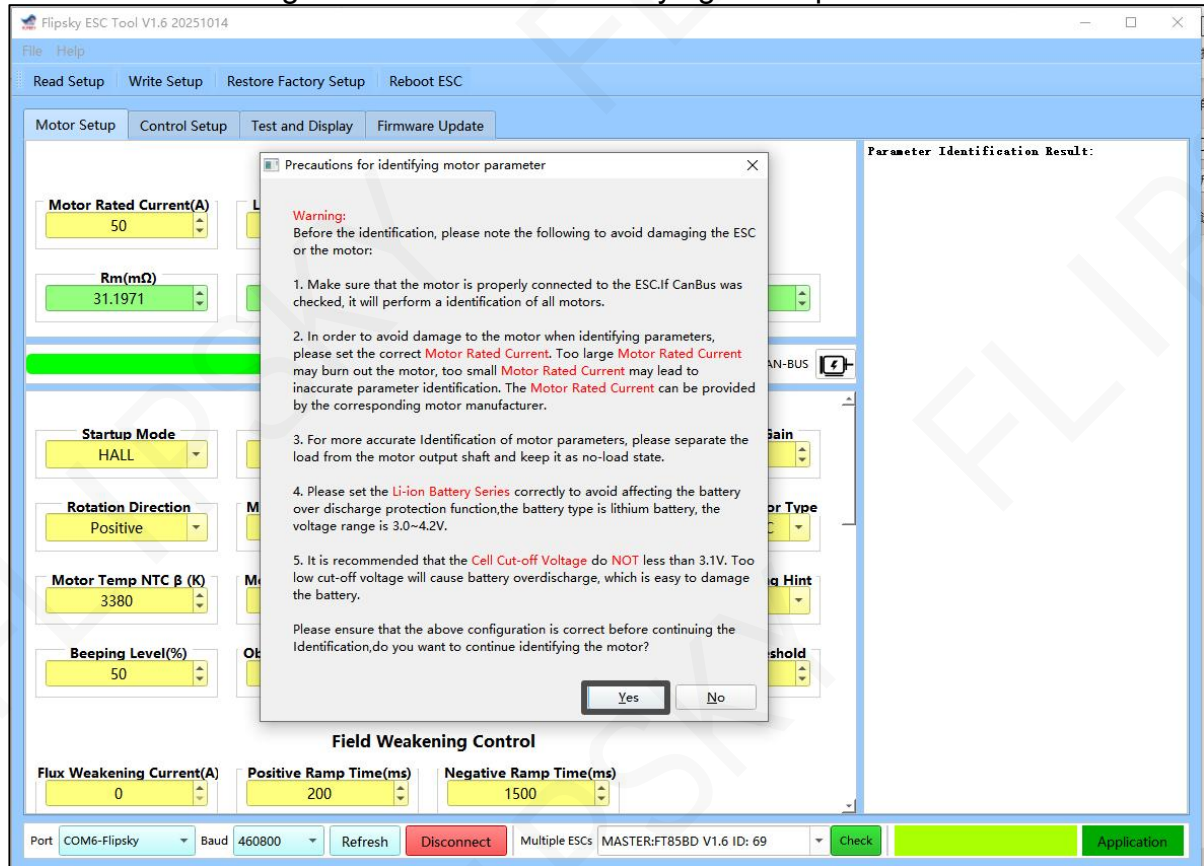
- Step 3: Click the parameter identification button(as Fig.6-4). The APP will show a warning dialog of precaution for identifying motor parameters. Click "Yes" to enter identification after confirmation.(as Fig.6-5)

Fig.6-4.Parameter identification button



This screenshot is identical to the one in Fig.6-3, showing the same parameter values. However, the button with the lightning bolt icon at the bottom right is now highlighted with a black border, indicating it is the 'parameter identification button' mentioned in the text.

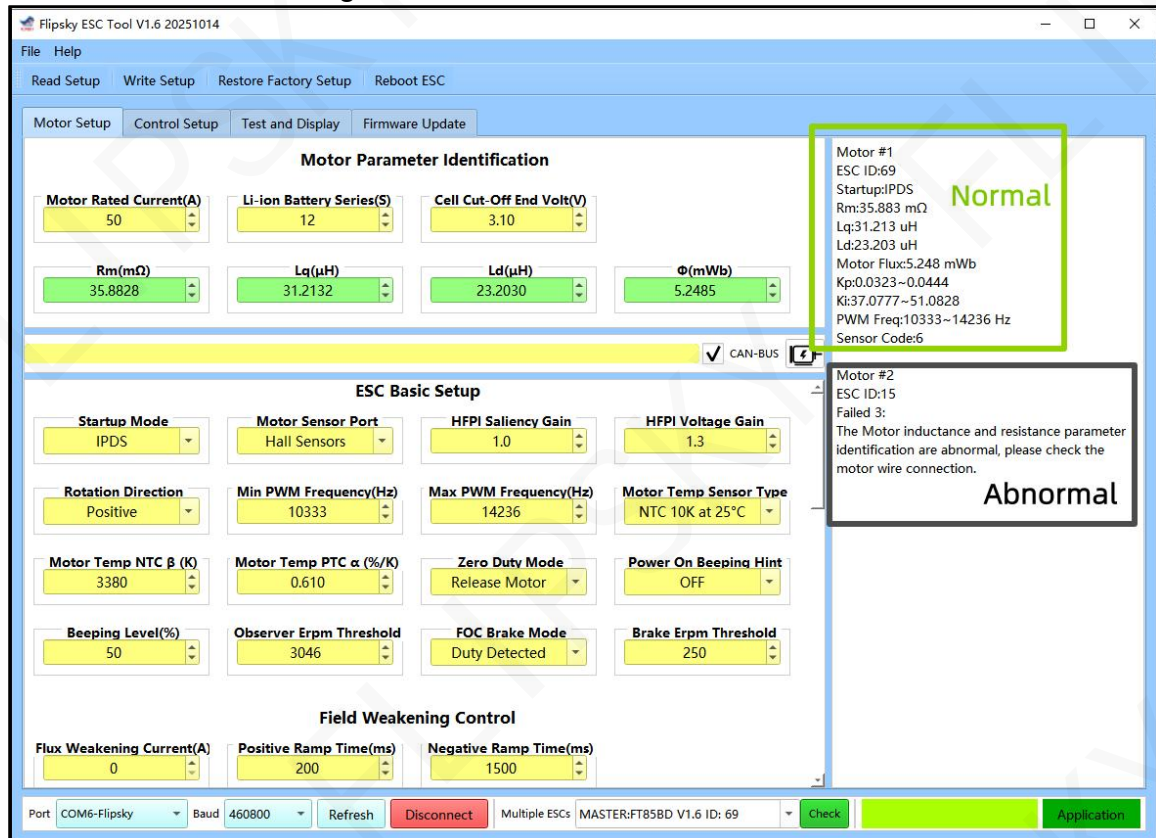
Fig.6-5.Precaution for identifying motor parameter



6.1.3 Motor Parameter Identification Results and Fault Codes

After identification, check the results on the right side of the home page.(as Fig.6-6) Check the result explanation to judge whether the identification is normal. If normal then you can continue the next step; if abnormal, find the causes through explanation.

Fig.6-6 Parameter identification results



● Motor Parameter Identification Results:

- ESC ID: ESC ID
- Startup: Motor Startup Mode:
 - ✧ IPDS: Initial Position Detection Startup
 - ✧ HFPI: High Frequency Pulse Injection
 - ✧ HALL: Hall Sensor Mode
 - ✧ ENCODER: Encoder Mode
 - ✧ OPEN: OPEN Mode
- Rm: Motor Phase Resistance (mΩ)
- Lq: Motor Q-Axis Inductance (uH)
- Ld: Motor D-Axis Inductance (uH)

- Motor Flux(Φ): Motor Flux Linkage (mWb)
- Kp、Ki: Dynamic variation range of Proportional(Kp) and Integral(Ki)
- PWM Freq: Dynamic variation range of PWM Frequency
- Hall Code:
 - ✧ HALL Sensor→Hall Code: 1
 - ✧ One Hall Phase Absence→Hall Code: 2
 - ✧ Two Hall Phase Absence→Hall Code: 3
 - ✧ Hall Three-Phase Short Circuit→Hall Code: 4
 - ✧ Overloading, Undetected HALL→Hall Code: 5
 - ✧ No HALL Sensor Connection→Hall Code: 6
- **Interpretation of motor parameters identification error:**
 - Failed 1: The battery S number is set incorrectly, and does not match the actual operating voltage. Please set the correct number of battery S.

NOTE: The battery S number shall match with the actual number, or it cannot be identified/activate the over discharged protection, affecting battery lifespan.
 - Failed 2: The ESC is still faulty, please check the error code.
 - Failed 3: The Motor inductance and resistance parameter identification are abnormal, please check the motor wire connection.
 - Failed 4: The motor hall sensor identify failed. please check the motor hall sensor.
 - Failed 5: The motor flux linkage identify failed. Maybe the motor load is too heavy.

6.2 ESC Adjustable Parameters Definition and Configuration

Some motor parameters like R_m , L_q , L_d , Φ etc. will be calculated automatically after the parameter identification finish which don't need to change manually. Some control parameters (e.g. Startup Mode, HFPI Saliency Gain, HFPI Voltage Gain, Max PWM Frequency, Min PWM Frequency, etc.) will be reasonably derived by the adaptive algorithm according to the motor characteristics and whether using the HALL sensor. If not necessary, these parameters do not need to be modified. Other parameters (e.g. Max Erpm, Max Current) can be manually set accordingly.

Adjustable parameters including: ESC Basic Setup, Field Weakening Control, Max Erpm Setup, Max Power Setup, Max Current Setup, Limitation, Encoder Setup, rpm PID setting and location PID setting, as Fig.6-7.

NOTE: Remember to click the upper left button "Write Setup" to save the configurations.

Fig.6-7 Adjustable Parameters

Flipsky ESC Tool V1.6 20251014

File Help

Read Setup **Write Setup** Restore Factory Setup Reboot ESC

Motor Setup Control Setup Test and Display Firmware Update

Motor Parameter Identification

Motor Rated Current(A)	Li-ion Battery Series(S)	Cell Cut-Off End Volt(V)
50	12	3.10
$R_m(m\Omega)$	$L_q(\mu H)$	$L_d(\mu H)$
33.4043	30.6763	19.2176
$\Phi(mWb)$		
5.2202		

☒ CAN-BUS

ESC Basic Setup

Startup Mode	Motor Sensor Port	HFPI Saliency Gain	HFPI Voltage Gain
IPDS	Hall Sensors	1.0	1.3
Rotation Direction	Min PWM Frequency(Hz)	Max PWM Frequency(Hz)	Motor Temp Sensor Type
Positive	9962	15127	NTC 10K at 25°C
Motor Temp NTC β (K)	Motor Temp PTC α (%/K)	Zero Duty Mode	Power On Beeping Hint
3380	0.610	Release Motor	OFF
Beeping Level(%)	Observer Erpm Threshold	FOC Brake Mode	Brake Erpm Threshold
50	3187	Duty Detected	250

Field Weakening Control

Flux Weakening Current(A)	Positive Ramp Time(ms)	Negative Ramp Time(ms)
0	200	1500

Port COM6-Flipsky Baud 460800 Refresh Disconnect Multiple ESCs MASTER:FT85BD V1.6 ID: 16 Check Application

6.2.1 Basic ESC Parameters Setup

Basic ESC parameters including "Startup Mode", "Motor Sensor Port", "HFPI Saliency Gain", "HFPI Voltage Gain Volt", "Rotation Direction", "Min PWM Frequency", "Max PWM Frequency", "Motor Temp sensor Type", "Motor Temp NTC β (K)", "Motor Temp PTC α (%/K)", "Zero Duty Mode", "Power on Beeping Hint", "Beeping Level(%)", "Observer ErpmThreshold", "FOC Brake Mode", "Brake Erpm Threshold". As Fig.6-8:

Fig.6-8.Basic ESC parameters

ESC Basic Setup			
Startup Mode IPDS	Motor Sensor Port Hall Sensors	HFPI Saliency Gain 1.0	HFPI Voltage Gain 1.3
Rotation Direction Positive	Min PWM Frequency(Hz) 9962	Max PWM Frequency(Hz) 15127	Motor Temp Sensor Type NTC 10K at 25°C
Motor Temp NTC β (K) 3380	Motor Temp PTC α (%/K) 0.610	Zero Duty Mode Release Motor	Power On Beeping Hint OFF
Beeping Level(%) 50	Observer Erpm Threshold 3187	FOC Brake Mode Duty Detected	Brake Erpm Threshold 250

● Startup Mode:

Three motor startup modes:

- ✧ IPDS: Initial Position Detection Startup (suitable for light loading sensorless mode such as underwater DPVs, multi-axis FPVs etc.)
- ✧ HFPI: High Frequency Pulse Injection (suitable for heavy loading sensorless mode such as E-Bike, E-Board, mini buggy etc.)
- ✧ HALL: Hall Sensor Startup(widely used, especially suitable for equipment with full load)
- ✧ ENCODER: Encoder Sensor Startup

NOTE: HFPI mode is not suitable for all motors. If there are serious vibrations, stalls etc. when using this mode, it means that this mode is not suitable for the motor. If the motor integrates HALL interface, we strongly recommend using HALL for smooth running and better performance.

● Motor Sensor Port:

Before setup, please choose the corresponding motor sensor port and remember to click "Write Setup".

- ✧ Hall sensor: Three Hall effect position sensors. A Hall effect component is installed every 60°/120° electrical Angle.
- ✧ ABI Encoder: An incremental encoder is a linear or rotary electric-mechanical device that has two output signals, A and B, which issue pulses when the device is moved.

NOTE: Since the incremental encoder cannot determine the initial position of the rotor, it is recommended to rotate the motor manually one circle for zero position calibration after the first start of the ESC, otherwise unexpected situations may occur when the control command is forcibly sent.

- ✧ AS5047(SPI): An absolute encoder maintains position information when power is removed from the encoder, which will immediately report changes and no need to do zero position calibration.

● HFPI Saliency Gain

Adjustable range: 0.5~4.0

Parameter Specification: The surface-mounted permanent magnet synchronous motor, because of its weak convex polarity, has a large noise in the angle measurement when using the HFPI mode, and this parameter is designed to improve the accuracy of the angle measurement. This value is automatically done by the algorithm after the motor parameters are recognized. The algorithm will calculate a reasonable value according to the identification results. **It's no need to change if not necessary.**

● HFPI Voltage Gain

Adjustable range: 0.2~3.0

Parameter Specification: Parameter description: This parameter setup is designed for high-frequency pulse voltage injection needs of motor D-axis in HFPI mode, which is determined by the motor specification and can be matched automatically after the motor parameter identification finish, so **it's no need to write manually if not necessary.**

● Rotation Direction

- ✧ Positive
- ✧ Negative

● Min PWM Frequency

Adjustable range: 2000~25000

Parameter Specification: This parameter can be matched automatically after the motor parameter identification finish and **it's no need to write manually if not necessary.**

- **Max PWM Frequency**

Adjustable range: 10000~25000

Parameter Specification: This parameter can be matched automatically after the motor parameter identification finish and **it's no need to write manually if not necessary.**

- **Motor Temp sensor Type**

- ✧ NTC 10K at 25°C
- ✧ NTC 10K at 25°C
- ✧ NTC 100K at 25°C
- ✧ PT1000PTC 1K at 100°C
- ✧ KTY83-122
- ✧ KTY83-152
- ✧ KTY84-130
- ✧ KTY84-150
- ✧ KTY84-151
- ✧ KTY84-152

Parameter Specification: NTC 10K temperature sensor with a temperature of 25°C is the most widely used one in applications but some others are also used in special application fields.

- **Motor Temp NTC β (K)**

Adjustable range: 1~10000

Parameter Specification: A thermistor's " β " value, or beta value, is an indication of the shape of the curve representing the relationship between resistance and temperature of an NTC thermistor. Please write the correct data per your motor. Flipsky's motor NTC β is 3380.

- **Motor Temp PTC α (%/K)**

Adjustable range: 0.5~100

Parameter Specification: The PTC type temperature sensor α coefficient value. Please select corresponding temperature sensor type.

- **Zero Duty Mode**

- ✧ Keep Output
- ✧ Release Motor

Parameter Specification: Zero Duty Mode. If set duty cycle 0% in Zero Duty Mode, users can choose the mode of Keep Output or Release Motor.

- **Power on Beeping Hint**

- ✧ OFF
- ✧ ON

Parameter Specification: When activate the function, the ESC will beep for 3 times after powering on.

NOTE: Motor parameter identification must be completed before turning on the function, otherwise the sound is abnormal.

- **Beeping Level(%)**

Adjustable range: 0~1

Parameter Specification: The larger of the value, the louder of the sound.

- **Observer Erpm Threshold**

Adjustable range: 1~250000

Parameter Specification: When the rpm is over than this value, switch to the observer erpm threshold to measure the rotor position and rpm.

- **FOC Brake Mode**

Available mode: Duty Cycle Detected、Erpm Detected、Duty Erpm Detected。

Parameter Specification:

- ✧ Duty Cycle Detected: When enter brake status and detect the Duty cycle is lower than 0.001, switch to the three-phase short-circuit status.
- ✧ Erpm Detected: When enter brake status and detect the Erpm is lower than Brake Erpm Threshold, switch to the three-phase short-circuit status.
- ✧ Duty Erpm Detected: When detect the duty cycle is lower than 0.001 and the absolute value of Erpm is lower than Brake Erpm Threshold in the meanwhile, switch to the three-phase short-circuit status.

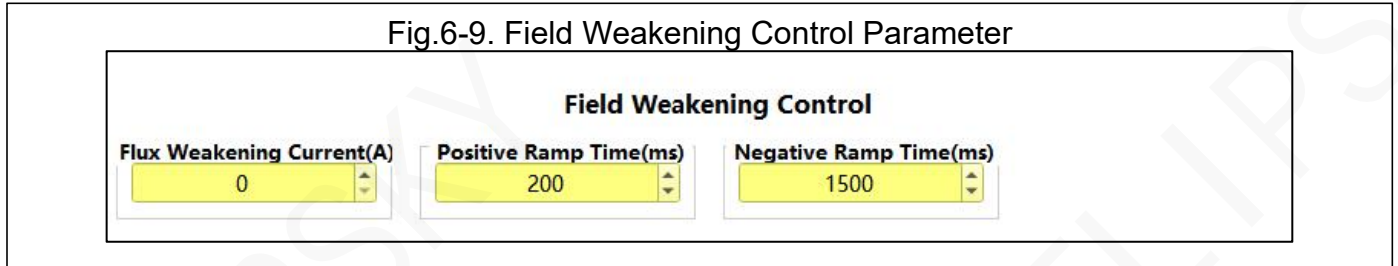
- **Brake Erpm Threshold**

Adjustable range: 0~8000

Parameter Specification: When in this brake control status, if the detected erpm is lower than this brake Erpm threshold, the ESC will switch to the three-phase short-circuit status.

6.2.2 Field Weakening Control Parameter Setup

Field weakening control parameters including "Flux Weakening Current(A)", "Positive Ramp Time(ms)", "Negative Ramp Time(ms)"; as Fig.6-9:



- **Flux Weakening Current**

Adjustable range: 0~500

Parameter Specification: the current applied to the D-axis in field weakening control. Increasing this value can increase the maximum speed limit of the motor. The higher the value, the higher the maximum speed, but at the same time, the higher the current consumption and the hotter the motor, it is recommended to set it lower than 1/4 of the Motor Current Max.

- **Flux Weakening Pos ramp time**

Adjustable range: 0~10000

Parameter Specification: When the Positive Ramp Time is set as 1000ms, the flux weakening current to 20A, the flux weakening Current will increase from 0A to 20A in 1000ms when the duty cycle reaches 90% or above.

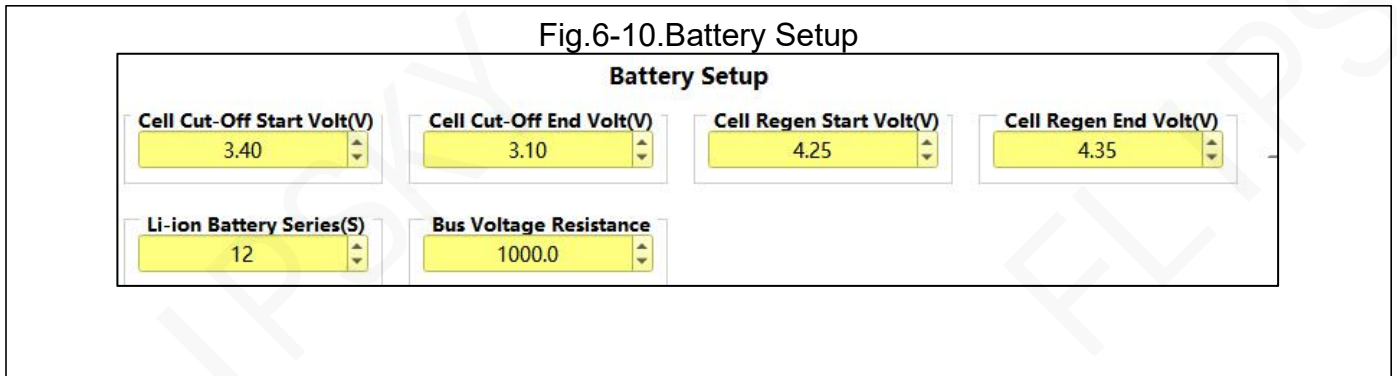
- **Flux Weakening Neg ramp time**

Adjustable range: 0~10000

Parameter Specification: When the Negative Ramp Time is set as 1000ms, and the current control is set to be 0, the flux weakening current will fall from 20A to 0A in 1000ms and ends the flux weakening control.

6.2.3 Battery Setup

Maximum motor rpm including "Cell Cut-Off start Volt(V)", "Cell Cut-Off End Volt(V)", "Cell Regen Start Volt(V)", "Cell Regen End Volt(V)", "Li-ion Battery Series(S)", "Bus Voltage Resistance". As Fig.6-10:



Cell Cut-off Start Volt(V)

Adjustable Range: 2.5~3.8V

Parameter Specification: Single-cell start cut-off voltage setting, when the voltage of a single cell falls below this setting value, the ESC will reduce the output power. NOTE: the output power will decrease gradually when the voltage is below this value.

● Cell Cut-off End Volt(V)

Adjustable Range: 2.5~3.8V

Parameter Specification: Single-cell end cut-off voltage setting, when the voltage of a single cell falls below this value, the ESC will stop outputting power.

NOTE: power output will be completely disabled when the voltage is below this value.

● Cell Regen Start Volt

Adjustable Range: 2.6~5V

Parameter Specification: The regenerative power will be decreased when the battery voltage exceeds this value.

NOTE: When the voltage exceeds this value, regenerative brake will decrease gradually.

● Cell Regen End Volt

Adjustable Range: 2.6~5V

Parameter Specification: When the voltage exceeds this value, the regenerative brake current will not be allowed.

NOTE: power output will be completely disabled when the voltage exceeds this value.

- **Li-ion Battery Series(s)**

Adjustable Range: 3~100S

Parameter Specification: Li-ion battery series setup, please set according to the using battery voltage(1S=4.2V).

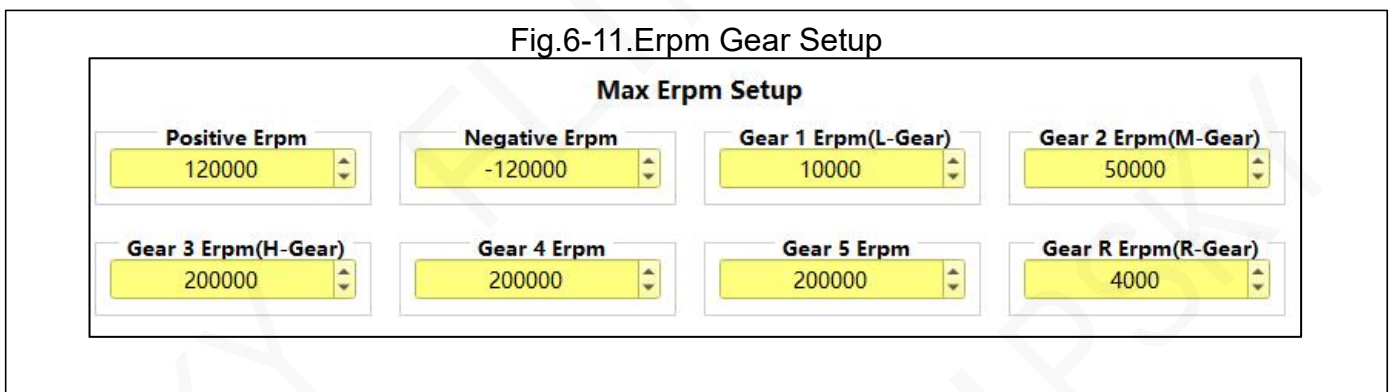
- **Bus Voltage Resistance**

Adjustable Range: 980~1020

Parameter Specification: Minor adjustments can improve the accuracy of the power supply voltage measured by the ESC (adjust based on the voltage reading from the multimeter or the voltage displayed on the switching power supply). Retain the default settings unless necessary.

6.2.4 Erpm Gear Setup

Max Erpm Setup including "Positive Erpm", "Negative Erpm", "Gear 1 Erpm(L-Gear)", "Gear 2 Erpm(M-Gear)", "Gear 3 Erpm(H-Gear)", "Gear 4 Erpm", "Gear 5 Erpm""R-Gear Max Erpm". As Fig.6-11:



- **Positive Erpm**

Adjustable Range: 500~250000

- **Negative Erpm**

Adjustable Range: -250000~-500

- **Gear 1 Erpm(L-Gear)**

Adjustable Range: 500~250000

- Gear 2 Erpm(M-Gear)

Adjustable Range: 500~250000

- Gear 3 Erpm(H-Gear)

Adjustable Range: 500~250000

- Gear 4 Erpm

Adjustable Range: 500~250000

NOTE: This gear is applicable for PAS control mode ONLY.

- Gear 5 Erpm

Adjustable Range: 500~250000

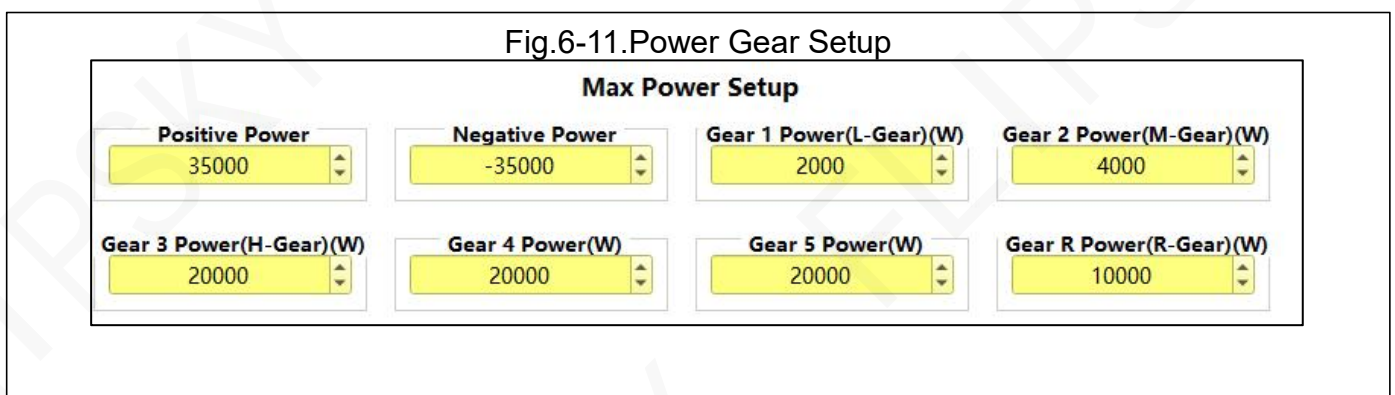
NOTE: This gear is applicable for PAS control mode ONLY.

- R-Gear Max Erpm

Adjustable Range: 500~250000

6.2.5 Power Gear Setup

Power gear setup including "Positive Power", "Negative Power", "Gear 1 Power(L-Gear)(W)", "Gear 2 Power(M-Gear)(W)", "Gear 3 Power(H-Gear)(W)", "Gear 4 Power(W)", "Gear 5 Power(W)", "R-Gear Max Power(W)". As Fig.6-11:



- Positive Power

Adjustable Range: 0~1000000

- Negative Power

Adjustable Range: -1000000~0

- Gear 1 Power(L-Gear)

Adjustable Range: 0~250000

- Gear 2 Power(M-Gear)

Adjustable Range: 0~250000

- Gear 3 Power(H-Gear)

Adjustable Range: 0~250000

- Gear 4 Power

Adjustable Range: 0~250000

NOTE: This gear is applicable for PAS control mode ONLY.

- Gear 5 Power

Adjustable Range: 0~250000

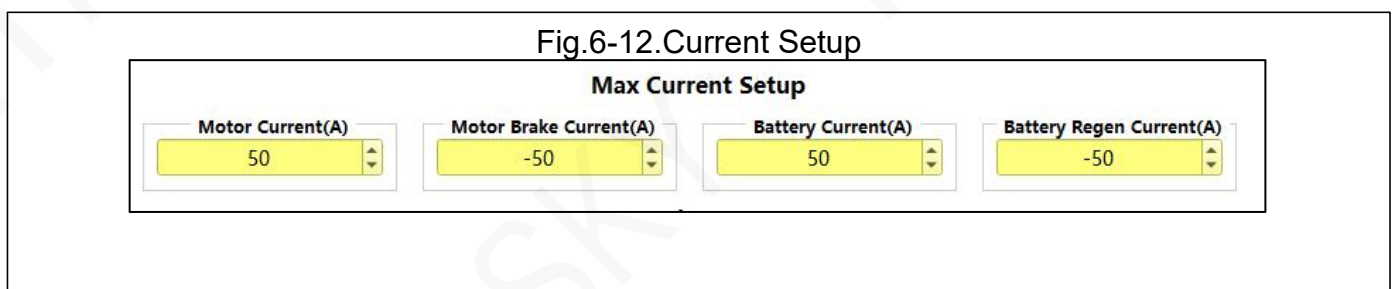
NOTE: This gear is applicable for PAS control mode ONLY.

- R-Gear Max Power

Adjustable Range: 0~250000

6.2.6 Current Setup

The max current setup includes maximum motor current and maximum battery current setup. The maximum motor current setup includes “Motor Current” and “Motor Brake Current”. The maximum battery current setup includes “Battery Current” and “Battery Regen Current” as Fig.6-12:



NOTE: The max(motor/current) current limit of every ESC of FT series is different.

- **Max Motor Current: the maximum continuous working current (A)**

Adjustable range: 0~550

- **Max Motor Brake Current: the maximum continuous braking current (A)**

Adjustable range: -550~0

NOTE: If using the DC power supply, the Max Motor Brake Current must be set to "0". Otherwise, the large back EMF will damage the ESC or even the power supply when the motor brakes in a high speed state.

- **Max Battery Current: the maximum continuous output current(A)**

Adjustable range: 0~550

NOTE: When setting Max Battery Current, it must be set according to the battery specification.

Set a battery with maximum continuous output current 50A as an example:

single ESC, Max Battery Current: 50A

dual ESC, Max Battery Current: $50/2=25A$

4WD ESC, Max Battery Current: $50/4=12.5A$

The total output current of the battery must not exceed the battery maximum continuous output current specification. Otherwise, the BMS will cut off the power supply and make the motor lose control, which is very dangerous. Moreover, please make sure that the Max Battery Current setting value does not exceed than that of the DC power supply to avoid ESC damage or other unpredictable conditions.

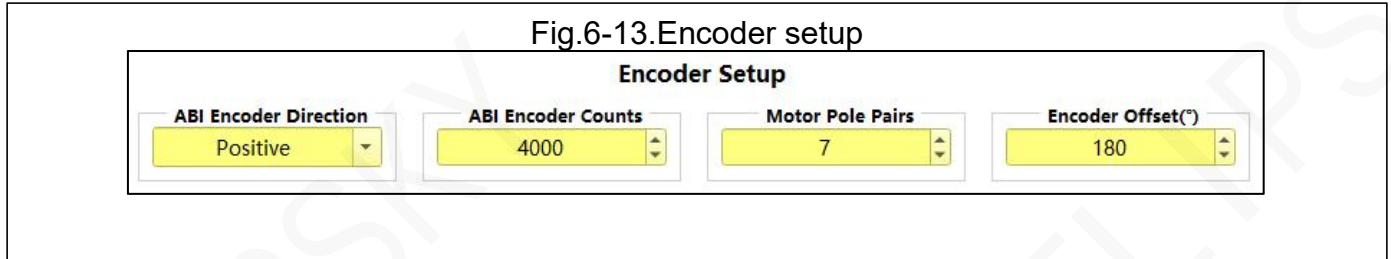
- **Max Battery Regen Current: the maximum continuous regenerative current of battery (A)**

Adjustable range: -550~0

NOTE: If using the DC power supply, the Max Battery Regen Current must be set to "0". Otherwise, the large back EMF will damage the ESC or even the power supply when the motor brakes in a high speed mode.

6.2.7 Encoder Setup

Encoder setup includes "ABI Encoder Direction", "ABI Encoder Counts", "Motor Pole Pairs", "Encoder Offset(°)" as Fig.6-13:



- **ABI Encoder Direction**

- ✧Positive
- ✧Negative

Parameter Specification: If the A and B phases are connected in reverse, the pulse count will be reversed. In this case, the encoder direction needs to be corrected.

- **ABI Encoder Counts**

Adjustable range: 0~100000

Parameter Specification: This parameter is 4 times the number of pulses output by the encoder when it rotates one circle. For example, if both phase A and phase B output 1000 pulses when they rotate one circle, they should be set to 4000 as 4 times of the frequency.

- **Motor Pole Pairs**

Adjustable range: 0~100

Parameter Specification: When the motor sensor port is set to encoder mode, this value will be calculated automatically after finishing the motor parameter identification. If no special application, no modification is required. NOTE: When starting in encoder mode, please ensure the correct setting of this value.

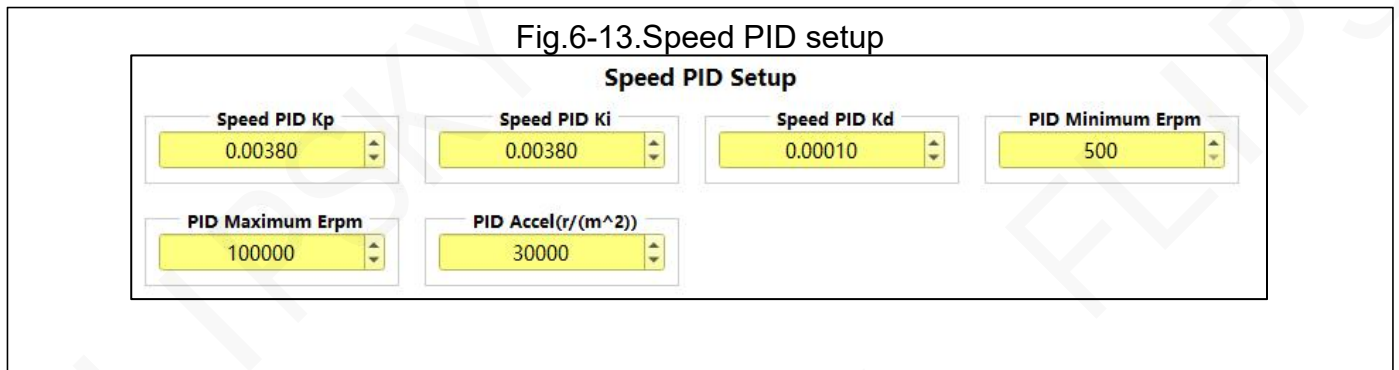
- **Encoder Offset(°)**

Adjustable range: 0~360°

Parameter Specification: It means the offset angle between the encoder and the motor rotor. When the motor sensor port is set to encoder mode, this value will be calculated automatically after finishing the motor parameter identification. No modification is required unless necessary.

6.2.8 Speed PID Setup

Speed PID setup includes "Speed PID Kp", "Speed PID Ki", "Speed PID Kd", "PID Minimum Erpm", "PID Maximum Erpm", "PID Accel ($r/(m^2)$)Erpm" as Fig.6-13:



- **Speed PID Ki: Speed PID controller integral gain coefficients.**

Adjustable range: 0~10000

If no special application, no modification is required.

- **Speed PID Kd: Speed PID controller differential coefficients**

Adjustable range: 0~10000

If no special application, no modification is required.

- **PID Maximum Erpm: Minimum speed for speed PID control.**

Adjustable range: 0~250000

Parameter Specification: When the ESC detects the Erpm higher than this max. value, it will maintain the maximum speed PID Maximum Erpm output.

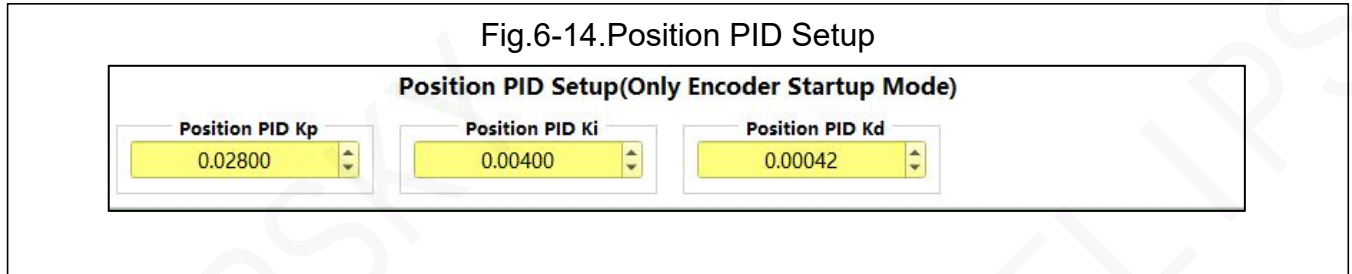
- **PID Accel ($r/(m^2)$)Erpm: Acceleration value for speed PID control.**

Adjustable range: -1~150000

Parameter Specification: The larger the value, the faster the speed changes. For example, when it is set to 30000 Erpm, it means that the speed changes by 30000 Erpm in 1 second. Which means that it takes 1 second only for the motor to increase Erpm from 0 to 30000 when a speed command of 30000 Erpm is given.

6.2.9 Position PID Setup(Encoder Startup Mode only)

Position PID Setup includes “Position PID Kp” , “Position PID Ki” and “Position PID Kd” as Fig.6-14 below:



- **Position PID Kp: Position PID controller proportional gain coefficients.**

Adjustable range: 0~10000

If no special application, no modification is required.

- **Position PID Ki: Position PID controller integral gain coefficients.**

Adjustable range: 0~10000

If no special application, no modification is required.

- **Position PID Kd: Position PID controller derivative coefficients.**

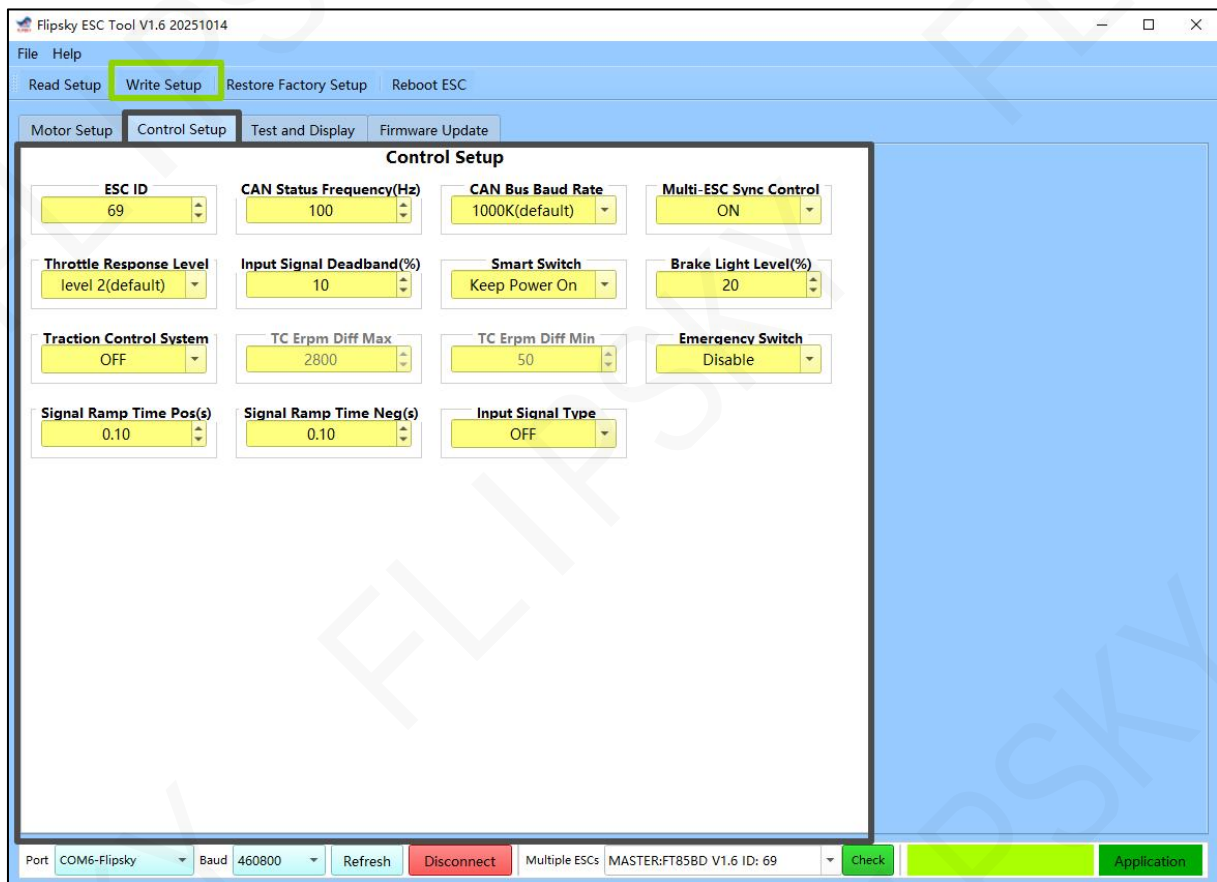
Adjustable range: 0~10000

If no special application, no modification is required.

7 Control Setup

Control setup parameter includes “ESC ID”, “CAN Status Frequency(Hz)”, “CAN Bus Baud Rate”, “Multi-ESC Sync Control”, “Throttle Accel Level”, “Input Signal Deadband”, “Smart Switch”, “Brake Light Level(%)”, “Traction Control Svstem”, “TC Erpm Diff Max”, “TC Erpm Diff Min”, “Emergency Switch”, “Signal Ramp Time Pos(s)”, “Signal Ramp Time Neg(s)” and “Input Signal Type” as Fig.7-1 as below:

Fig7-1.Control setup



Setup Steps are as follow:

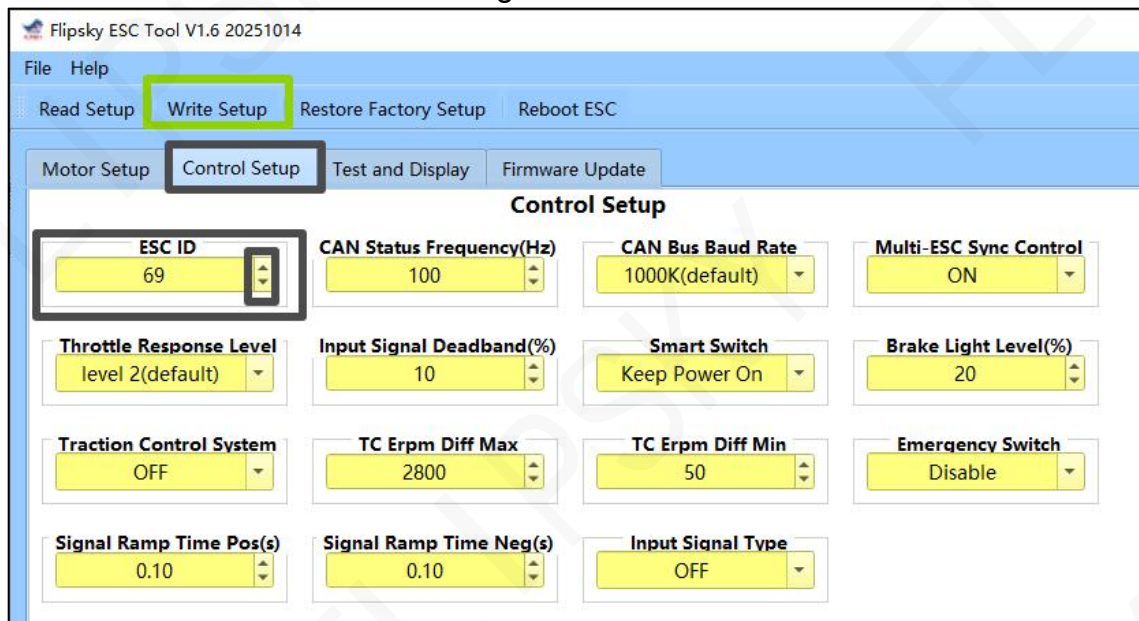
- Click “Control Setup” to enter the control setup interface
- Set the needed parameters
- Click “Write Setup” on the upper left to save the configuration

7.1 ESC ID Setup

ESC ID: ESC ID(Fig.7-2), adjustable range: 1~254. For distinguish the ESCs on the same CAN bus. Please make sure the uniqueness of the ESC ID since the same ESC IDs result in parameter identification failures or abnormal control. ESC ID 0 (special command number) and ESC ID 255 (special broadcast command) are not available for setup.

ESC ID setup steps are in Chapter 4.2.1.

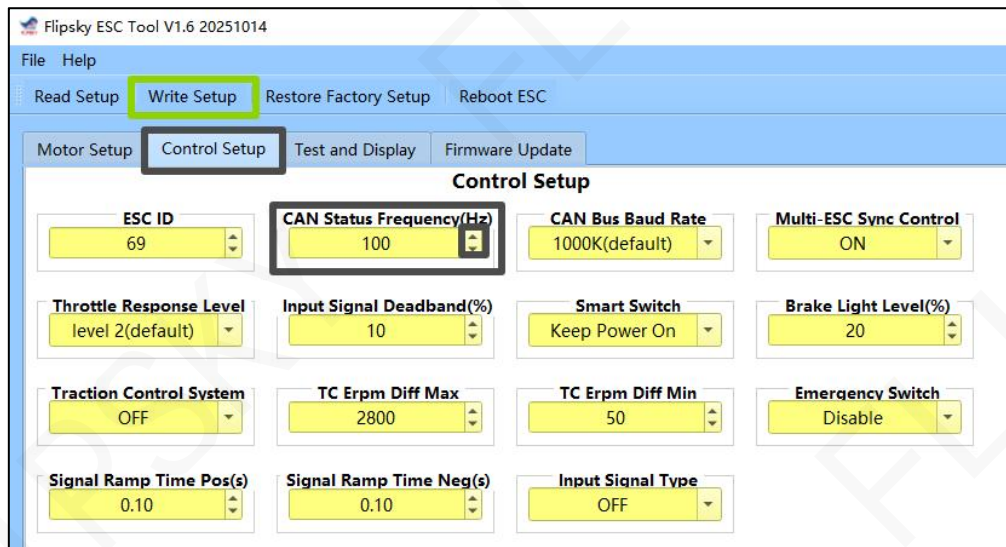
Fig.7-2.ESC ID



7.2 CAN Status Frequency Setup

CAN Status Frequency(Hz): Each ESC will automatically return real-time data through the CAN bus, where the real-time data includes motor current, battery current, motor erpm, duty cycle, MOSFET temperature, motor temperature and battery voltage data. All of the above real-time data will be returned and sent to the CAN bus at regular intervals, and the frequency of sending can be set via CAN Status Frequency as Fig.7-3 below.

Fig.7-3.CAN status frequency

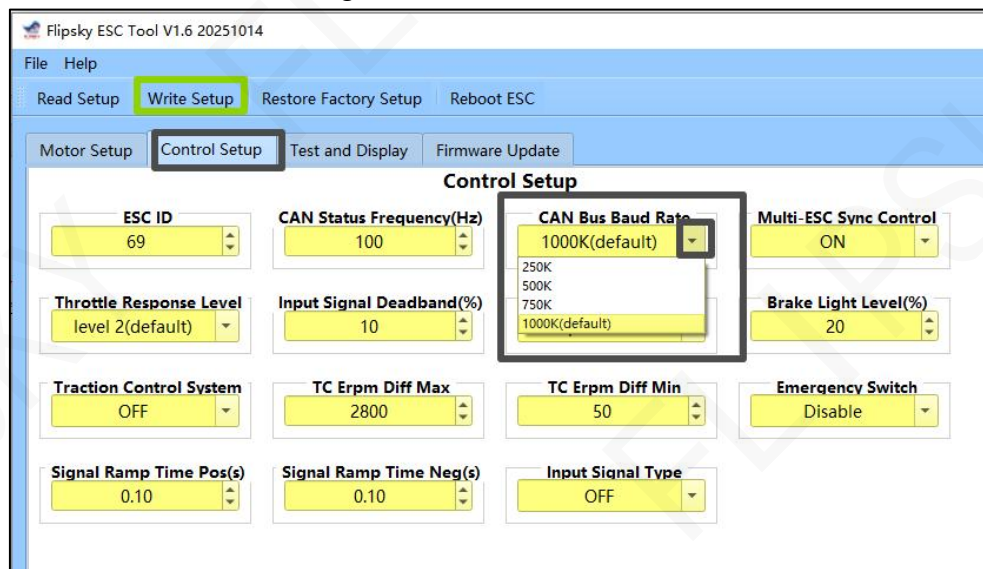


7.3 CAN Bus Baud Rate Setup

CAN Bus Baud Rate(Fig.7-4): Please make sure the consistency of CAN baud rate for the same CAN devices.

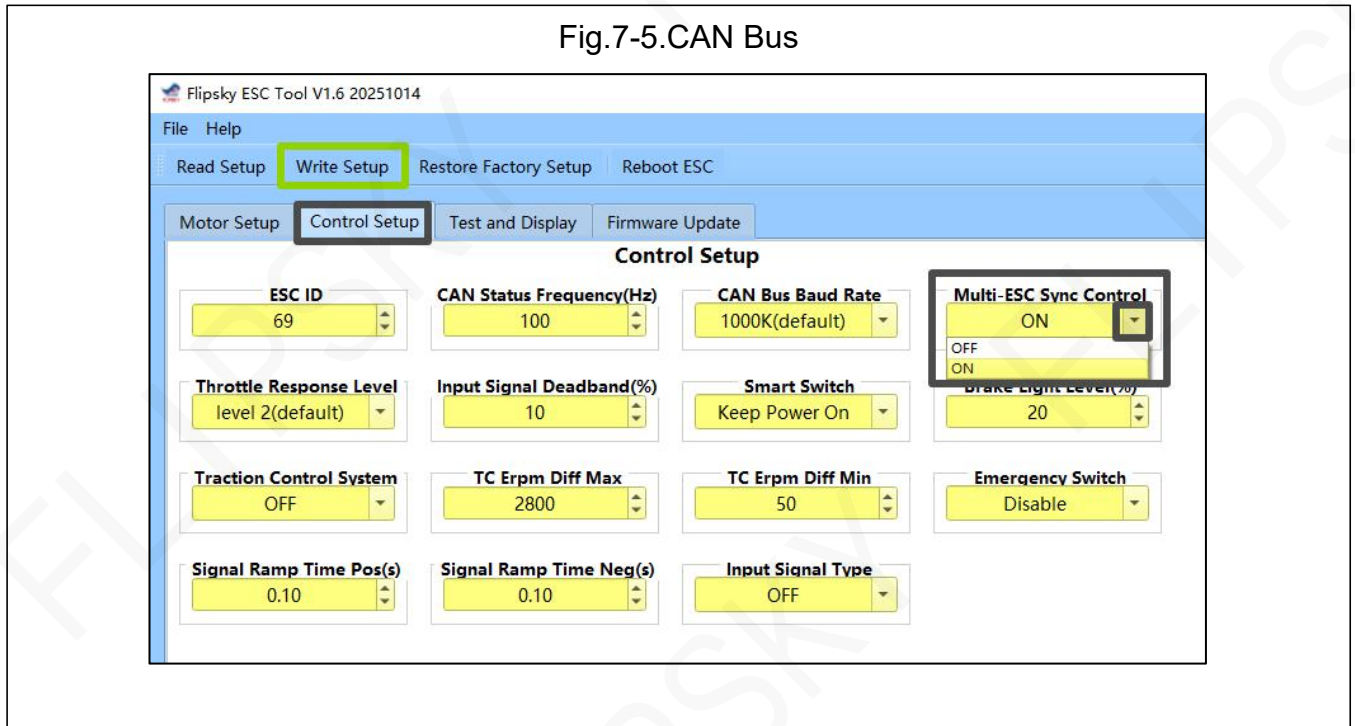
You can view the CAN bus baud rate setup in chapter 4.2.2.

Fig.7-4.CAN bus baud rate



7.4 CAN Bus Multi-ESC Sync Control Setup

Multi-ESC Sync Control: synchronous control for multiple ESCs via CAN as Fig.7-5:



Setup Steps:

- Click “Control Setup” to enter the control setup interface
- Click "Multi-ESC Sync Control" to select whether synchronous control of the CAN ESCs or not. (ON - with synchronous control, OFF - not)
- Click the upper left menu "Write Setup" to save the setup

Parameter Specification:

When the ESC is used as master and “Multi-ESC Sync Control” is set to ON, the input signal will synchronize the master ESC and all slave ESCs mounted on the CAN bus; when “Multi-ESC Sync Control” is set to OFF, the input signal controls only the master ESC.

When the ESC acts as a slave and “Multi-ESC Sync Control” is set to ON, the slave ESC will accept control commands sent from the CAN bus; when “Multi-ESC Sync Control” is set to OFF, the slave ESC will block the control commands sent from the CAN bus.

If you want multiple ESCs to be controlled synchronously, set all ESC’s “Multi-ESC Sync Control” as “ON”.

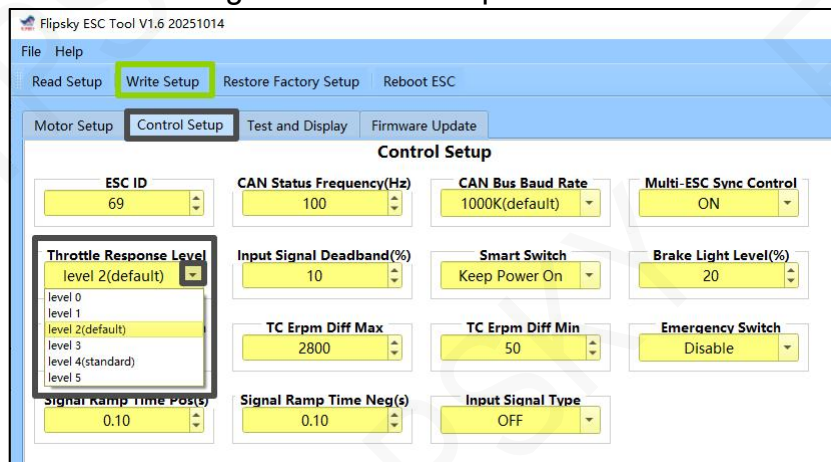
However, if you want to control each ESC separately in UART, PPM or ADC control type, set all ESC’s “Multi-ESC Sync Control” as “OFF”.

If use your own control module as the master board and want to control the ESCs via CAN-Bus, set all ESC's "Multi-ESC Sync Control" as "ON", and the "Input Signal Type" as "OFF".

7.5 Throttle Response Level Setup

Throttle Response Level: Level 0~Level 5(as Fig.7-6):

Fig.7-6.Throttle Response Level



Setup Steps:

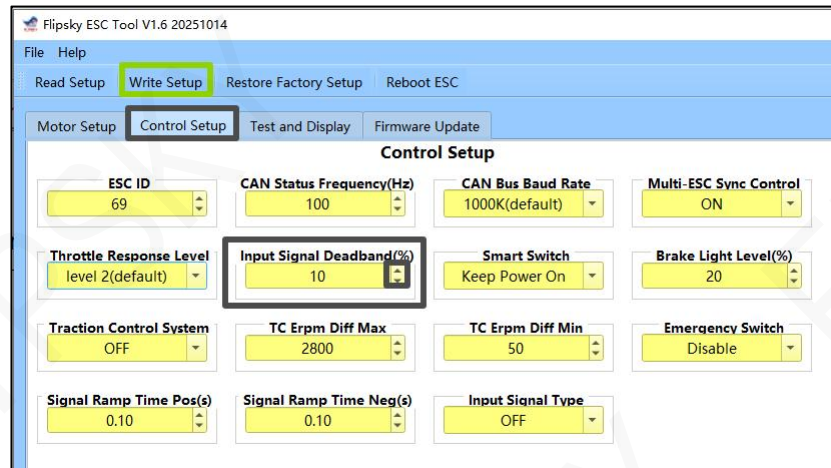
- Click "Control Setup" to enter the setup interface
- Click "Throttle Response Level" to select the appropriate throttle response level
- Click "Write Setup" on the upper left to save the configuration

Parameter Specification: The higher of the level, the more sensitive of the throttle and brake response. Level 4 is the standard level which has no advance or lag on all input signals and is suitable for common users

7.6 Input Signal Deadband Setup

Input Signal Deadband: adjustable range: 1%~25%. as Fig.7-7:

Fig.7-7. Input signal deadband



Setup Steps:

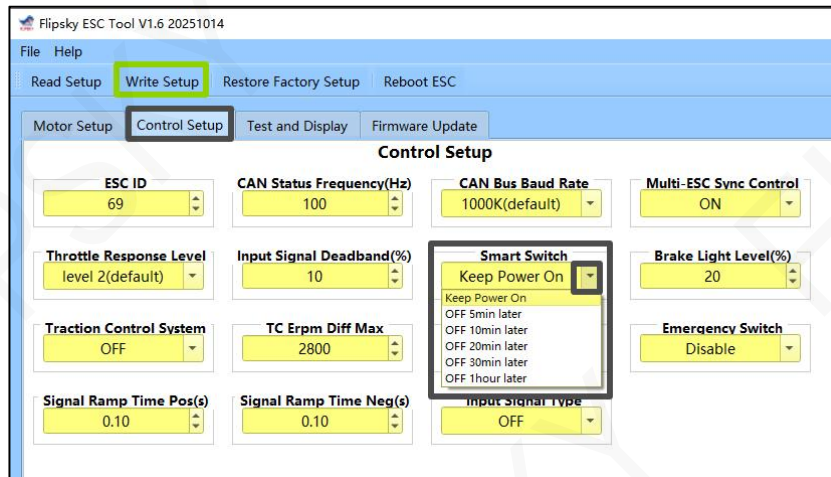
- Click “Control Setup” to enter the control setup interface
- Click “Input Signal Deadband” to adjust suitable input signal deadband(%)
- Click “Write Setup” on the upper left to save the configuration

Parameter Specification: The ESC starts the current output only when the throttle input signal percentage is over than that of the input deadband setup . It's recommended to use the default value 10%. Regarding the input control type PPM and UART, the deadband percentage can be small or even 0% as the remote controller has already set it, for example, the Flipsky remote controller VX4 with 5% deadband already. It's strongly recommended to keep the default setting as 10% under ADC control.

7.7 Smart Switch

Smart Switch: This function only support FT85BD、FT60BD and FT85KS of FT series ESC. You can set it through Smart Switch as Fig.7-8:

Fig.7-8. Smart Switch setup

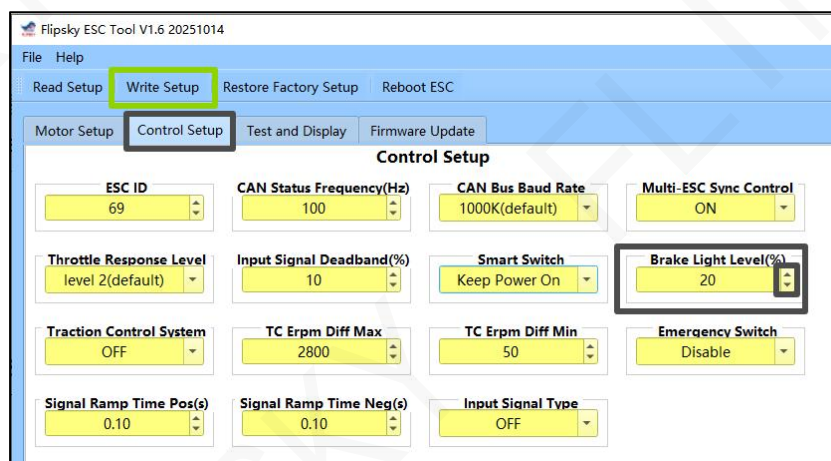


Parameter Specification: two modes: “Keep Power On” stands for always on without power-off. The second mode means power off after a specific time, for example, “OFF 5min later” supports auto shutoff without drive signal after 5 minutes later.

7.8 Brake Light Level Setup

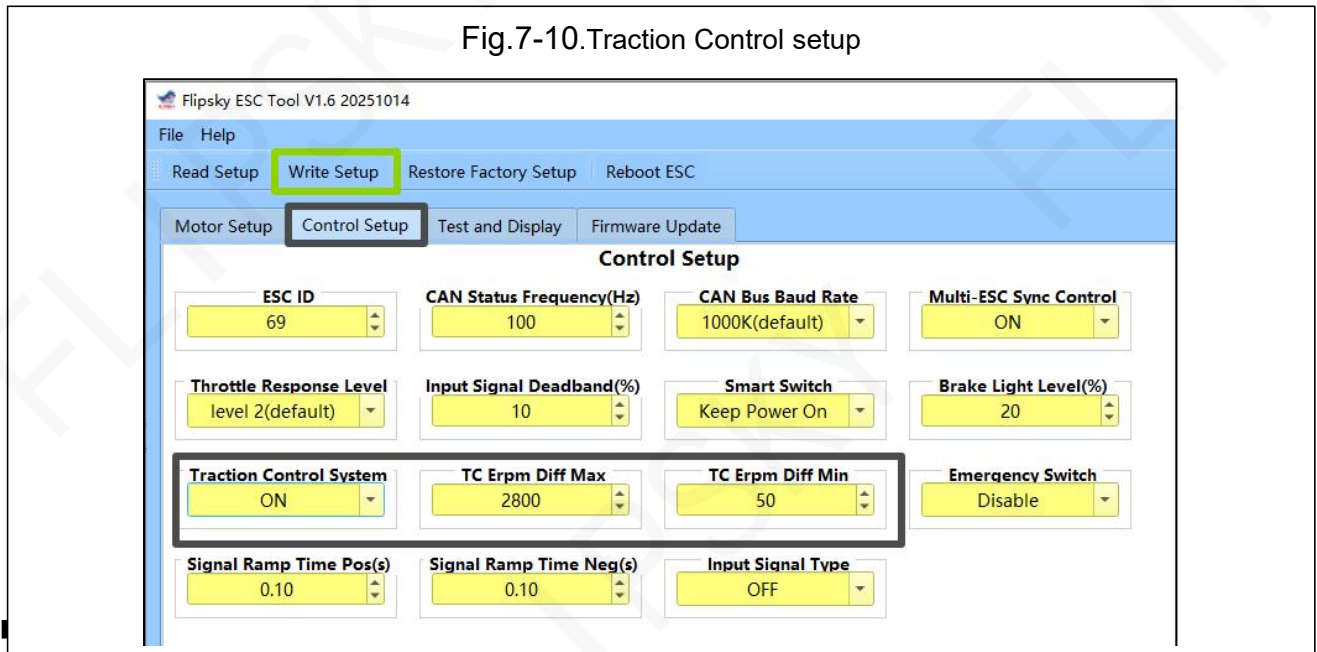
Brake Light Level: The brake light brightness setup in static. Adjustable range: 0~100%, default as “20%”.

Fig.7-9.Brake light brightness setup in static



7.9 Traction Control Setup

TCR Control: Traction control is realized when multiple ESCs are jointly controlled, and the algorithm can be used to prevent the drive wheels from skidding when multiple ESCs accelerate the drive motors at the same time. This mode is only for current control (does not support current hybrid bi-directional control in PPM mode). Traction control setup as Fig.7-10:



- ✧ ON
- ✧ OFF

Parameter Specification: This mode is mainly for preventing the drive wheel from slipping when ESK8 accelerates and maintains its stability.

■ TC Erpm Diff Max

Adjustable range: 0~10000

Parameter Specification: Maximum erpm difference threshold. Different motors correspond to different maximum erpm difference thresholds, which are typically set to 5% of the maximum erpm at full voltage. For example, a 6374 motor with a 7-pole KV of 160 would have a maximum erpm of about 56,500 at full voltage of 50.4V and 12S, so this value is set to $56,500 * 0.05 = 2825$.

■ TC Erpm Diff Min

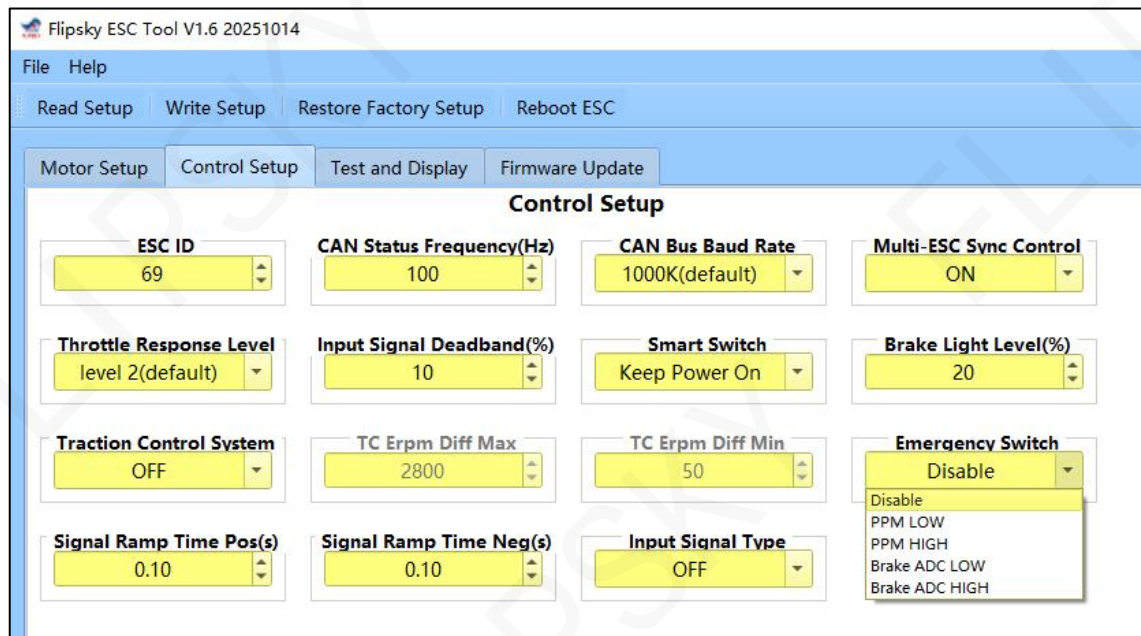
Adjustable range: 0~1000

Parameter Specification: Minimum Erpm difference threshold. When the speed difference is lower than this value, the TCR control will be cut off.

7.10 Emergency Switch Setup

Emergency Switch : Mainly used for cutting off power. You can set any of signal below to activate this function, as Fig. 7-11:

Fig 7-11.Emergency Switch Setup



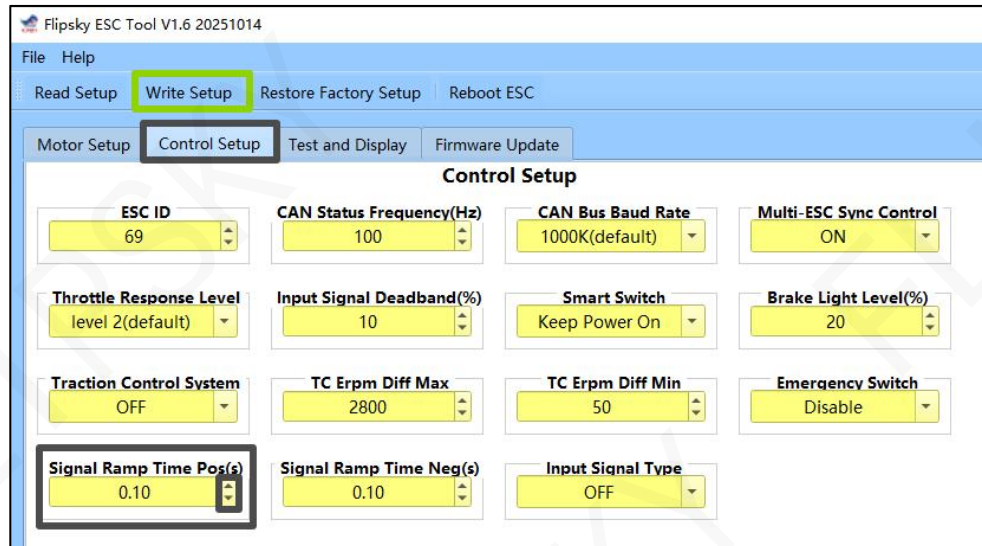
Parameters Specification:

- Disable : Disable emergency switch function
- PPM LOW : When select “PPM LOW”, if the PPM signal is detected as low level, it will cut off the power output.
- PPM HIGH : When select “PPM HIGH”, if the PPM signal is detected as high level, it will cut off the power output.
- Brake ADC LOW : When select “Brake ADC LOW ”, if the Brake ADC signal is detected as low level, it will cut off the power output.
- Brake ADC HIGH : When select “Brake ADC HIGH ”, if the Brake ADC signal is detected as high level, it will cut off the power output.

7.11 Signal Ramp Time Pos(s) Setup

Signal Ramp Time Pos(s): Adjustable range: “0.1~100s”; default as 0.1s. As Fig.7-12:

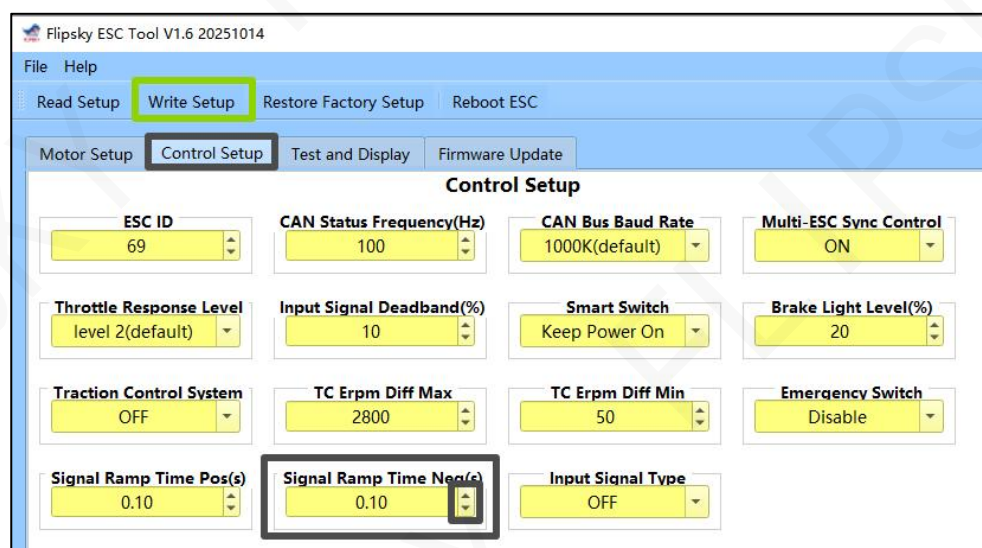
Fig.7-12.Signal Ramp Time Pos(s) Setup



7.12 Signal Ramp Time Neg(s) Setup

Signal Ramp Time Neg(s): Adjustable range: “0.1~100s”; default as 0.1s. As Fig.7-13:

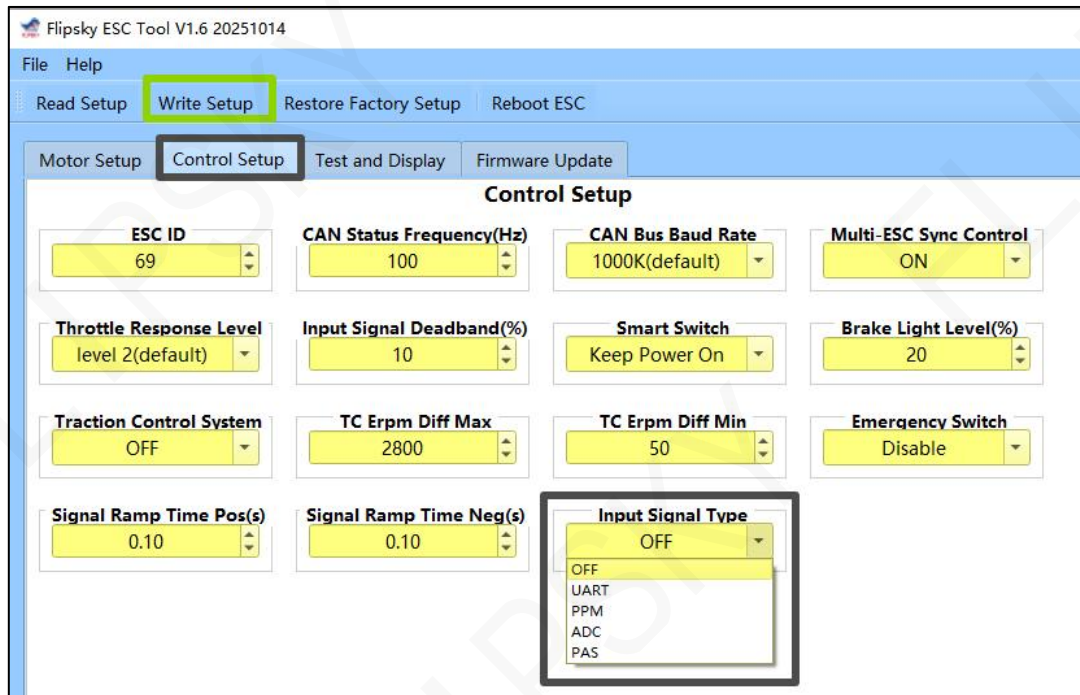
Fig.7-13. Signal Ramp Time Neg(s) Setup



7.13 Input Signal Type Setup

Input Signal Type:OFF/UART/PPM/ADC/PAS. Default as OFF(as Fig.7-14). View the chapter below for details.

Fig.7-14.Input signal type setup



7.13.1 OFF

When “Input Signal Type” is set to OFF(Fig.7-15), all input signals will be shielded. You can only control the ESC through Current, Duty, Brake and other buttons in Test and Display interface(Fig.7-16). View button operations in Chapter 8.1.1.

Fig.7-15.OFF

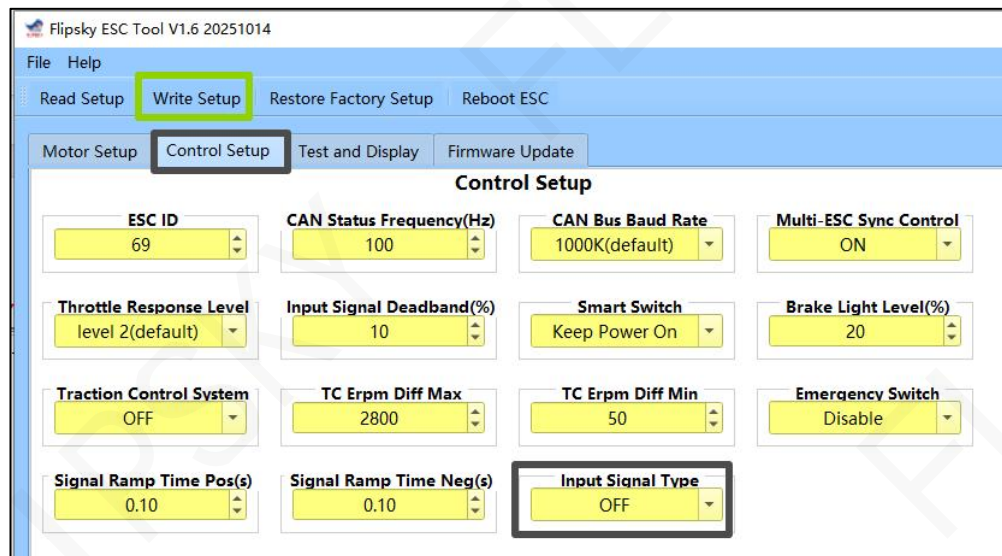
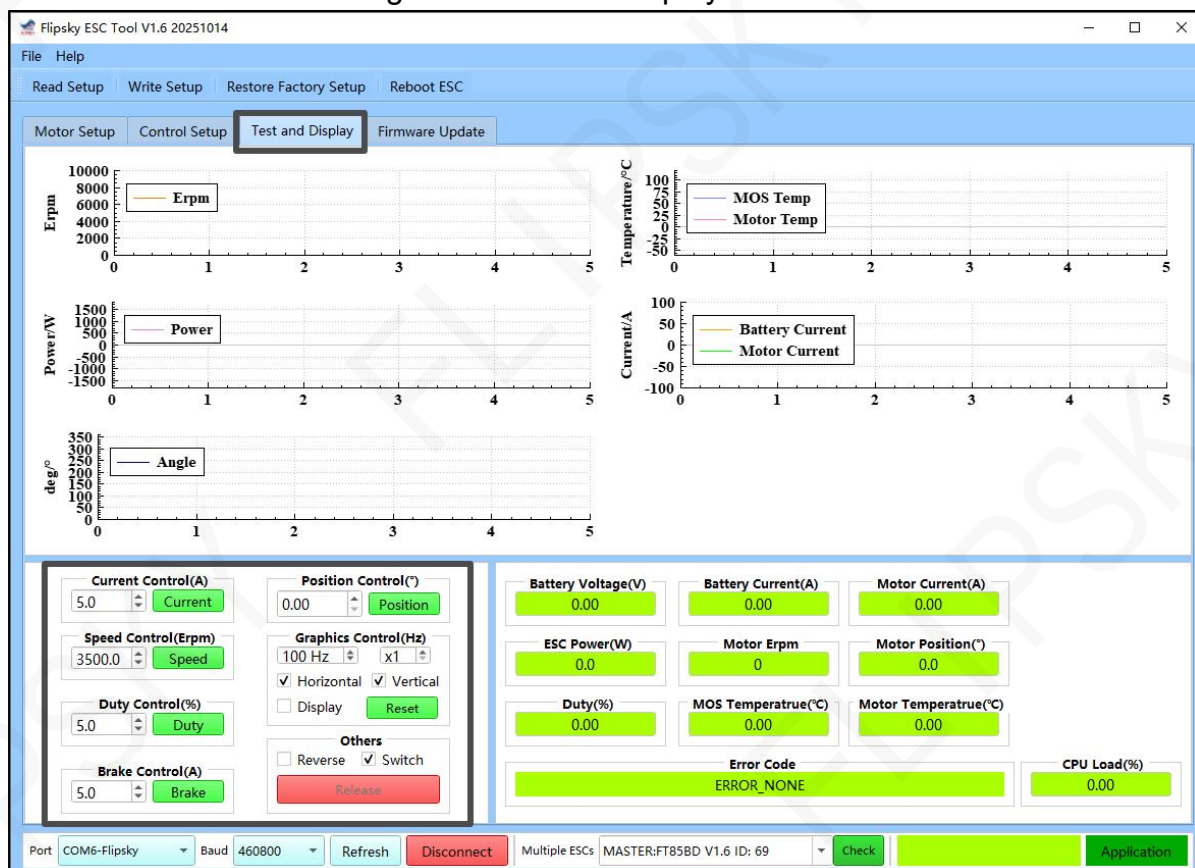


Fig.7-16.Test and Display interface



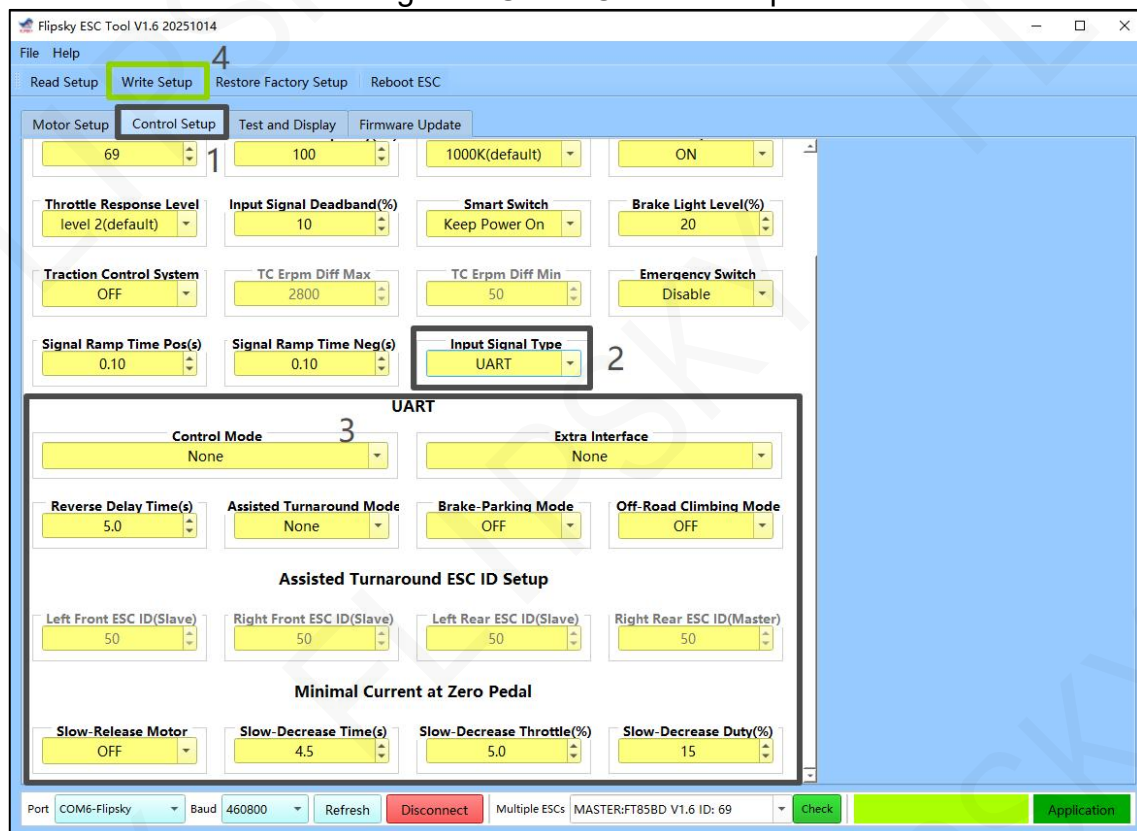
7.13.2 UART

7.13.2.1 UART Parameter Specification and Setup

UART control mode can support all FT series remotes.(VX1, VX1Pro, VX2, VX3, VX3Pro, VX4, VX4Pro, VX5) Support limit its maximum rpm or max power in different gears.

Setup steps are as Fig.7-17 below when using UART signal to control the ESC:

Fig.7-17.UART Control Setup



NOTE: Step 3 and Step 10 is must, other parameters can be set according to your needs.

- Step 1: Click "Control Setup" to enter the control setup interface
- Step 2: Select UART in "Input Signal Type"
- Step 3: Select suitable UART Control Mode per your needs(view details in 7.13.2.2)
- Step 4: Set UART Reverse Delay Time per your needs(view details in 7.13.2.3)
- Step 5: Set Extra Interface per your needs(view details in 7.13.2.4)
- Step 6: Set Minimal Current at Zero Pedal per your needs(view details in 7.13.2.5)
- Step 7: Set Assisted Turnaround Mode and Assisted Turnaround ESC ID Setup per your needs (view details in 7.13.2.6)

- Step 8: Set Brake Parking Mode per your needs(view details in 7.13.2.7)
- Step 9: Set Off-Road Climbing Mode per your needs(view details in 7.13.2.8)
- Step 10: Click “Write Setup” on the upper left again, write and save the configuration to ESC.

7.13.2.2 UART Control Mode Setup

User can select applicable UART Control Mode as Fig.7-18:

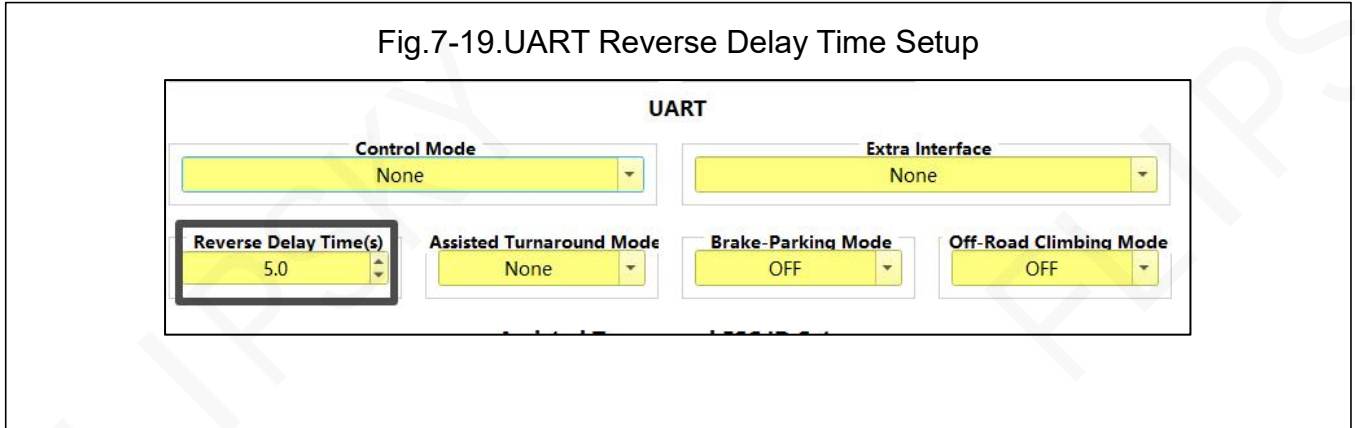
Fig.7-18.UART Control Mode



- None
- Current Gear Reverse Brake: Current control mode: gear speed limit, reverse with brake, reverse
- Current Gear Brake: Current control mode: gear speed limit, reverse with brake
- Current Reverse Brake: Current control mode: reverse with brake, reverse
- Current Brake: Current control mode: reverse with brake
- Duty Gear Brake Reverse: Duty cycle control mode: gear speed level limit, reverse with brake, reverse
- Current Bidirectional Gear: Current control mode: bilateral control, gear speed limit

7.13.2.3 UART Reverse Delay Time Setup

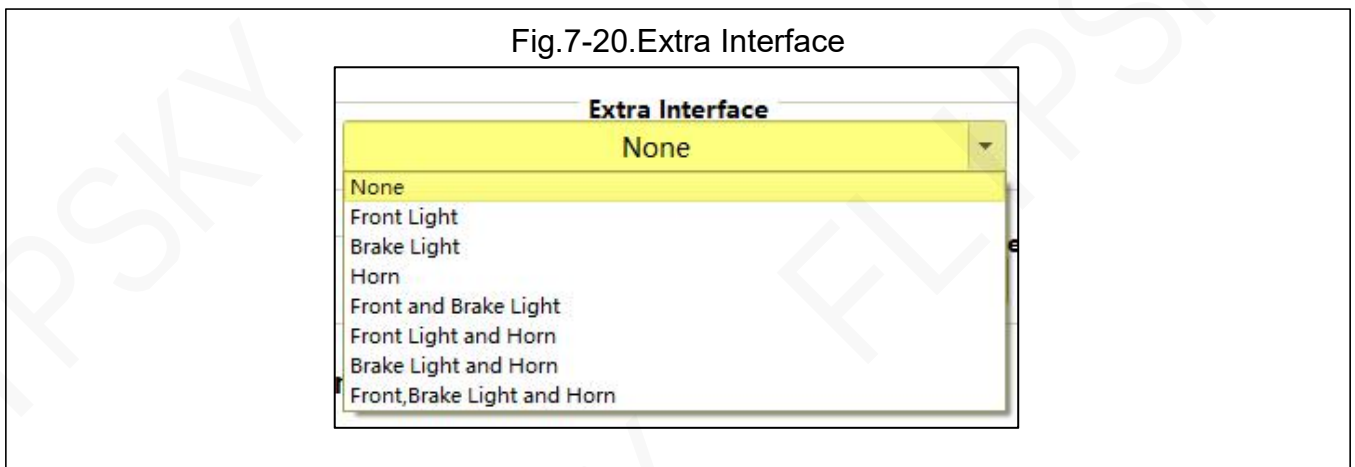
The UART control mode supports reverse function, so you can set the UART Reverse Delay Time per your needs. The adjustable range: 0.0s~10s, and the default is 5s as shown in Fig. 7-19:



Parameter Specification: When the motor is stopped, pull the throttle back to the maximum and then start to rotate in the reverse direction. It can reach the maximum reverse speed within the set time period(as Fig.7-19, it's 5s as set)

7.13.2.4 Extra Interface Setup

This function only support FT85BD, FT60BD and FT85KS of FT series ESC. Support front light, brake light and horn control under UART control. Support VX2, VX3, VX4, VX4Pro and VX5 remote to control front light, brake light and horn. User can set through Extra Interface accordingly as Fig.7-20:



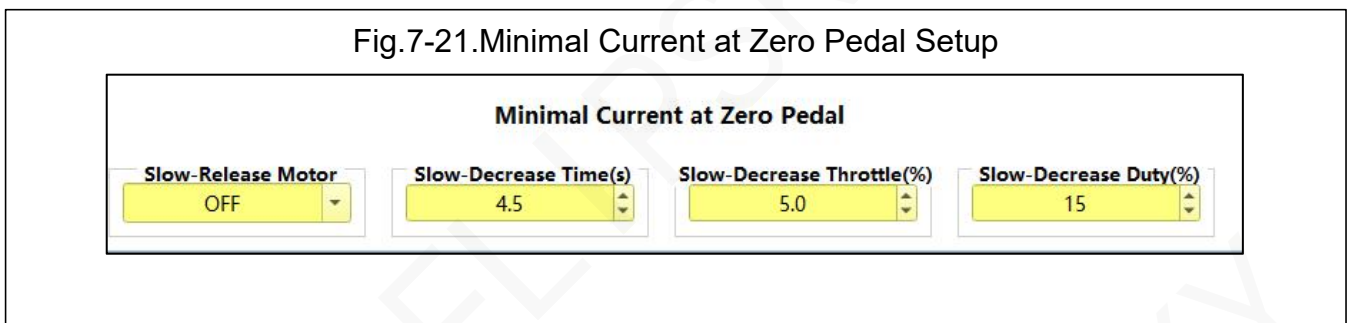
- None
- Front Light
- Brake Light

- Horn
- Front and Brake Light
- Front and Horn
- Brake Light and Horn
- Front, Brake Light and Horn

7.13.2.5 Minimal Current at Zero Pedal Setup

This function enables FT series ESCs to maintain a certain current output after releasing the throttle under the UART input signal. In certain skateboards with higher gear ratios ($>3.5:1$), such as off-road models, a braking-like resistance may occur after releasing the throttle, which is caused by the gear ratio. This feature maintains a certain level of current output even after the throttle is released via the remote control, which could counteract the resistance, enabling the skateboard to enter a near-zero-resistance state.

The setup steps are as Fig. 7-21 below:



User can set below parameters accordingly.

Parameter Specification:

- **Slow-Release Motor:** Enable this function or not, default: OFF
- **Slow-Decrease Time:** Current output releasing time, default: 4.5s
- **Slow-Decrease Throttle:** Current output throttle percentage, default:5% ($0.05 \times \text{Motor Current}$)
- **Slow-Decrease Duty:** When the motor duty cycle is lower than this value, this function is off. Only when the duty cycle is higher than this value will this function be effective.

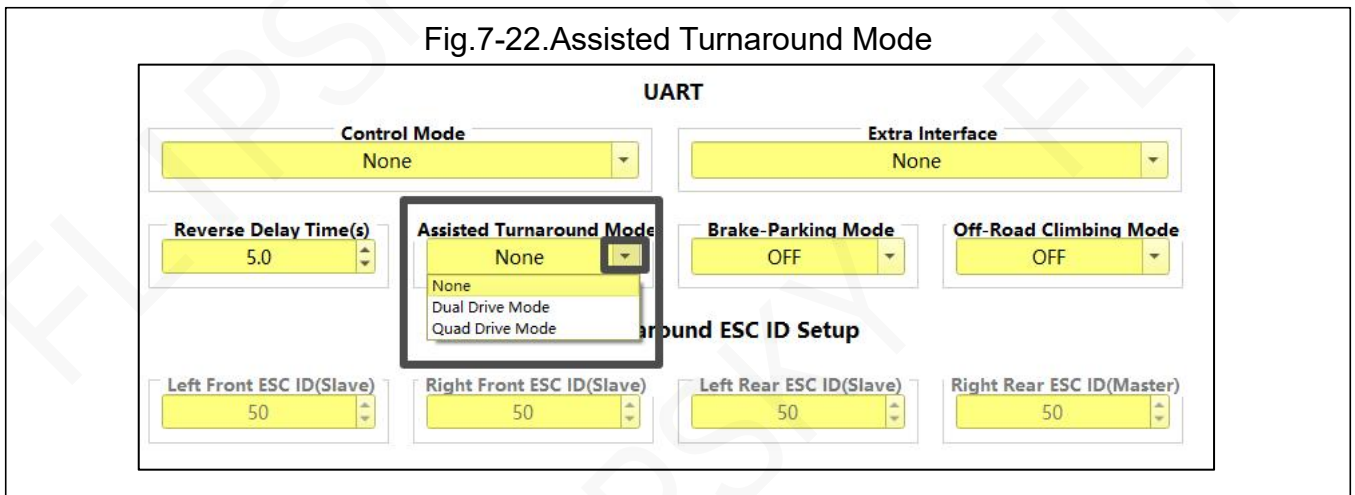
NOTE: This function is not available under the control mode of Duty Gear Brake Reverse (Duty cycle control mode: gear speed limit, reverse with brake, reverse)

7.13.2.6 Assisted Turnaround Mode and ESC ID Setup

Assisted Turnaround: This assisted turnaround function is applicable for UART control only. It only supports for remote VX4(Firmware: V1.4. or above), VX4Pro(Firmware: V1.0 or above), VX2(Firmware: V1.9), VX3(Firmware: V1.9) and VX5 currently. Please check details in the user manual.

Setup Steps:

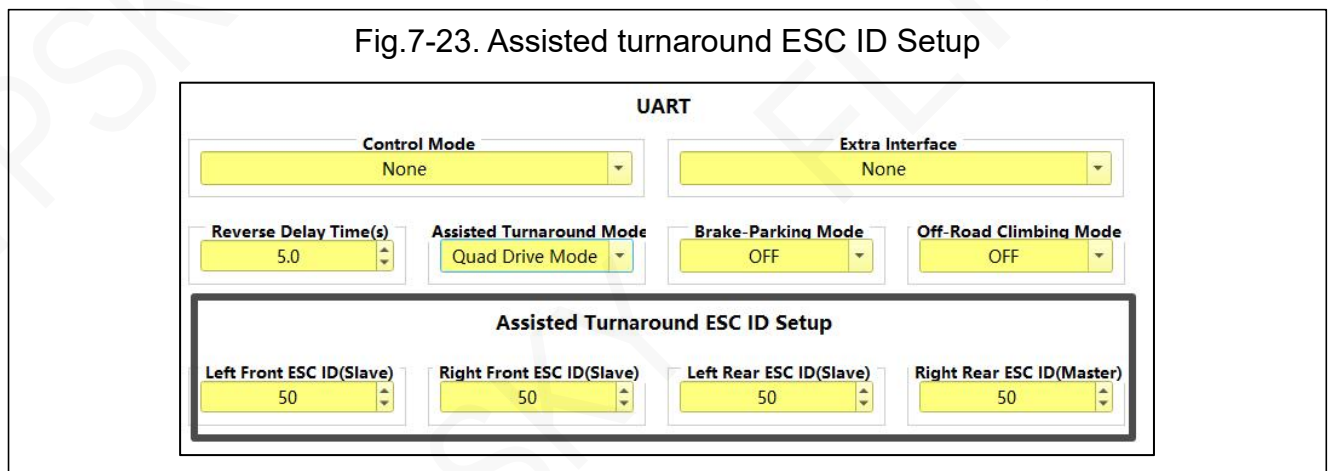
- Step 1: Select Assisted Turnaround Mode accordingly as Fig.7-22:



Parameter Specification:

- Assisted Turnaround Mode:
 - ✧ None: Turn off assisted turnaround mode.
 - ✧ Dual Drive Mode: Dual drive turnaround mode. Uses only two ESC drives. One forward and one reverse.
 - ✧ Quad Drive Mode: 4WD turnaround mode. All four ESCs drive at the same time. Forward on the same side, reverse on the other side.

- Step 2: Set assisted turnaround ESC ID as Fig.7-23:



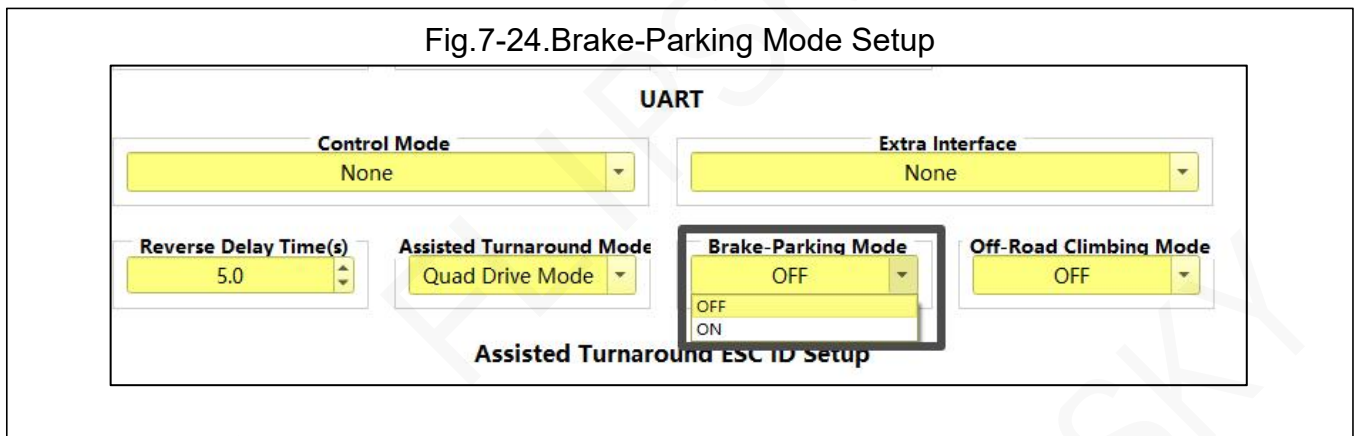
- Assisted Turnaround ESC ID Setup:
 - Left Front ESC ID(Slave): Set in 4WD.
 - Right Front ESC ID(Slave): Set in 4WD.
 - Left Rear ESC ID(Slave): Set in 4WD.
 - Right Rear ESC ID(Master): Set in 4WD.

NOTE:

- (1) Dual Drive Mode: Set “Left Rear ESC ID(Slave)” and “Right Rear ESC ID(Master)”.
- (2) Quad Drive Mode: Set “Left Front ESC ID(Slave)”, “Right Front ESC ID(Slave)”, “Left Rear ESC ID(Slave)” and “Right Rear ESC ID(Master)”.

7.13.2.7 Brake-Parking Mode Setup

Brake-Parking Mode: Pull the remote controller throttle gear/bar 0.5S to enter into Brake-Parking Mode after the motor stop running. The greater percentage of pulling back the throttle gear, the bigger of the parking force. Setup as shown in Fig.7-24:



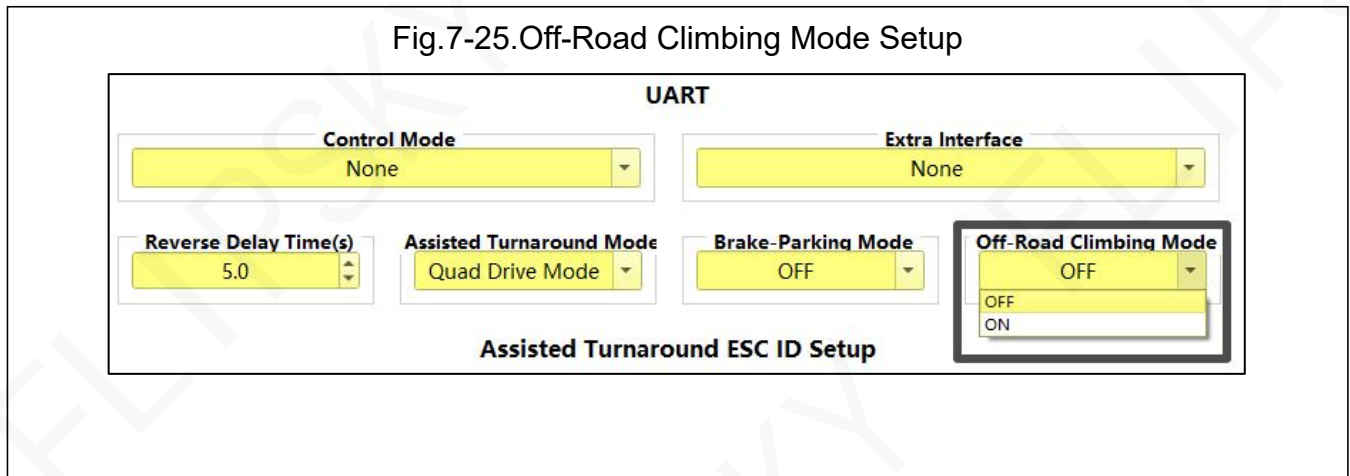
NOTE: This mode injects current into the motor d-axis to make the coil generate magnetic force and attract the rotor to achieve the parking effect, so current will be consumed when parking. The maximum d-axis current can reach Motor Current Max when parking. It is not recommended to park in this mode for a long time, otherwise the motor will get hot and lead to overload current limiting phenomenon.

When in Reverse Mode like Current Gear Reverse Brake or Current Gear Brake mode, max pulling the throttle gear / bar will end the Brake-Parking Mode but enter into Reverse Mode instead.

Currently, this mode supports for remote VX4(firmware: V1.4. or above), VX4Pro(firmware : V1.0 or above), VX2(firmware: V1.9), VX3(firmware: V1.9) and VX5 only. Please check user manuals for details.

7.13.2.8 Off-Road Climbing Mode

Off-Road Climbing Mode: In this mode, all CAN ESCs will be switched to duty cycle control and allow users to pass off-road passing gravel roads or climbing high slopes easily by controlling speed of the four wheels synchronously via remote controller. Setup as shown in Fig.7-25:



NOTE:

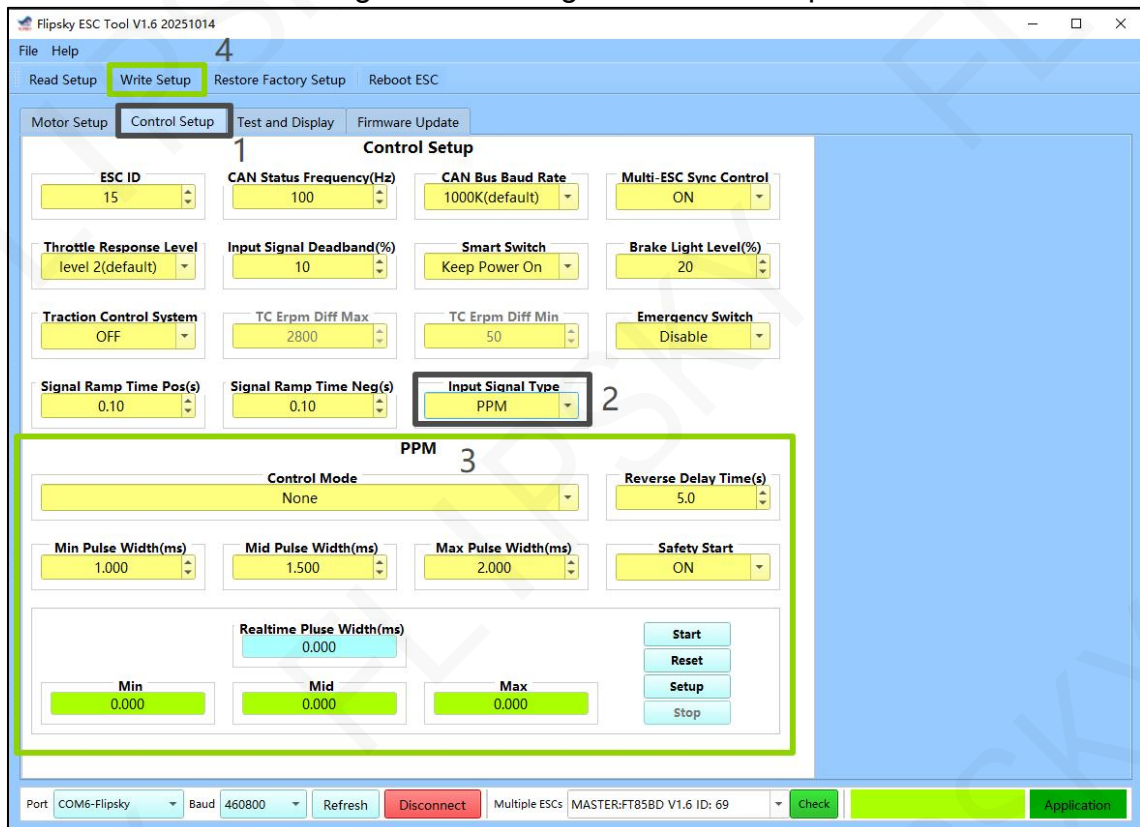
1. After using this mode, the reverse mode will be disabled.
2. Currently, this mode supports for remote VX4(firmware: V1.4. or above), VX4Pro(firmware: V1.0 or above), VX2(firmware version:V1.9), VX3(firmware:V1.9) and VX5 only. Please check user manual for details.

7.13.3 PPM

7.13.3.1 PPM Parameter Specification and Setup

The PPM signal pulse width usually ranges from 1ms to 2ms, so theoretically the PPM control frequency supports up to 500Hz. Considering the MCU clock accuracy, it is recommended to use the PPM control frequency less than 400Hz. Setting steps as Fig.7-26 below:

Fig.7-26.PPM Signal Control Setup



- Step 1: Click "Control Setup" to enter the setup interface
- Step 2: Switch the "Input Signal Type" to "PPM".
- Step 3: Calibrate when use PPM as input signal for the first time(check details in chapter 7.13.3.2)

NOTE: Calibrate the PPM signal for one side of the dual ESC controller is workable. The other side's Input Signal Type set "OFF".

- Step 4: Select the appropriate PPM Control Mode as you need (see details in 7.13.3.3)
- Step 5: Set the Reverse Delay Time (see details in 7.13.3.4)
- Step 6: Click the upper left menu "Write Setup" to save the setup.

7.13.3.2 PPM Signal Calibration Steps

Calibrate PPM signal first when use it as input signal for the first time. Please plug the PPM signal cable to the ESC PPM port before calibration. Take VX1 remote of Flipsky as an example:

- Click "Start" to begin the calibration as Fig.7-27 below:

Fig.7-27.Enter PPM signal calibration

The screenshot shows the PPM signal calibration interface. At the top, a label 'Realtime Pluse Width(ms)' is above a light blue box displaying '1.505'. Below this, there are three green boxes labeled 'Min', 'Mid', and 'Max', each displaying '1.505'. To the right of these boxes is a vertical stack of four buttons: 'Start' (highlighted with a black border), 'Reset', 'Setup', and 'Stop'.

NOTE: No PPM signal can be detected if the remote receiver hasn't been plugged with the ESC PPM wire when you click "Start" as Fig7-28:

Fig.7-28.No PPM signal

The screenshot shows the PPM signal calibration interface with no signal detected. The 'Realtime Pluse Width(ms)' box displays '0.000'. The 'Min', 'Mid', and 'Max' green boxes also display '0.000'. The 'Start' button is highlighted with a black border.

- Connect the remote receiver with the ESC PPM wire in this case. If the PPM signal data seems abnormal (as shown in Fig.7-29), please click "Reset". (Refer to Fig.7-27).

Fig.7-29.Abnormal PPM signal

The screenshot shows the PPM signal calibration interface with an abnormal signal. The 'Realtime Pluse Width(ms)' box displays '1.505'. The 'Min' green box displays '1.005', the 'Mid' green box displays '1.256', and the 'Max' green box displays '1.506'. The 'Reset' button is highlighted with a black border.

- Pull and push the remote throttle gear to refresh the minimum, middle and maximum PPM signal values as Fig.7-30, Fig.7-31:

Fig.7-30.Refresh the maximum PPM signal

Min Pulse Width(ms): 1.030
Mid Pulse Width(ms): 1.505
Max Pulse Width(ms): 1.988
Safety Start: ON

Realtime Pluse Width(ms): 1.986

Min: 1.506
Mid: 1.747
Max: 1.988

Start
Reset
Setup
Stop

Fig.7-31.Refresh the minimum PPM signal

Min Pulse Width(ms): 1.030
Mid Pulse Width(ms): 1.505
Max Pulse Width(ms): 1.988
Safety Start: ON

Realtime Pluse Width(ms): 1.034

Min: 1.032
Mid: 1.510
Max: 1.988

Start
Reset
Setup
Stop

- Click “Setup” to write the current values into Min Pulse Width, Mid Pulse Width and Max Pulse Width separately as Fig.7-32:

Fig.7-32.Write current PPM signal value

Min Pulse Width(ms): 1.030
Mid Pulse Width(ms): 1.505
Max Pulse Width(ms): 1.988
Safety Start: ON

Realtime Pluse Width(ms): 1.506

Min: 1.032
Mid: 1.506
Max: 1.988

Start
Reset
Setup
Stop

- Click "Stop" to complete the PPM calibration as Fig.7-33:

Fig.7-33.PPM signal calibration ends

The screenshot shows the 'PPM' calibration window. At the top, 'Control Mode' is set to 'None'. Below it, 'Reverse Delay Time(s)' is 5.0. There are three input fields for pulse widths: 'Min Pulse Width(ms)' at 1.030, 'Mid Pulse Width(ms)' at 1.505, and 'Max Pulse Width(ms)' at 1.988. A 'Safety Start' dropdown is set to 'ON'. In the center, 'Realtime Pulse Width(ms)' is 1.506. At the bottom, there are three green boxes labeled 'Min' (1.030), 'Mid' (1.506), and 'Max' (1.988). On the right side, there are four buttons: 'Start', 'Reset', 'Setup', and 'Stop'. The 'Stop' button is highlighted with a black border, indicating it is the final step to complete calibration.

7.13.3.3 PPM Control Mode Setup

User can select suitable PPM Control Mode accordingly as Fig.7-34:

Fig.7-34.PPM Control Mode

The screenshot shows the 'PPM' control mode selection window. The 'Control Mode' dropdown menu is open, showing a list of options: 'None', 'Current(forward:1.0~2.0ms)', 'Current Brake(brake:1.0~1.5ms,forward:1.5~2.0ms)', 'Current Reverse(reverse:1.0~1.5ms,forward:1.5~2.0ms)', 'Current Reverse Brake(reverse/brake:1.0~1.5ms,forward:1.5~2.0ms)', 'Duty(forward:1.0~2.0ms)', 'Duty Reverse(reverse:1.0~1.5ms,forward:1.5~2.0ms)', 'Duty Reverse Brake(reverse/brake:1.0~1.5ms,forward:1.5~2.0ms)', 'Duty Reverse Slow(reverse:1.0~1.5ms,forward:1.5~2.0ms)', and 'Speed(forward:1.0~2.0ms)'. The 'None' option is currently selected. To the right, 'Reverse Delay Time(s)' is 5.0 and 'Safety Start' is 'ON'. At the bottom, there are three green boxes labeled 'Min' (1.032), 'Mid' (1.506), and 'Max' (1.988). On the right side, there are four buttons: 'Start', 'Reset', 'Setup', and 'Stop'.

- None
- Current
 - Current control mode, unidirectional control, forward only:1.0~2.0ms,
- Current Brake
 - Current control mode, bidirectional control, brake:1.0~1.5ms,forward:1.5~2.0ms
- Current Reverse
 - Current control mode, bidirectional control, no brake, reverse:1.0~1.5ms, forward:1.5~2.0ms
- Current Reverse Brake
 - Current control mode, bidirectional control, brake/reverse:1.0~1.5ms, forward:1.5~2.0ms.

- Duty(forward:1.0~2.0ms)

Current control mode, bidirectional control, brake/reverse:1.0~1.5ms, forward:1.0~2.0ms.

- Duty Reverse

Duty cycle control mode, bidirectional control, backward:1.0~1.5ms, forward:1.5~2.0ms.

- Duty Reverse Brake

Duty cycle control mode, bidirectional control, brake/reverse:1.0~1.5ms, forward:1.5~2.0ms.

- Duty Reverse Slow

Duty cycle control mode, bidirectional control, reverse:1.0~1.5ms, forward:1.5~2.0ms.

NOTE: No back EMF is generated when the accelerator is released.

- Speed

PID control mode, unidirectional control, forward only:1.0~2.0ms

- Speed Reverse

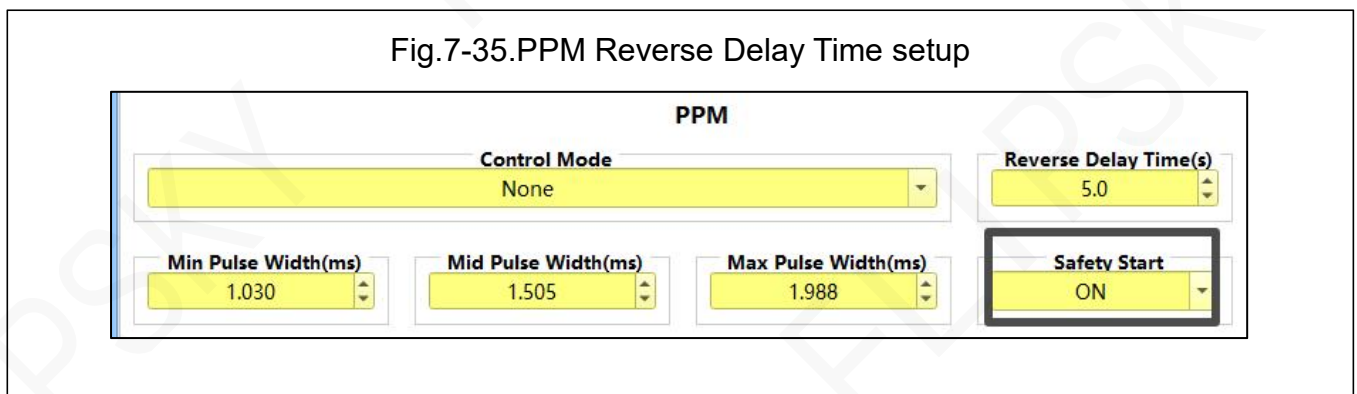
PID control mode, bidirectional control, reverse:1.0~1.5ms, forward:1.5~2.0ms

- Current Hybrid Bidirectional With Brake

Bidirectional current mixed control, reverse/brake: 1.0~1.5ms, forward/brake: 1.5~2.0ms

7.13.3.4 PPM Reverse Delay Time Setup

The PPM control mode supports reverse function, so you can set the UART Reverse Delay Time as you need. Adjustable range: 0.0s~10s, and the default is 5s as Fig.7-35:



Parameter Specification: When the motor is stopped, pull the throttle back to the maximum and start to rotate in the reverse direction. The reverse speed can reach the maximum within the set time period (as shown in Fig.7-32, it's 5 seconds as this set).

7.13.3.5 Safety Start Setup

When this parameter is set as ON, if the PPM throttle receives abnormal signal when ESC is connected, it will prevent the motor from accidental activation. Please calibrate the PPM signal again. After a certain period of power-up of ESC, the throttle signal should be zero. Only then will the PPM signal enter a valid control state as Fig.7-36:

Fig.7-36.Safety Start Setup

The screenshot displays the 'PPM' configuration window. It contains the following settings:

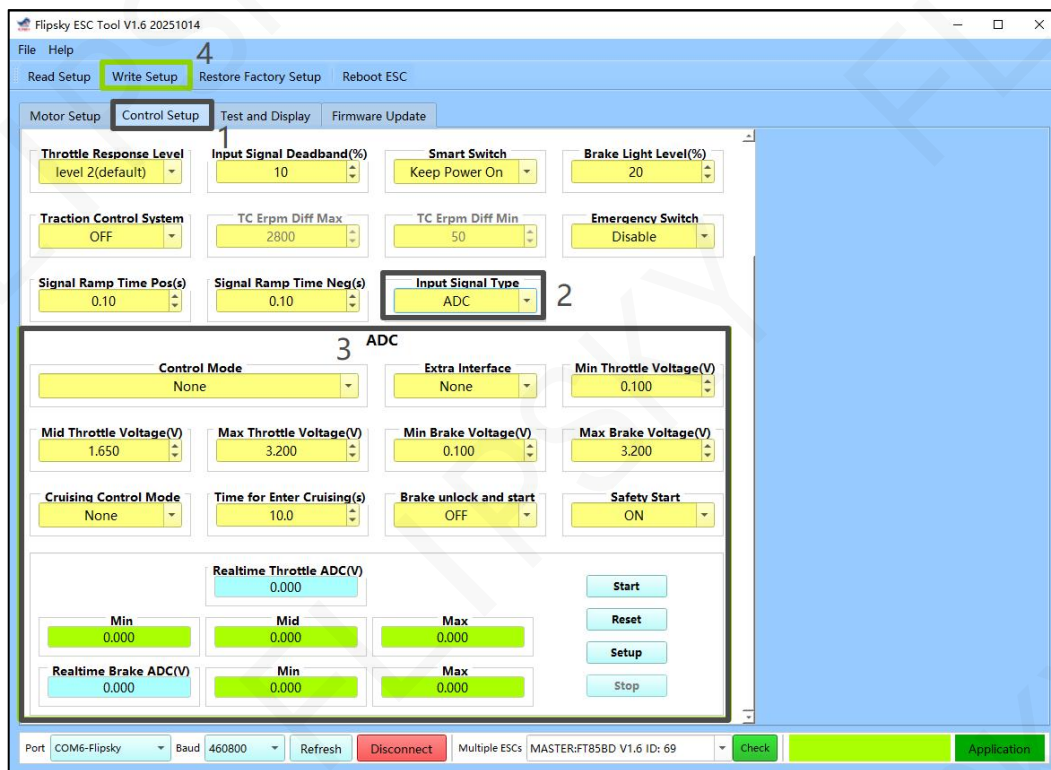
- Control Mode:** None
- Reverse Delay Time(s):** 5.0
- Min Pulse Width(ms):** 1.030
- Mid Pulse Width(ms):** 1.505
- Max Pulse Width(ms):** 1.988
- Safety Start:** ON (highlighted with a black box)

7.13.4 ADC

7.13.4.1 ADC Parameter Specification and Setup

ADC control type voltage range: 0V~3.3V. Please make sure the ADC input voltage lower than 3.3V to avoid the ESC 3.3V chip damage. Setting steps are as Fig.7-37. below:

Fig.7-37.ADC signal setup



- Step 1: Click “Control Setup” to enter the setup interface
- Step 2: Switch the "Input Signal Type" to "ADC"
- Step 3: Do ADC throttle calibration when use it for the first time. (view details in Chapter 7.13.4.2.)

NOTE: When using a DUAL ESC, only select the signal control type calibration for the signal master or slave under control, and set the signal type of the other master/ slave to OFF.

- Step 4: Select the appropriate ADC Control Mode per your needs (view details in chapter 7.13.4.3)
- Step 5: Set Extra Interface per your needs (view details in chapter 7.13.4.4)
- Step 6: Select the Cruising Control Mode per your needs (view details in chapter 7.13.4.5)

- Step 7: Set the Time for Enter Cruising per your needs (view details in chapter 7.13.4.5)
- Step 8: Set Brake Unlock and Start per your needs (view details in chapter 7.13.4.6)
- Step 9: Set Safety Start per your needs (view details in chapter 7.13.4.7)
- Step 10: Click the upper left menu "Write Setup" to save the setup.

7.13.4.2 ADC Input Throttle Calibration

Do ADC throttle calibration when use it as input signal for the first time as follows:

- Click "Start" to begin the calibration as Fig.7-38 below:

Fig.7-38.Enter ADC input throttle calibration

Realtime Throttle ADC(V) 0.773			Start Reset Setup Stop
Min 0.772	Mid 0.773	Max 0.774	
Realtime Brake ADC(V) 0.841			
Min 0.841	Max 0.844		

- Twist the ADC throttle to refresh the min, mid and max throttle signal as Fig.7-39 below:

Fig.7-39.Refresh ADC throttle signal

Realtime Throttle ADC(V) 2.420			Start Reset Setup Stop
Min 0.770	Mid 1.599	Max 2.428	
Realtime Brake ADC(V) 0.842			
Min 0.841	Max 0.845		

- Press the ADC brake to refresh the min and max brake value as Fig.7-40 below:

Fig.7-40. Refresh ADC brake signal

Realtime Throttle ADC(V) 0.770			Start Reset Setup Stop
Min 0.769	Mid 1.599	Max 2.430	
Realtime Brake ADC(V) 2.555			
Min 0.841	Max 2.558		

- Click "Setup" to write the min and max ADC signal value to Min Throttle Voltage, Mid Throttle Voltage, Max Throttle Voltage, Min Brake Voltage and Max Brake Voltage separately as Fig.7-41 below:

Fig.7-41. Write current ADC configurations

The screenshot shows the 'ADC' configuration window. At the top, there are dropdown menus for 'Control Mode' (set to 'None') and 'Extra Interface' (set to 'None'). Below these are four input fields for voltage thresholds: 'Min Throttle Voltage(V)' (0.768), 'Mid Throttle Voltage(V)' (1.599), 'Max Throttle Voltage(V)' (2.430), 'Min Brake Voltage(V)' (0.841), and 'Max Brake Voltage(V)' (2.558). Further down, there are settings for 'Cruising Control Mode' (None), 'Time for Enter Cruising(s)' (10.0), 'Brake unlock and start' (OFF), and 'Safety Start' (ON). At the bottom, there are real-time ADC values for throttle (0.770) and brake (0.845), along with their respective min, mid, and max values. On the right side, there are five buttons: 'Start', 'Reset', 'Setup' (highlighted with a black border), and 'Stop'.

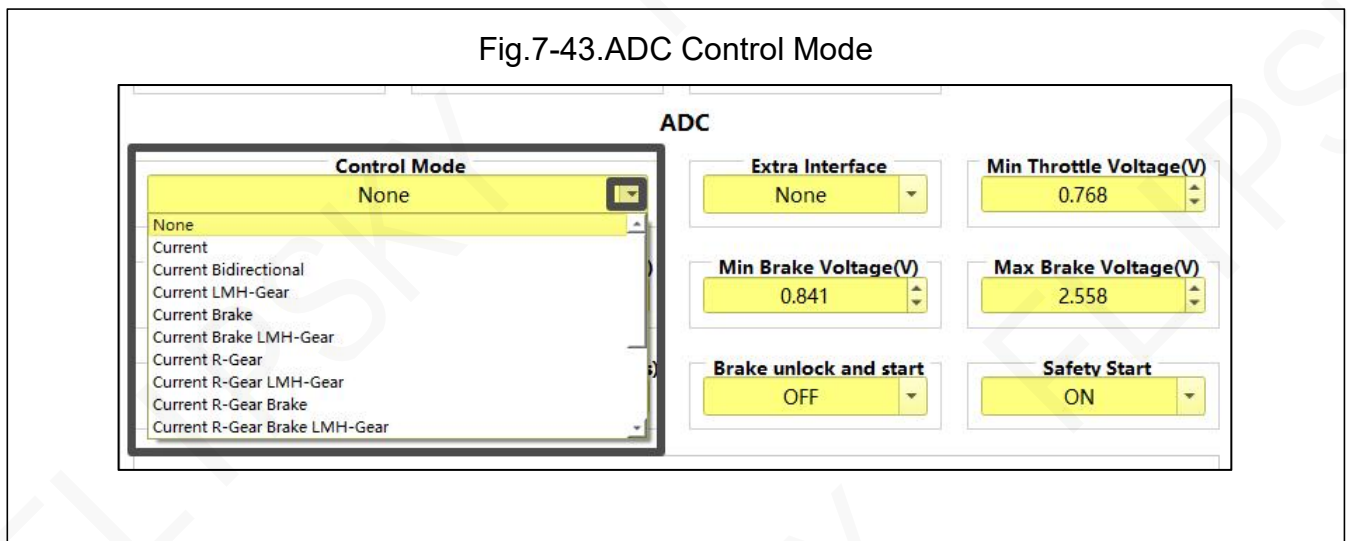
- Click "Stop" to finish the calibration as Fig.7-42 as below:

Fig.7-42. ADC input setup complete

This screenshot shows the same ADC configuration window as Fig.7-41, but with the 'Stop' button highlighted with a black border, indicating the completion of the calibration process. The values and settings remain the same as in the previous figure.

7.13.4.3 ADC Control Mode Setup

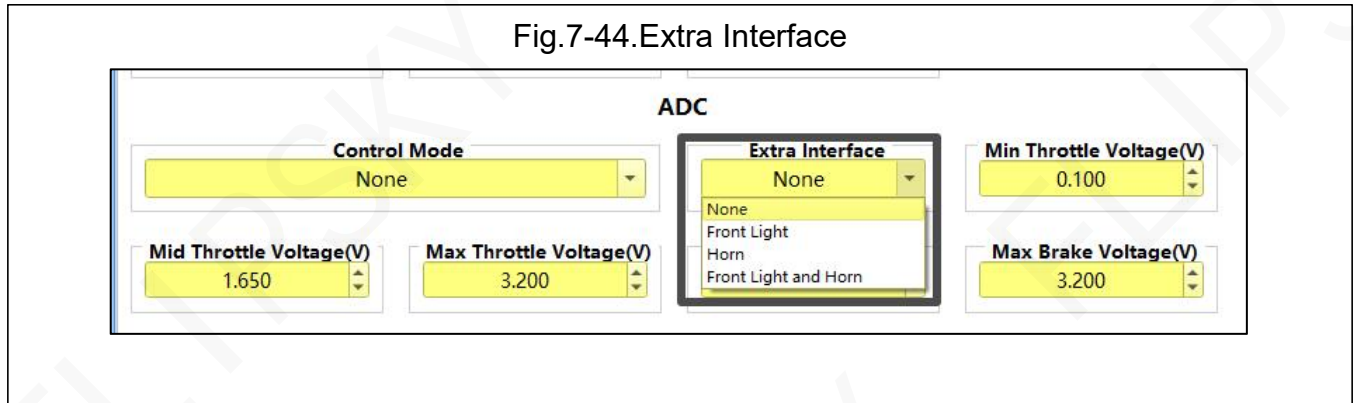
Select the appropriate ADC Control Mode per your needs as Fig.7-43 below:



- None: No response to any ADC input signal.
- Current: Current control mode, ADC throttle only without brake.
- Current Bidirectional: Current control mode, enable Throttle ADC and bidirectional current control
- Current LMH-Gear: Current control mode, only enable ADC throttle and L\|M\H speed gear.
- Current Brake: Current control mode, enable ADC throttle and brake.
- Current Brake LMH-Gear: Current control mode, enable Throttle ADC, Brake ADC and L\|M\H speed gear
- Current R-Gear: Current control mode, enable Throttle ADC and reverse gear.
- Current R-Gear LMH-Gear: Current control mode, enable Throttle ADC, reverse gear and L\|M\H speed gear
- Current R-Gear Braker: Current control mode, enable Throttle ADC, reverse gear and Brake ADC
- Current R-Gear Brake LMH-Gear: Current control mode, enable Throttle ADC, reverse gear , Brake ADC and L\|M\H speed gear
- Duty: Duty cycle mode, ADC throttle only without brake
- Duty Bidirectional: Bidirectional duty cycle control mode, ADC throttle only without brake
- Duty R-Gear: Reverse duty cycle control mode, Throttle ADC only without brake
- Speed: PID control mode, Throttle ADC only without brake
- Speed Bidirectional: Bidirectional PID control mode, Throttle ADC only without brake
- Speed R-Gear: Reverse PID control mode, Throttle ADC only without brake

7.13.4.4 Extra Interface Setup

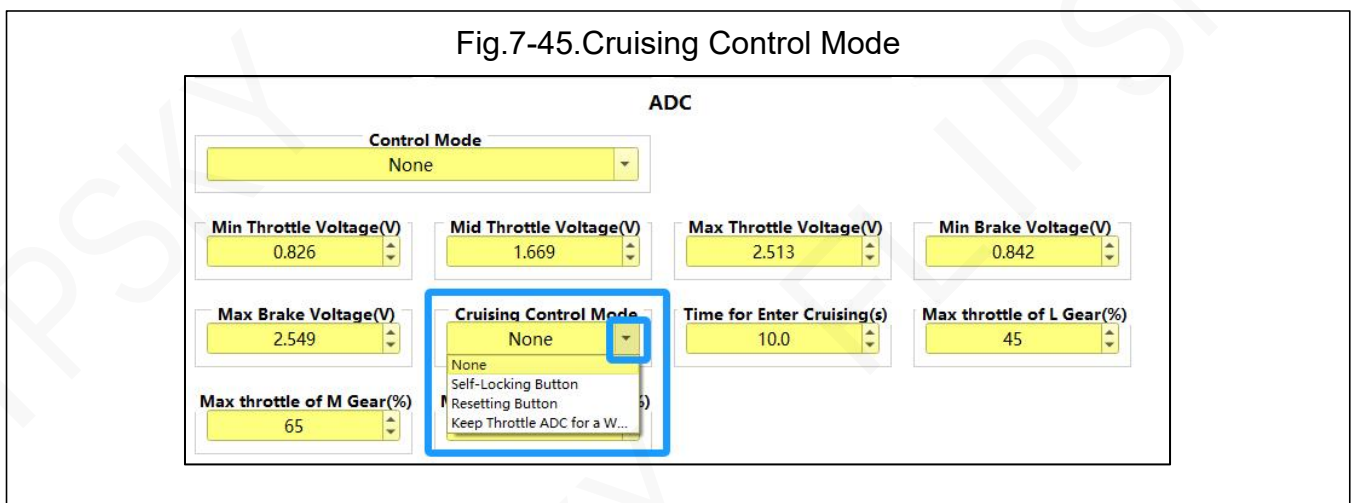
This function only support FT series ESCs with extra interface of 12V(front light and horn). It can switch on/off physically, setting the front light and horn interfaces to a continuous output state when controlled by ADC signal. User can set via Extra Interface per your needs, as Fig.7-44:



- None
- Front Light
- Horn
- Front and Horn

7.13.4.5 Cruising Control Mode Setup

Cruising Control Mode: three modes available, select different mode accordingly as Fig.7-45:



Parameter Specification:

- None

Turn off Cruising Control Mode

■ Self-Locking Button

Using self-locking button to activate the cruising control. Connect the PPM and GND pins for the self-locking button. Press the switch button to enter into cruise mode, and then re-press the button to bump out to end the cruise mode.

NOTE:

- The cruise mode can be entered only in status of throttle signal detected or motor running with rpm
- The cruise mode will stop if receive a brake signal in ADC control mode

■ Resetting Button

Using resetting button to activate the cruising control. Connect the PPM and GND pins for the self-locking button. Press once to enter into the cruise mode, and press again to end the cruise mode.

NOTE:

- The cruise mode can be entered only in status of throttle signal detected or motor running with rpm
- The cruise mode will stop if receive a brake signal in ADC control mode.

■ Keep Throttle ADC for a While

Entering the cruising control by keeping Throttle ADC for a while. Set the time via Time for Enter Cruising. Keep the throttle at desired level for a period of time and release, and then you will enter the cruise mode and stay. If any throttle change or brake signal is detected, the mode will be exited.

NOTE:

- The cruise mode will stop if receive a brake signal in ADC control mode.

◆ Time for Enter Cruising

If the cruise control mode is set to Keep Throttle ADC for a While, you can set the Time for Enter Cruising per your needs. Adjustable range: 3s to 30s, and the default is 10s as shown in Fig.7-46:

Fig.7-46.Time for Enter Cruising

ADC			
Control Mode None			
Min Throttle Voltage(V) 0.826	Mid Throttle Voltage(V) 1.669	Max Throttle Voltage(V) 2.513	Min Brake Voltage(V) 0.842
Max Brake Voltage(V) 2.549	Cruising Control Mode None	Time for Enter Cruising(s) 10.0	Max throttle of L Gear(%) 45
Max throttle of M Gear(%) 65	Max throttle of H Gear(%) 100		

7.13.4.6 Brake Unlock and Start Setup

Brake Unlock and Start: Unlock the throttle handle via the brake handle. When this function is enabled, ESC startups, please press the brake handle to activate the throttle, otherwise you will not be able to control it. This function prevents accidental activation of the throttle handle, which could trigger unintended acceleration and cause unexpected mechanical startup.

图 7-47.Brake Unlock and Start Setup

ADC			
Control Mode Current R-Gear Brake LMH-Gear	Extra Interface None	Min Throttle Voltage(V) 0.768	
Mid Throttle Voltage(V) 1.599	Max Throttle Voltage(V) 2.430	Min Brake Voltage(V) 0.841	Max Brake Voltage(V) 2.558
Cruising Control Mode None	Time for Enter Cruising(s) 10.0	Brake unlock and start OFF	Safety Start ON

7.13.4.7 Safety Start Setup

Safety Start: When this parameter is set as ON, if the ADC throttle receives abnormal signal when ESC is connected, it will prevent the motor from accidental activation. Please calibrate the ADC signal again. After a certain period of power-up of ESC, the ADC signal should be zero. Only then will the ADC signal enter a valid control state, as Fig.7-48:

Fig7-48.Safety Start Setup

ADC			
Control Mode Current R-Gear Brake LMH-Gear	Extra Interface None	Min Throttle Voltage(V) 0.768	
Mid Throttle Voltage(V) 1.599	Max Throttle Voltage(V) 2.430	Min Brake Voltage(V) 0.841	Max Brake Voltage(V) 2.558
Cruising Control Mode None	Time for Enter Cruising(s) 10.0	Brake unlock and start OFF	Safety Start ON

7.13.5 PAS

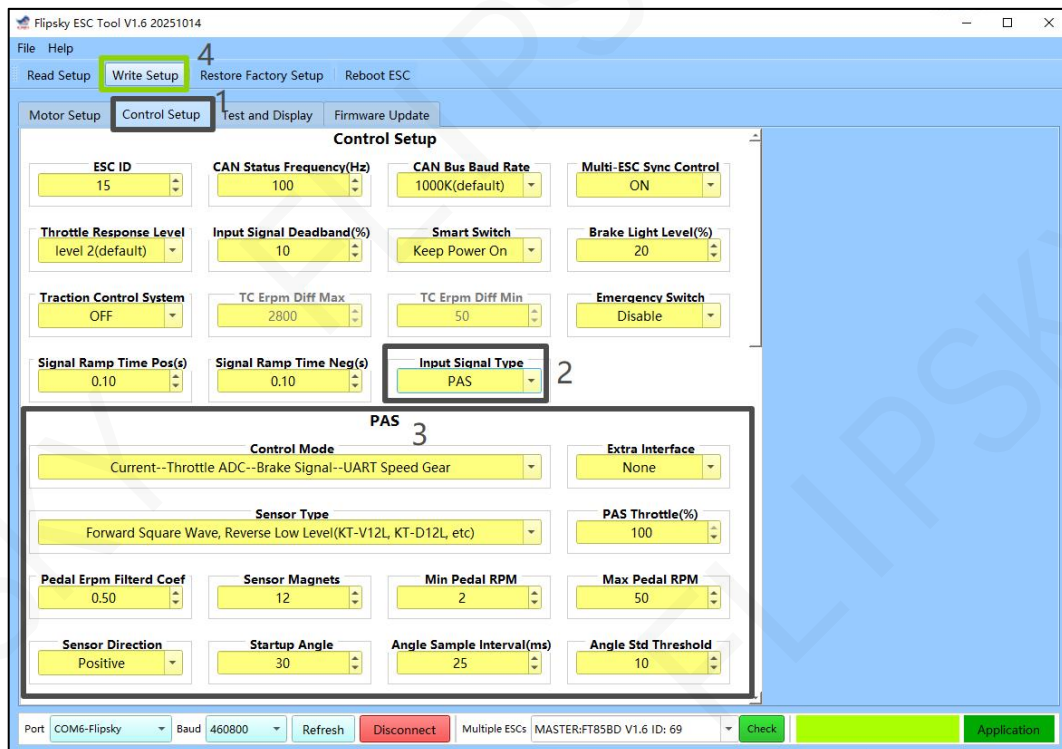
7.13.5.1 PAS Parameter Specification and Setup

PAS Function Description:

- (1) It's compatible with most of PAS cadence sensors in the market, which mainly include below three types(View in Chapter 7.13.5.3)
- (2) Enable FT ESCs to be compatible with Flipsky TFT screen display which supports real-time data like speed, voltage etc. read and PAS speed gear levels setup.
- (3) Support connect brake signal function: The ESC will cut off power output immediately once receiving the brake signal; When the control mode set to enable Brake Signal, pin GEAR_R and pin GEAR_L are used to detect the brake signal and are available at GND. That is, when the pin GEAR_R or the pin GEAR_L detects GND, the ESC will cut off the power output immediately, and disable the throttle gear.

Setup steps are as Fig.7-49 below:

Fig.7-49.PAS Input Signal Setup



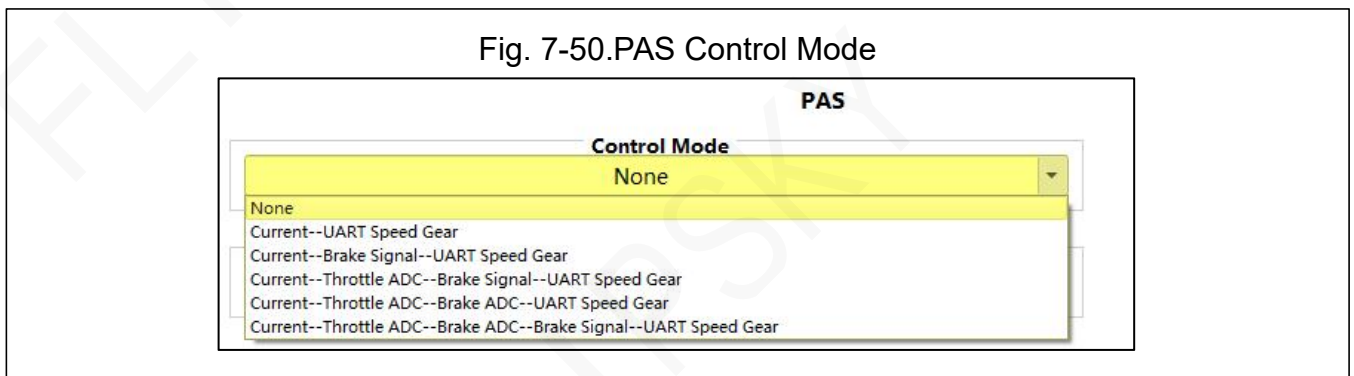
- Step 1: Click “Control Setup” to enter the setup interface
- Step 2: Switch the "Input Signal Type" to "PAS"
- Step 3: When using PAS as the input signal for the first time, please do ADC calibration if

you firstly use PAS as input signal.(view details in 7.13.4.2);

- Step 4: Select suitable PAS Control Mode per your needs (view details in 7.13.5.2)
- Step 5: Select suitable PAS Sensor Type per your needs (view details in 7.13.5.3)
- Step 6: Set the PAS parameters per your needs (view details in 7.13.5.4)
- Step 7: Set the Boost Current For Startup per your needs (view details in 7.13.5.4);
- Step 8: Set the Minimal Current at Zero Pedal per your needs (view details in 7.13.5.4);
- Step 9: Set the Extra Interface per your needs (view details in 7.13.4.4)
- Step 10: Click the upper left menu "Write Setup" to save the configuration.

7.13.5.2 PAS Control Mode Setup

Select suitable PAS Control Mode per your needs as Fig.7-50:



■ None: No response to any PAS input signal

■ Current--UART Speed Gear:

Current control mode: UART Speed Gear--Use ESC UART port to connect the Flipsky TFT screen display to switch different speed gears

■ Current--Brake Signal--UART Speed Gear:

Current control mode: Brake Signal--The ESC will cut off power output immediately once receiving the brake signal; UART Speed Gear--Use ESC UART port to connect the Flipsky TFT screen display to switch different speed gears.

■ Current--Throttle ADC--Brake Signal--UART Speed Gear:

Current control mode: enable throttle ADC control; Brake Signal--The ESC will cut off power output immediately once receiving the brake signal; UART Speed Gear--Use ESC UART port to connect the Flipsky TFT screen display to switch different speed gears.

■ Current--Throttle ADC--Brake ADC--UART Speed Gear:

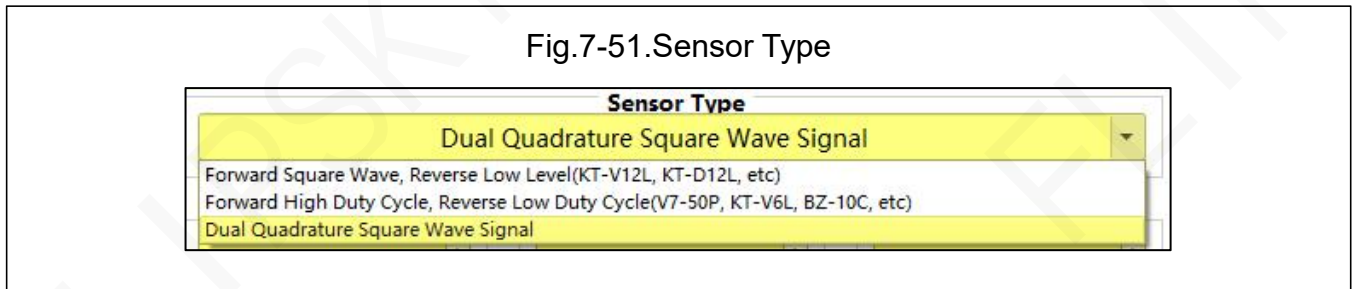
Current control mode: enable throttle ADC and Brake ADC control; UART Speed Gear--Use ESC UART port to connect the Flipsky TFT screen display to switch different speed gears.

■ Current--Throttle ADC--Brake ADC--Brake Signal--UART Speed Gear:

Current control mode: enable throttle ADC and Brake ADC control; UART Speed Gear--Use ESC UART port to connect the Flipsky TFT screen display to switch different speed gears.

7.13.5.3 PAS Sensor Type Setup

Select suitable PAS Sensor Type per your needs as Fig.7-51 below:



■ Forward Square Wave, Reverse Low Level(KT-V12L, KT-D12L, etc):

Output square wave when rotates in forward direction; output Low Level when rotates in reverse.

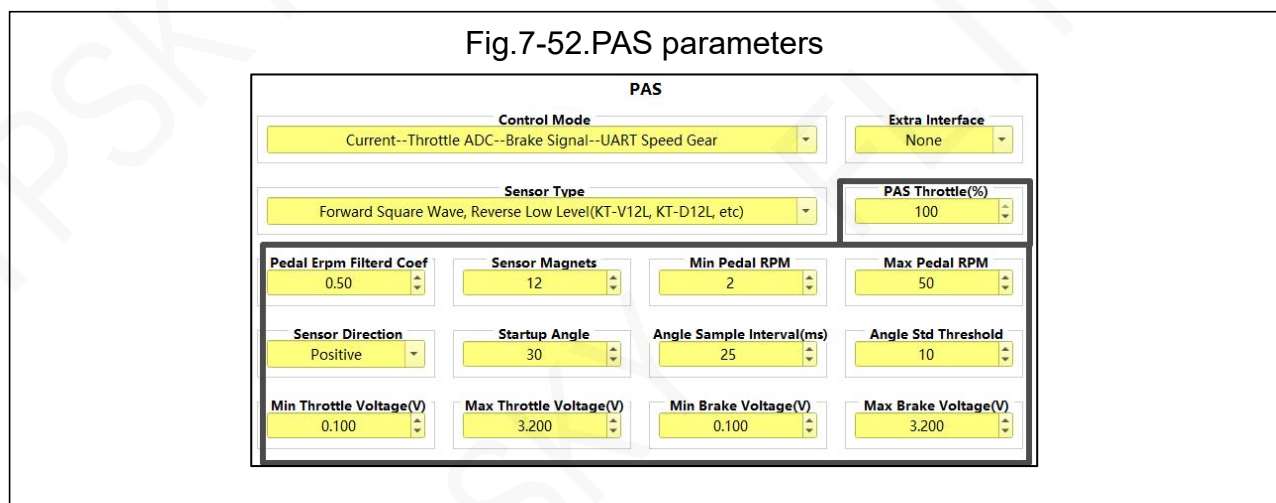
■ Forward High Duty Cycle, Reverse Low Duty Cycle(V7-50P, KT-V6L, BZ-10C, etc):

Output High duty cycle rectangular wave when rotates in forward direction; output Low duty cycle rectangular wave when rotates in reverse.

■ Dual Quadrature Square Wave Signal

7.13.5.4 PAS Parameters Setup

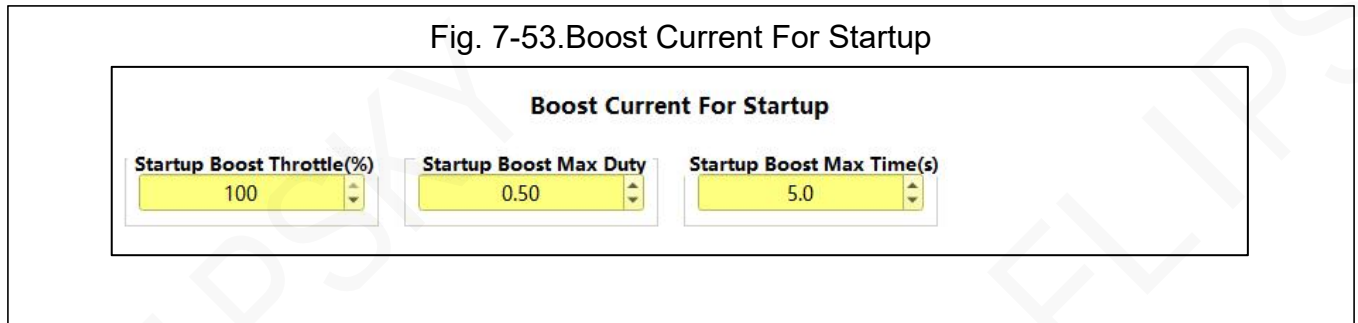
PAS parameters include: "PAS Throttle", "Pedal Erpm Filterd Coef", "Min Pedal RPM", "Max Pedal RPM", "Sensor Direction", "Sensor Magnets", "Startup Angle", "Angle Sample Interval(ms)", "Angle Std Threshold" as Fig.7-52:



- **PAS Throttle:** PAS max current output percentage (adjustable range: 1-100%).
The maximum PAS current output will be limited within this setting percentage of the motor current.
- **Pedal Erpm Filterd Coef:** adjustable range:0~1;
- **Sensor Magnets:** magnet number of PAS sensor, adjustable range:1~32.
e.g., 12 magnets generate 12 pulses per revolution.
- **Min Pedal RPM:** the minimum PAS pedal RPM (adjustable range: 1-200RPM).
The ESC starts current output when the pedal RPM reaches the mini setting value. Otherwise, the current output is zero.
- **Max Pedal RPM:** the maximum PAS pedal RPM (adjustable range: 1-500RPM).
The ESC current output will stop increasing if the pedal RPM reaches the max setting value. Once this value is exceeded, the ESC output current will remain at the maximum value (%) of "PAS Current Throttle".
- **Sensor Direction:** positive/negative, applicable for dual square signal ONLY.
- **Startup Angle:** adjustable range: 1~360°
When PAS sensor detects the angle exceeds this value, the output function will be activated.
- **Angle Sample Interval(ms):** adjustable range: 0~30000ms
Used to sample the PAS angle data over a specific time period and calculate the standard deviation based on it, to confirm whether the pedal is in an operational state.
- **Angle Std Threshold:** adjustable range: 1~255
When the standard deviation is less than this value, it means the user has stopped pedaling, which also signifies that the PAS current output will simultaneously cease.

7.13.5.5 Boost Current For Startup

The parameters of this function are used for improving startup performance, including: “Startup Boost Throttle(%)”, “Startup Boost Max Duty”, “Startup Boost Max Time(s)”; as Fig.7-53:



- **Startup Boost Throttle(%)**: adjustable range:0~100%

When the parameter set as 100%, indicating it will startup with max motor current.

- **Startup Boost Max Duty**: adjustable range:0~1

The startup booster power output will stop when the motor loading exceeds this value.

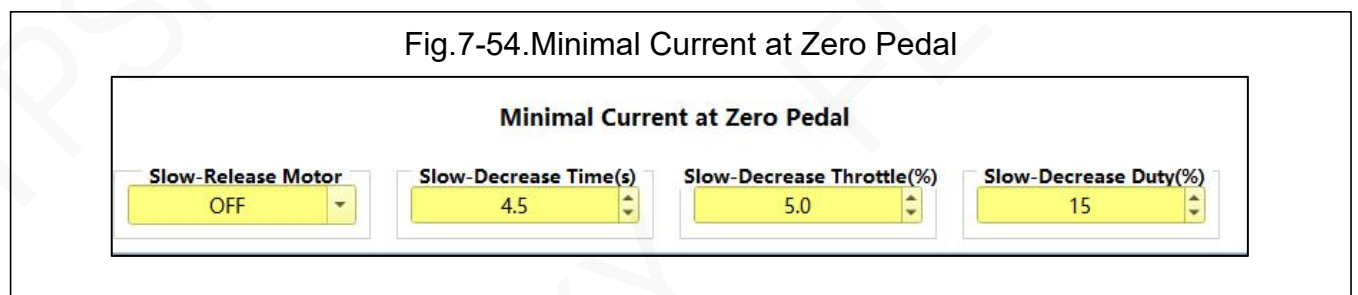
- **Startup Boost Max Time(s)**: adjustable range:0~100

It will stop once the startup boost reaches a certain time.

7.13.5.6 Minimal Current at Zero Pedal Setup

This function enables the FT series ESC to maintain a certain level of current output even after releasing the throttle pedal when receiving a PAS input signal.

Setup steps are as Fig.7-54:



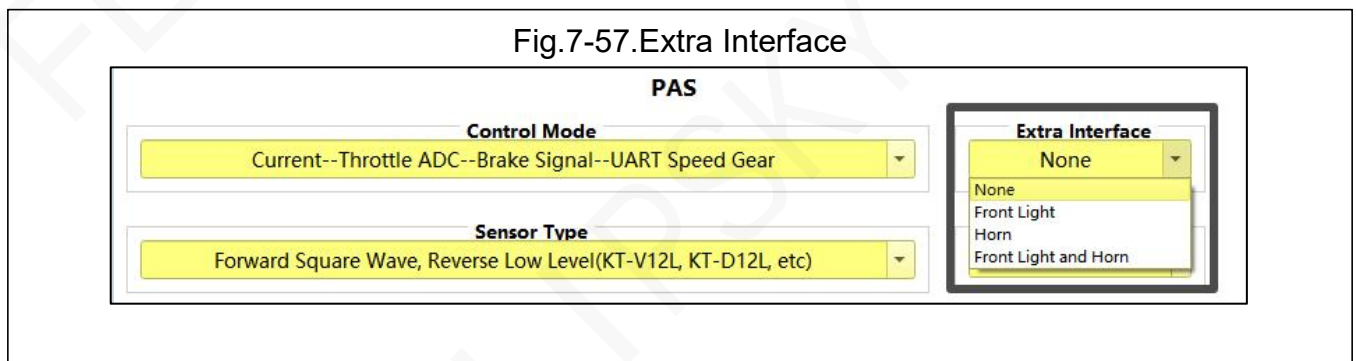
User can set according to the parameter specification and actual needs.

Parameter Specification:

- **Slow-Release Motor:** use the function or not, default as OFF;
- **Slow-Decrease Time:** set the output current continuous time of it, default as 4.5s;
- **Slow-Decrease Throttle :** set output throttle, default as 5%, which means the output current is $0.05 \times \text{Motor Current}$;
- **Slow-Decrease Duty:** Only when the motor duty exceeds this setting value this function is available.

7.13.5.7 Extra Interface Setup

This function only support FT series ESCs with extra interface of 12V(front light and horn). It can switch on/off physically, setting the front light and horn interfaces to a continuous output state when controlled by PAS signal. User can set via Extra Interface per your needs, as Fig.7-57:



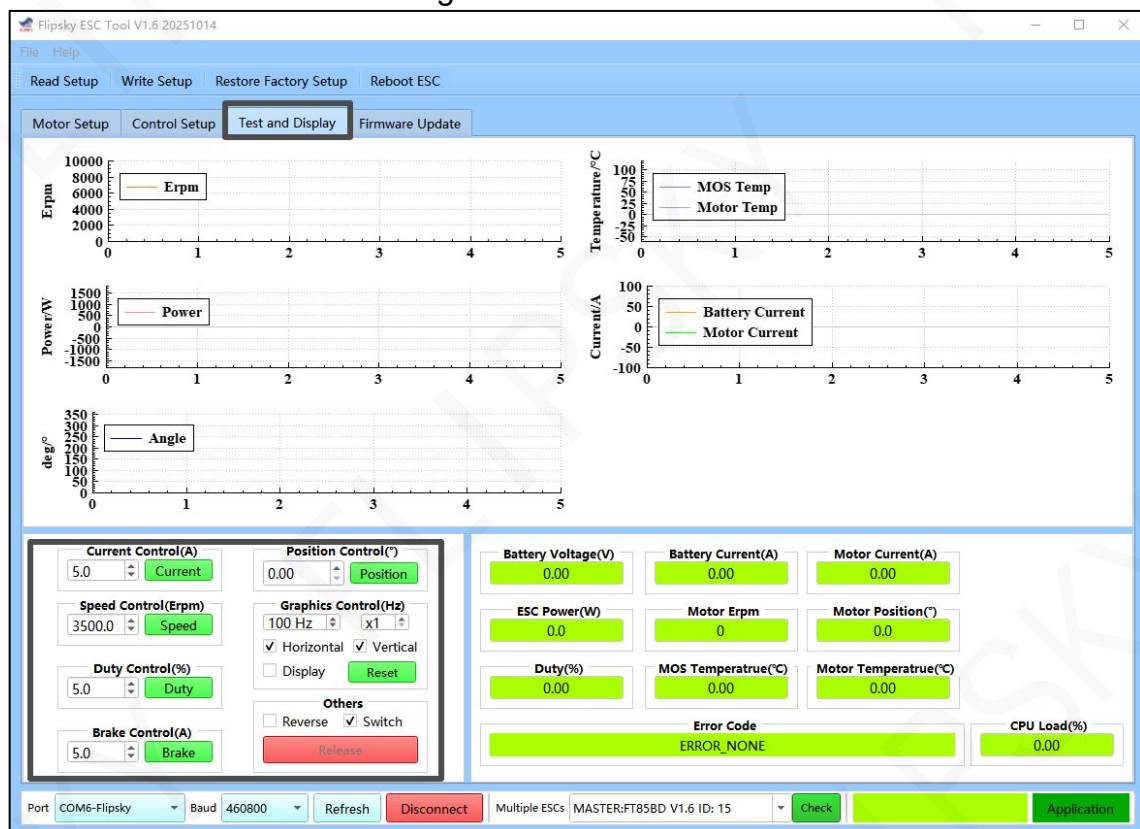
- None
- Front Light
- Horn
- Front and Horn

8 Test and Display

8.1 Flipsky ESC Tool Button Control

Switch the "Input Signal Type" to "OFF" if you want to control the ESC on the Test and Display interface of Flipsky ESC Tool, only then can you control the ESC by Current, Duty, Speed, Position, Brake and other buttons as Fig.8-1:

Fig.8-1.Control button

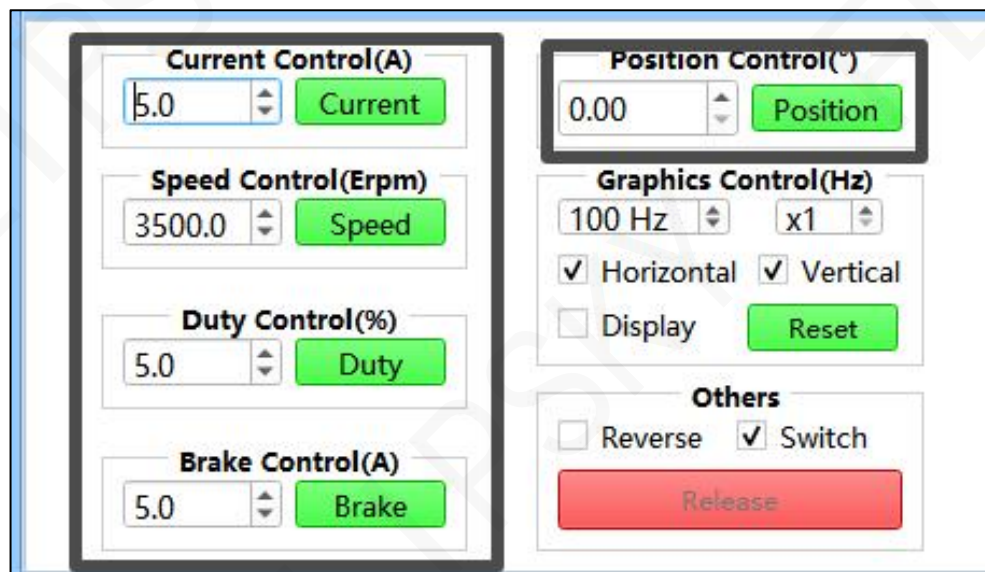


8.1.1 Button Control Specification

Button control setup:

- Drive: Click the "Current"-current control mode (torque control mode), "Speed"- speed control mode, "Duty"-duty cycle control mode (voltage control mode) and "Position"-position control mode buttons to drive. During the driving process, the current, speed, duty cycle and position angle can be adjusted according to needs as shown in 8-2:

Fig.8-2.Four control modes



NOTE: Position Control is applicable in **Encoder Control Mode** only.

- Reverse the motor: Click "Reverse" to change the motor running direction as Fig.8-3.

Fig.8-3.Reverse direction



- Brake: Click "Brake" to enter the brake setup as Fig.8-4:

Fig.8-4.Brake

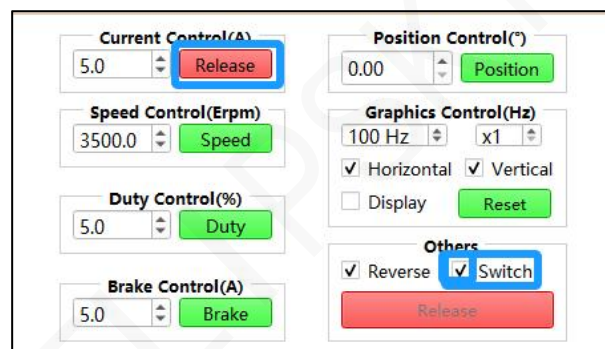


NOTE: If you use the DC power supply, please do not brake in high speed. Otherwise, the large back EMF will damage the ESC or even the power supply when the motor brakes in a high speed mode.

- Stop: Click "Release" to stop the motor running. Two methods available:

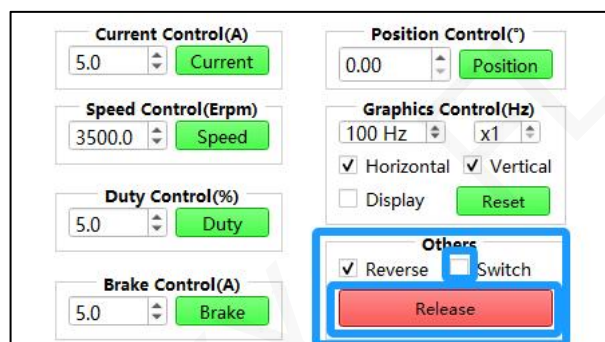
Click "Switch" and the "Current" under Current Control, then the "Current" will switch to "Release". Click the "Release" to stop the motor rotating as Fig.8-5:

Fig.8-5.Stop method 1



If not tick the "Switch" and it's "Current" under Current Control, the motor can only be stopped through the "Release" under the "Others" as Fig.8-6:

Fig.8-6. Stop method 2

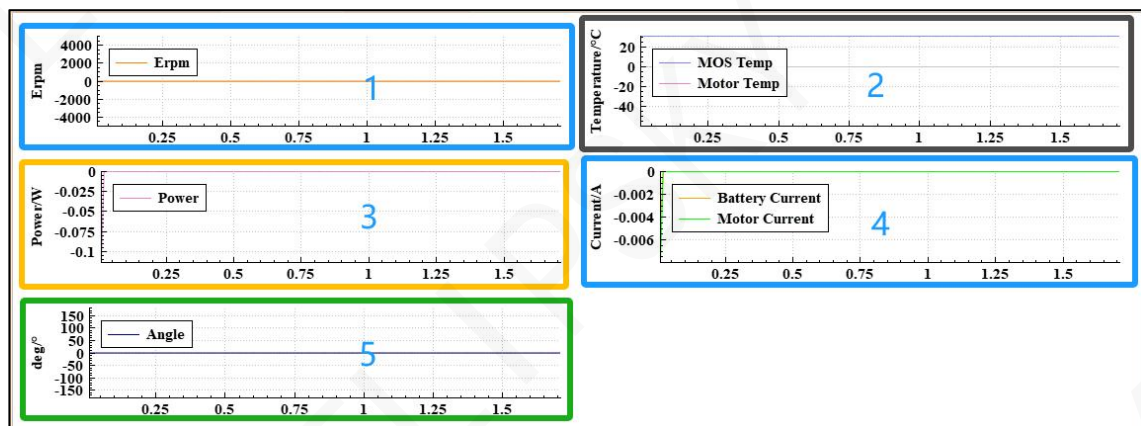


8.2 Data Analysis

8.2.1 Data Curve Display and Setup

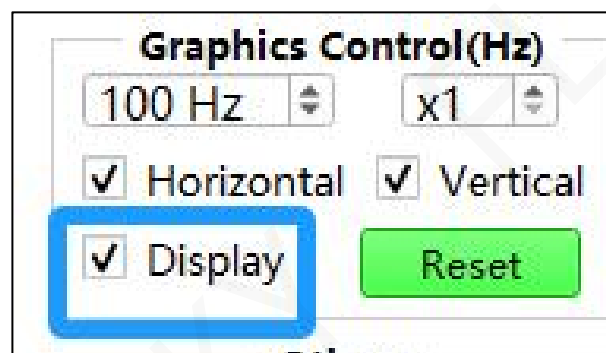
- There are 5 real-time data graphs as Fig.8-7 below:
 - Erpm curve
 - ESC MOSFET and motor temperature curve (Temperature/°C)
 - Power curve (Power/W)
 - Battery and motor current curve(Current/A)
 - Motor deflection angle curve(deg/°)

Fig.8-7.Real-time data curve display interface



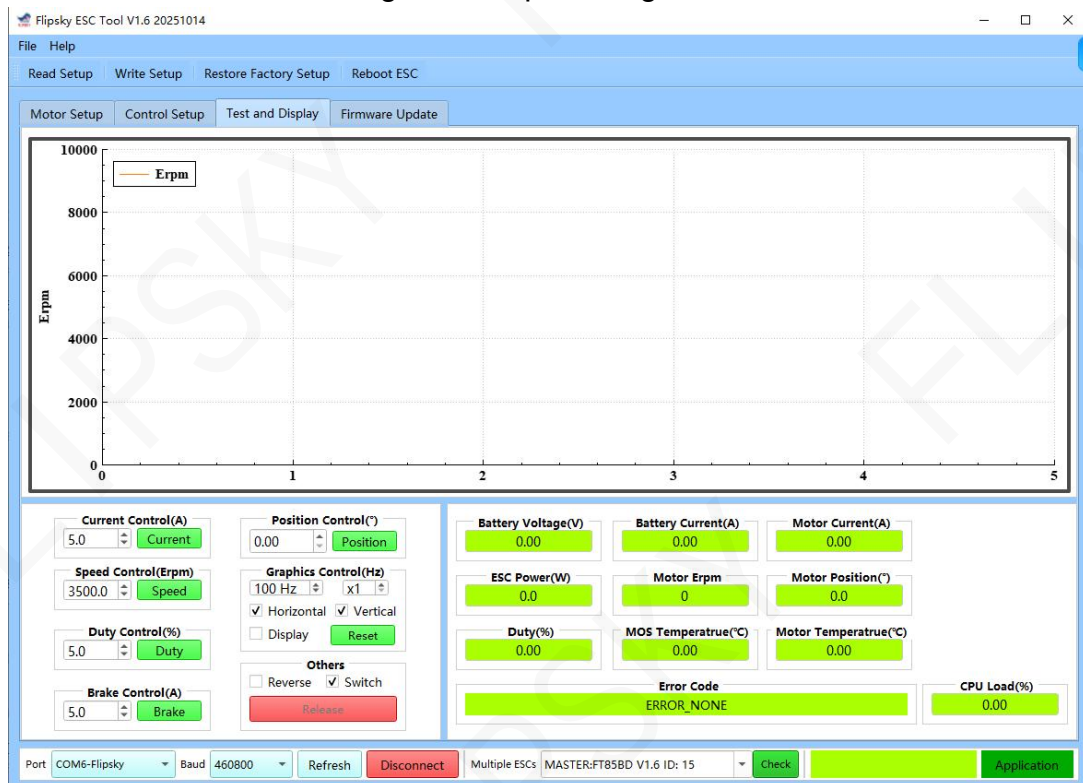
- Data Curve Display Setup Steps:
 - Activate the graph: Tick "Display" to activate graphics data display as Fig.8-8:

Fig.8-8.Graph display setup



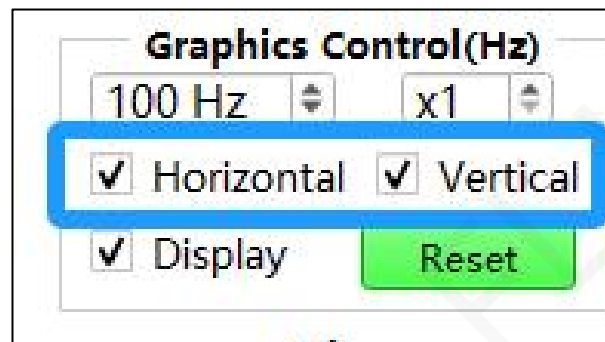
- Graph enlargement: Move the mouse cursor in the graph and left-click twice to enlarge it, and restore it in the same way as Fig.8-9 below:

Fig.8-9. Graph enlargement



- Magnify/reduce the graph horizontal and vertical axis values: Tick both "Horizontal" and "Vertical" (as Fig.8-10), move the mouse cursor into the graph and then scroll the mouse wheel to magnify/reduce.

Fig.8-10



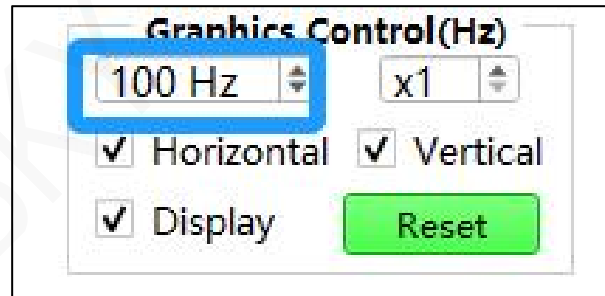
NOTE:

- (1) Tick "Horizontal" only if you just want to increase or decrease the values of horizontal axis.
- (2) Tick "Vertical" only if you just want to increase or decrease the values of vertical axis.
- (3) Tick both "Horizontal" and "Vertical" if you want to increase or decrease the values of

vertical and horizontal axis at the same time.

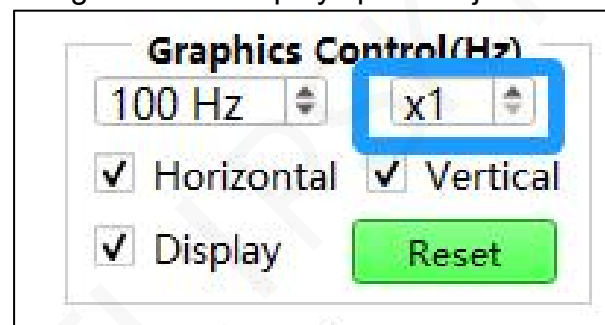
- The sampling frequency setup: default 100Hz, which means that the data samplings are 100 times per second. Adjustable setup range: 50Hz~1000HZ as Fig.8-11:

Fig.8-11. The sampling frequency setup of graphics



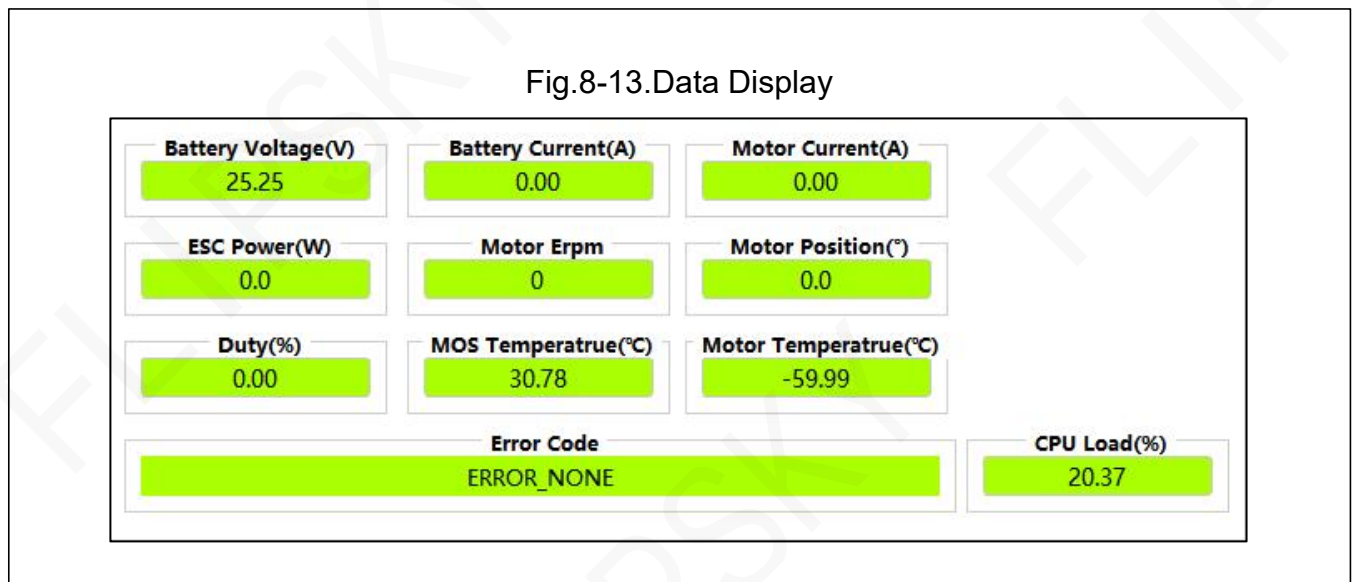
- Curve play speed setup: default "x1", adjustable range 1~10 times as8-12:

Fig.8-12. Curve play speed adjustment



8.2.2 Real-time Data Display

In order to facilitate the observation of the data and display the curve graph data in a more intuitive way, we increase some data that the curve graph does not have to better understand the equipment status as Fig.8-13:

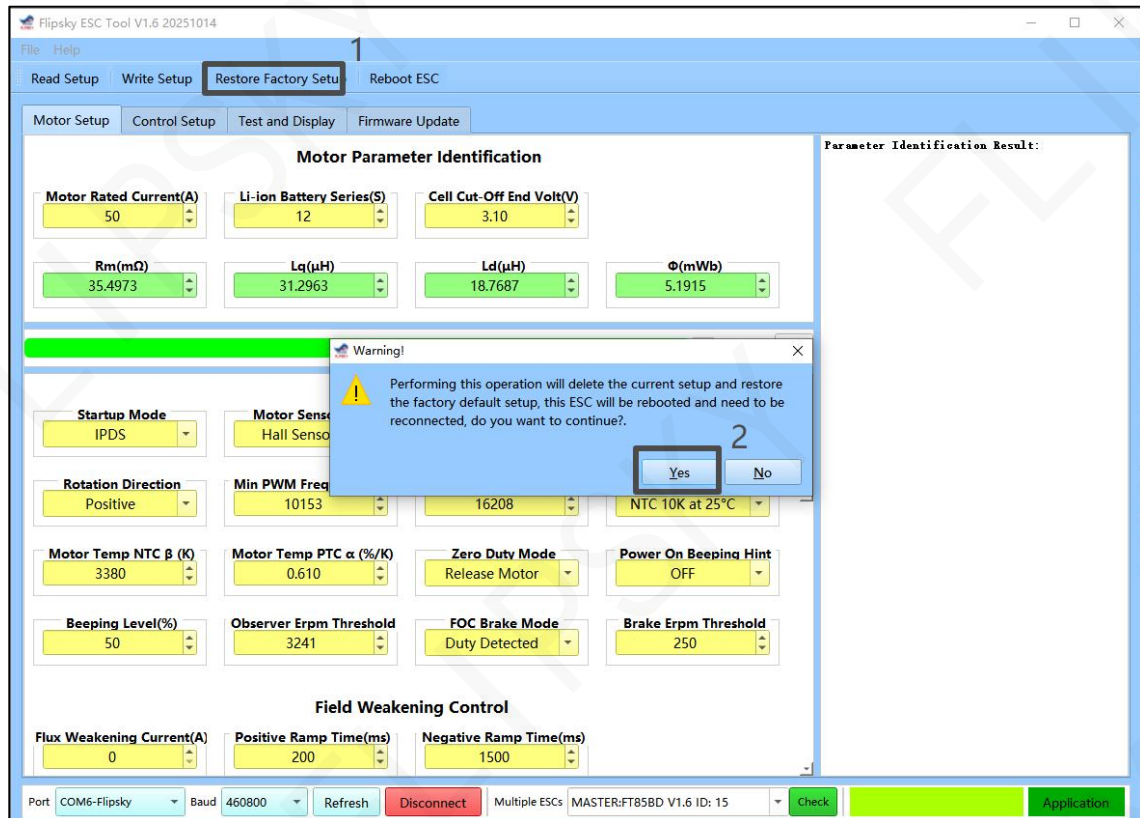


- ESC Power(W): ESC power display, corresponding to the graphic data
- Battery Voltage(V): Battery voltage display
- Battery Current(A): Battery current display, corresponding to the graphic data
- Motor Current(A): Motor current display, corresponding to the graphic data
- Motor Erpm: Motor erpm display, corresponding to the graphic data
- Motor Position: Motor deflection angle position display, corresponding to the graphic data
- Duty(%): Duty cycle percentage display
- MOS Temperature (°C): ESC MOSFET temperature display, corresponding to the graphic data
- Motor Temperature (°C): Motor working temperature display, corresponding to the graphic data
- CPU Load(%): CPU loading percentage display
- Error Code: Fault code display

9 Restore Factory Default Setup

If meet with some problems like parameters setup failure, ESCs connection failure, motor parameter identification failure etc., please try to restore the factory default setup as Fig.9-1 below:

Fig.9-1.Restore factory default setup



10 Version

Sheet 1. File Version

Date	Version	Contents Changed
2023/8/12	V1.0	
2023/11/8	V1.1	1. Fixed the problem that the Hall parameter cannot pass when the load is too heavy. 2. Optimize the parameter adjustment procedure. All interfaces are shielded during parameter adjustment to prevent other signals from affecting parameter adjustment. 3. Optimize the dead zone setting mode in ADC mode, and fix the problem that some gamepads may not start smoothly when used; 4. Rectify the fault that the electrical connection to the upper computer fails.
2024/01/19	V1.2	FT ESC Function Upgrade: 1. Add dual-drive ESC FT85BD 2. Add Smart Switch function supporting slide power on and auto turn off(support FT85BD only) 3. Add Extra Interface setup to enable front light, brake light and horn etc. for ESC FT85BD under UART control mode. Support VX4 to control the function of front lights, brake lights and horn; this function only supports FT85BD; set via Extra Interface parameters. 4. Add the function of FT series ESC under UART input signal to maintain a certain current output after releasing the throttle. In some high gear ratio (>3.5:1) skateboards, such as off-road skateboards, after releasing the throttle, there will be some resistance similar to braking, but in fact the resistance is caused by the high gear ratio. This function can maintain a certain current output after the remote releases the

		throttle, and this current can counteract these resistances, and be able to make the skateboard in an almost zero resistance release state. 5. Slow-Release Motor: Whether use this function, default as OFF 6. Slow-Decrease Time: Used for setting the continuous time of output current, default as 4.5s 7. Slow-Decrease Throttle: Used for setting the output throttle, default as 5%, which means the output current is $0.05 \times \text{Motor Current}$. 8. Slow-Decrease Duty: Only when the duty cycle is higher than this value, this function is available. 9. Add the Duty control mode of FT series ESCs in ADC mode, as well as bidirectional current and bidirectional duty cycle control modes. 10. Add Zero Duty Mode, when the user gives 0% duty cycle, the user can choose whether to keep output or release motor. 11. Repair the problem that some hub motors can not measure the magnetic chain when the inertia is relatively large. 12. Repair the problem that the maximum speed cannot limit the gear speed. Flipsk ESC Tool Function Upgrade: 1. Repair the abnormal problem that the curve cannot display the waveform corresponding to the waveform name. 2. Optimize the version judgment to prevent user from using unmatched firmware and causing abnormal problems. 3. Repair the problem that sometimes real-time data cannot be displayed normally after the Flipsky ESC Tool turns on.
2024/04/22	V1.3	FT ESC Function Upgrade: 1. Add FT ESC FT85RS 2. Optimize the motor flux linkage detection for some small motors with large resistance. 3. Optimize real-time data acquisition protocol <u>(Notes: the remote VX4 needs to update firmware V1.3.0, otherwise the real-time data will display abnormally and the parameter adjustment results will display abnormally.)</u>

		4. The minimum set voltage of lithium battery can be set to 2.6V to support lithium iron phosphate battery. 5. Add PAS Function: (1) It's compatible with most of PAS cadence sensors in the market, which mainly include below three types: a. Forward Square Wave, Reverse Low Level like KT-V12L, KT-D12L etc. b. Forward High Duty Cycle, Reverse Low Duty Cycle like V7-50P, KT-V6L, BZ-10C etc. c. Dual quadrature square wave signal output: the A-phase signal leads the B-phase signal when rotating in the forward direction, while the A-phase signal lags behind the B-phase signal when rotating in the backward / reverse direction. (2) Enable FT ESCs to be compatible with Flipsky TFT screen display which supports real-time data like speed, voltage etc. read and PAS speed gear levels setup. (3) Support Brake Signal function: The ESC will cut off power output immediately once receiving the brake signal; When the control mode set to enable Brake Signal, pin GEAR_R and pin GEAR_L are used to detect the brake signal and are active at Vol. That is, when the pin GEAR_R or the pin GEAR_L detects a Vol(GND), the ESC will cut off the power output immediately, and disable the throttle gear. Flipsky ESC Tool Function Upgrade: 1. Add FT85RS ESC 2. Add PAS function
2024/8/24	V1.4	FT ESC Function Upgrade: 1. Add FT60BD ESC 2. Optimize the motor voltage detection accuracy by adding motor "Bus Voltage Resistance" setup. 3. Add Flux Weakening Current Control. 4. Add encoder function. Enable encoder control setup like AS5047 (SPI)

		and ABI Encoder. 5. Add speed control PID mode. Maintain the default parameters in most situations. 6. Add position control PID mode (for encode control status only). Maintain the default parameters in most situations. 7. Add more types of motor temperature sensors such as NTC-10K-25℃, NTC-100K-25℃, PT1000, PTC-1K-100℃, KTY83 series, KTY84 series etc. 8. Add ESC power on beeping alert ON/OFF setup. Also, the beeping voice level percentage can be adjusted. 9. Add Brake Light Level setup to enable the light brightness to be adjusted during the braking process (available for FT60BD and FT85BD only at present). 10. In UART control mode, add Assisted Turnaround Mode setup like fixed-circle turnaround for 2WD mode and tank turnaround for 4WD mode. 11. In UART control mode, add Brake-Parking Mode ON/OFF setup to support parking mode by pulling remote controller brake gear/bar (achieved by transferring current into the motor D axis) 12. In UART control mode, add Off-Road Climbing Mode ON/OFF setup to switch duty cycle control and allow users to pass off-road climbing process easily by controlling speed of the four wheels synchronously 13. In ADC control mode, add Cruising Control Mode setup in three different situations. (1) Self-locking switch (2) Reset switch (3) Throttle hold Flipsky ESC Tool Function Upgrade: 1. Add FT60BD ESC. 2. Add erpm control and position control modes in Test and Display interface; add position data display.
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2025/5/11	V1.5	FT ESC Function Upgrade 1. Add firmware of FT85RS with enclosure. Enclosure version and water cooling version use the same firmware 2. Add firmware of FT85GS with enclosure 3. Add firmware of DR60BS UAV ESC 4. Optimize the brake control algorithm and added three new brake modes control: (1) When entering the brake state and the duty cycle is detected to be lower than 0.001, it switches to three-phase short circuit state; (2) When entering the braking state and the detected erpm is lower than the Brake Erpm Threshold value, it will switch to the three-phase short-circuit state; (3) When entering the braking state, it will switch to the three-phase short-circuit state when it detects the duty cycle is lower than 0.001 and the absolute value of erpm is lower than the Brake Erpm Threshold value at the same time; 5. Optimize Flux Weakening Control Algorithm, add new Flux Weakening Control Current Positive Ramp Time and Negative Ramp Time. Users can adjust the Flux Weakening Current Response Time by the value of Positive Ramp Time and Negative Ramp Time. For example, when Positive Ramp Time is set to 1000ms and Flux Weakening Current is 20A, then when the duty cycle reaches 90% or more, the Flux Weakening Current will increase from 0A to 20A within 1000ms, and when Negative Ramp Time is set to 1000ms, when the current Control command is set to 0, then the weak magnetic current will decrease from 20A to 0A within 1000ms, and then end the weak magnetic control. 6. Add multiple ESCs jointly control to realize TRC control. This algorithm can be used to prevent the driving wheels from slipping when multiple ESCs accelerate the driving motors at the same time, and this mode is only for the current control (does not support the current mixing bi-directional control in the PPM mode); 7. Add CAN Status Frequency parameter. Each ESC will automatically return real-time data through the CAN bus, which includes motor current, battery current, motor electronic speed, duty
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		cycle, MOSFET temperature, motor temperature, battery voltage data. The real-time data above will be sent back to the CAN bus at regular intervals. The sending frequency can be set via CAN Status Frequency. 8. Add unidirectional speed PID control, bidirectional speed PID control and bidirectional current mixing control under PPM mode. The maximum speed value of the speed PID control mode in PPM mode can be set by the parameter PID Maximum Erpm. For bidirectional current hybrid control, the control logic is as follows: (1) When the rocker has no movement and is centered (PPM input signal is Mid Pulse Width value), there is no current motor power output; (2) When you push the throttle forward (PPM input signal > Mid Pulse Width value), then the motor turns positively; (3) When you push the throttle forward, the motor is rotating forward, if you pull back the rocker now (the input signal < Mid Pulse Width value), then the motor enters the braking state; (4) When the motor enters the braking state, after the rocker is centered, pull back the throttle again, the motor will enter the reverse drive state. Only when you push the throttle forward again to make the motor rotate positively, and then pulls the throttle backward, then the motor will re-enter the braking state. Bi-directional Current Mixing Control Mode is suitable for RC Car. 9. Add unidirectional speed PID control, bidirectional speed PID control and bidirectional speed PID control with reverse gear in ADC mode. 10. Add maximum throttle percentage for different gears in ADC mode, the maximum throttle percentage for L, M and H gears can be set by the parameters Max throttle of L Gear, Max throttle of M Gear and Max throttle of H Gear respectively.
2025/11/25	V1.6	FT ESC Function Upgrade: 1. Add waterproof ESC FT85CS_WP and high-voltage ESC FT110BS.

2. Add 5 speed gears with the top 2 reserved for PAS mode.
3. Add 5 speed gears and reverse power gear with the top 2 reserved for PAS mode.
4. Add 2 encoders: AS5600_IIC and TLE5012_SPI.
5. Add new motor startup mode Open Mode. In some cases, e.g., light-load starts for underwater propeller motor, the motor might work more smoothly in Open Mode.
6. Add the max&min working voltage for ESC, MOSFET max over-current , MOSFET over-temperature current limit, and motor over-temperature current limit parameters setup.
7. Add battery parameters setup, including Cell Cut-Off start Volt(V), Cell Cut-Off End Volt(V), Cell Regen Start Volt(V)", "Cell Regen End Volt(V), Li-ion Battery Series(S), Bus Voltage Resistance.
8. Add motor_conf and control_conf. User can save the configuration and burn it to the new ESC, improving quantity production frequency.
9. Add Emergency Switch. User can connect to the PPM/ADC Brake pin for emergency stop.
10. Add Safety Start in PPM mode. This function can be shutdown. When user activate this function, the input signal should be 0 after a certain period of power-up of ESC. Only then will the PPM signal enter a valid control state. When turn it down, after powering on the ESC, when PPM signal is detected then you could startup.
11. Add safety start function in ADC mode. This function can be shutdown. When user activate this function, the input signal should be 0 after a certain period of power-up of ESC. Only then will the ADC signal enter a valid control state. When turn it down, after powering on the ESC, when PPM signal is detected then you could startup. Disabling this function enables the ESC for one-touch power-on.
12. Add 12V extra interface function in ADC mode. Support output including brake light, front light and horn. This function support ESC with 12V output including FT85BD, FT85KS, FT60BD.

		13. Add brake unlock function in ADC mode. Please press the brake handle to activate the throttle. This function prevents accidental activation of the throttle handle, which could trigger unintended acceleration and cause unexpected mechanical startup. 14. Remove the multiple gears in ADC mode. All gears of input signals are set in motor interface, taking place of current gear by power gears. 15. Optimize dual-hall sensor in PAS mode. Add throttle startup boost parameters, including startup boost throttle(%), max startup throttle rpm, max duty and the duration of throttle. 16. Add idle current function of zero throttle in PAS mode. The ESC will maintain current output for a while to prevent the motor from sudden stop. This also avoids gear chattering when the throttle is reapplied. 17. Add parameters of sensor direction, startup angle, angle sample interval(ms) and angle std threshold of dual-hall sensor in PAS mode.
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